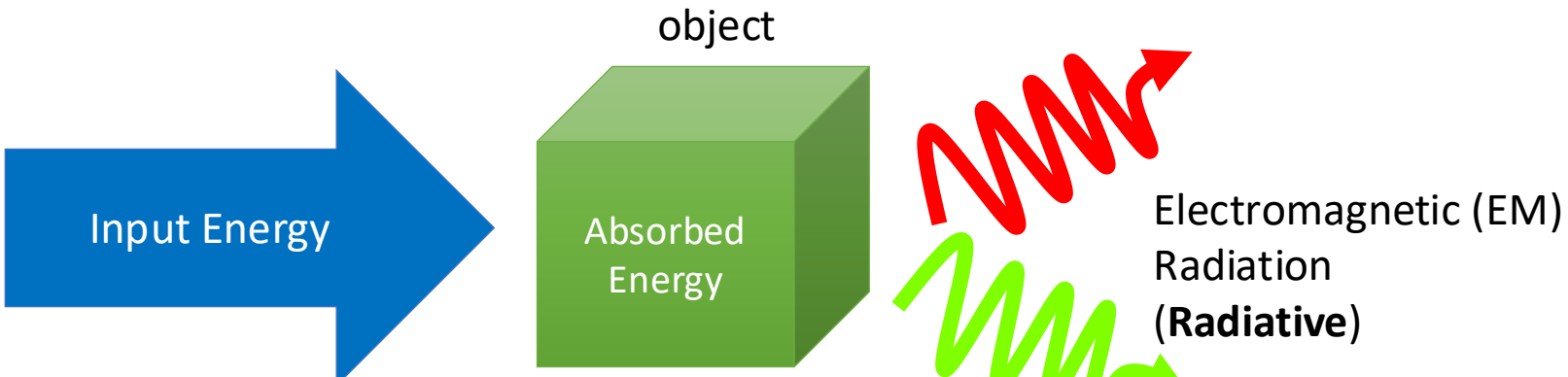
- **Part 1: Optical Imaging**
  - Generation of light
  - Plane wave propagation: geometrical optics
  - Optical Imaging modalities
    - Low scattering (microscopy, OPT, SPIM)
    - High scattering (DOT,FMT)
- **Part 2: Innovation**
  - 6 teams
  - Work on Canvas
  - Work on SWOT matrix (Strengths, Weaknesses, Opportunities, Threats)
  - Elevator Pitch
- **Part 3: Neuroimaging**
- **Part 4: Biomedical analysis technologies**
  - Elisa
  - FACS
- **Part 5: Biomedical Optics**
- **Part 6: Imaging Trivial**
- **Part 7: Presentation of Innovation Teams**

Properties of light and Lenses

Reminder of how light is generated

How is Electromagnetic Radiation Produced?



In order to understand how this electromagnetic (EM) radiation is produced, we need to understand the basic energy levels present in atoms and molecules. By giving these atoms or molecules enough input energy, we might convert that energy either into non-radiative or radiative energy. In this latter case, we would have EM radiation. If the wavelength of this radiation is in the **400nm-780nm** range, we will then have **visible** light.

How is Electromagnetic Radiation Produced?



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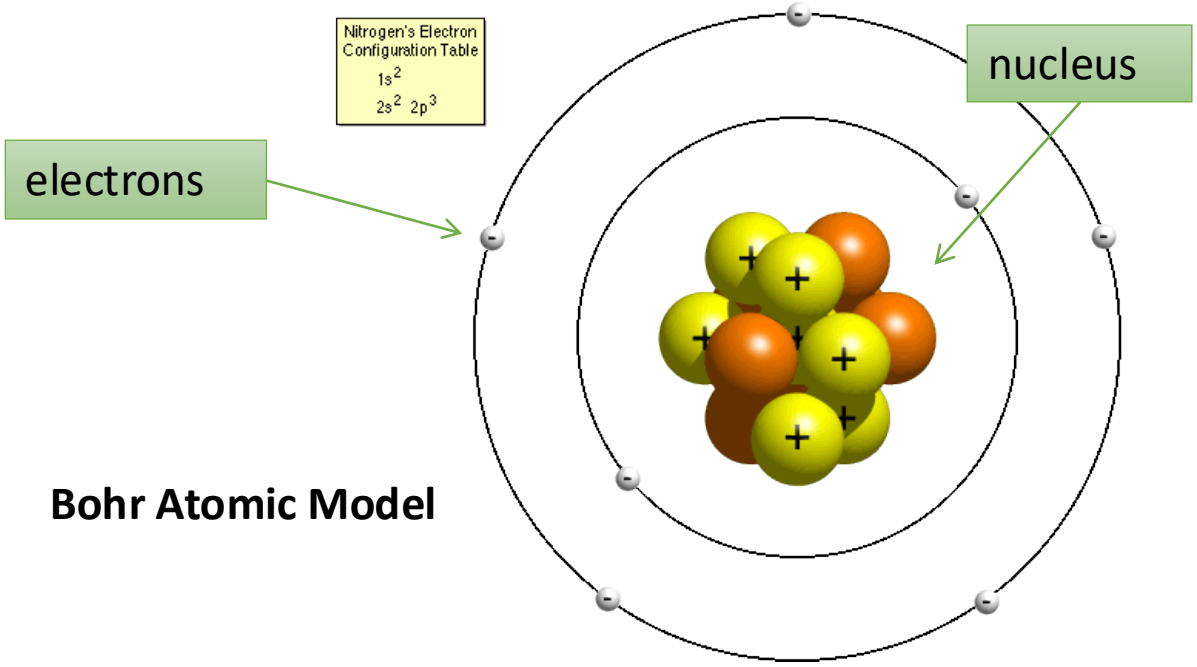
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Energy Levels of an atom or molecule

The term is used for the energy levels of electrons in atoms or molecules, which are bound by the electric field of the nucleus. The energy spectrum of a system with energy levels is said to be quantized. Roughly speaking, a molecular energy state is the sum of the electronic, vibrational, rotational, nuclear, and translational components, such that:

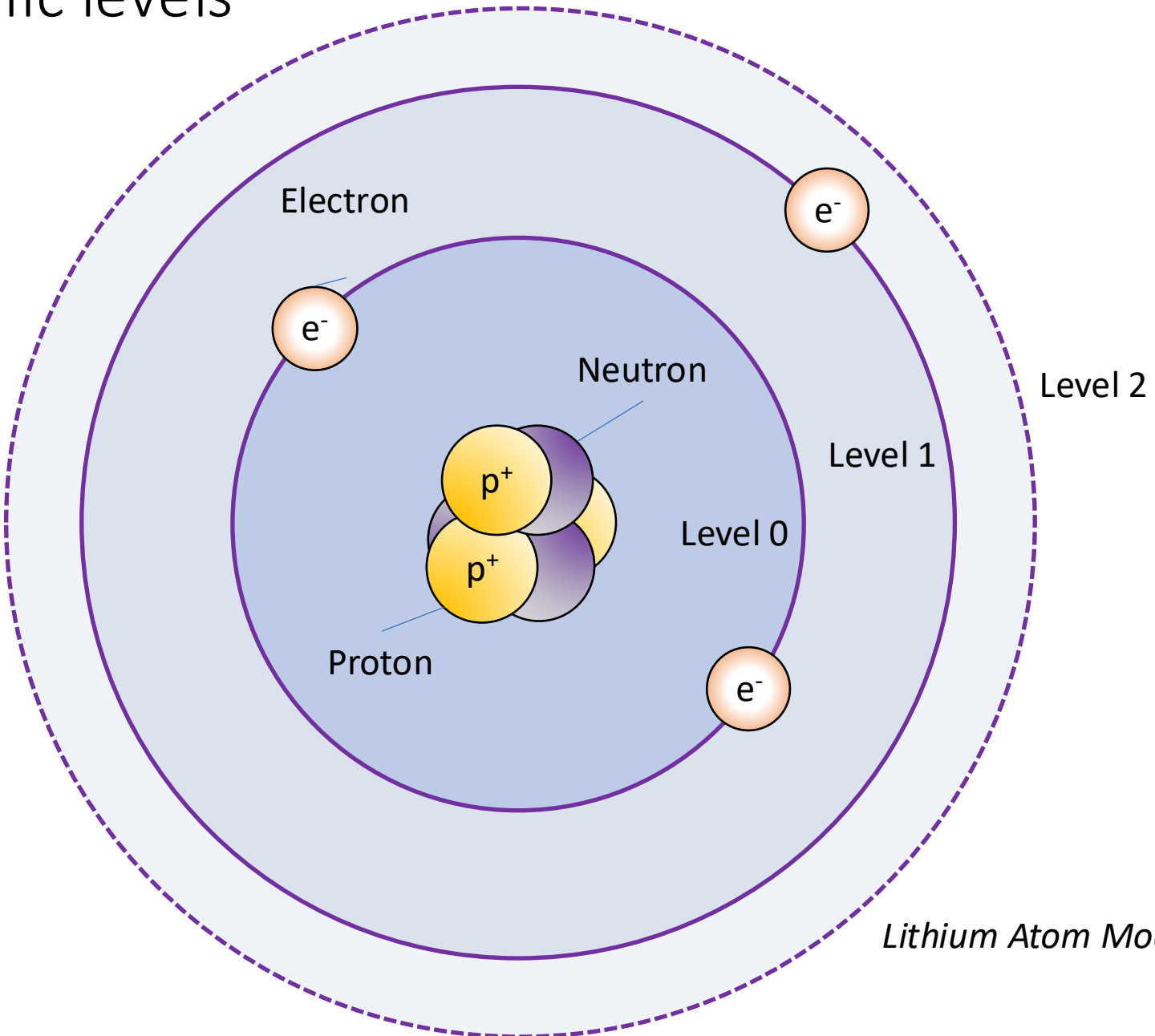
$$E = E_{electronic} + E_{vibrational} + E_{rotational} + E_{nuclear} + E_{translational}$$

What this means is that if we give the system (the atom or molecule) enough energy, it will reach a higher energy – excited - level. This will happen by promoting an electron to an excited level.



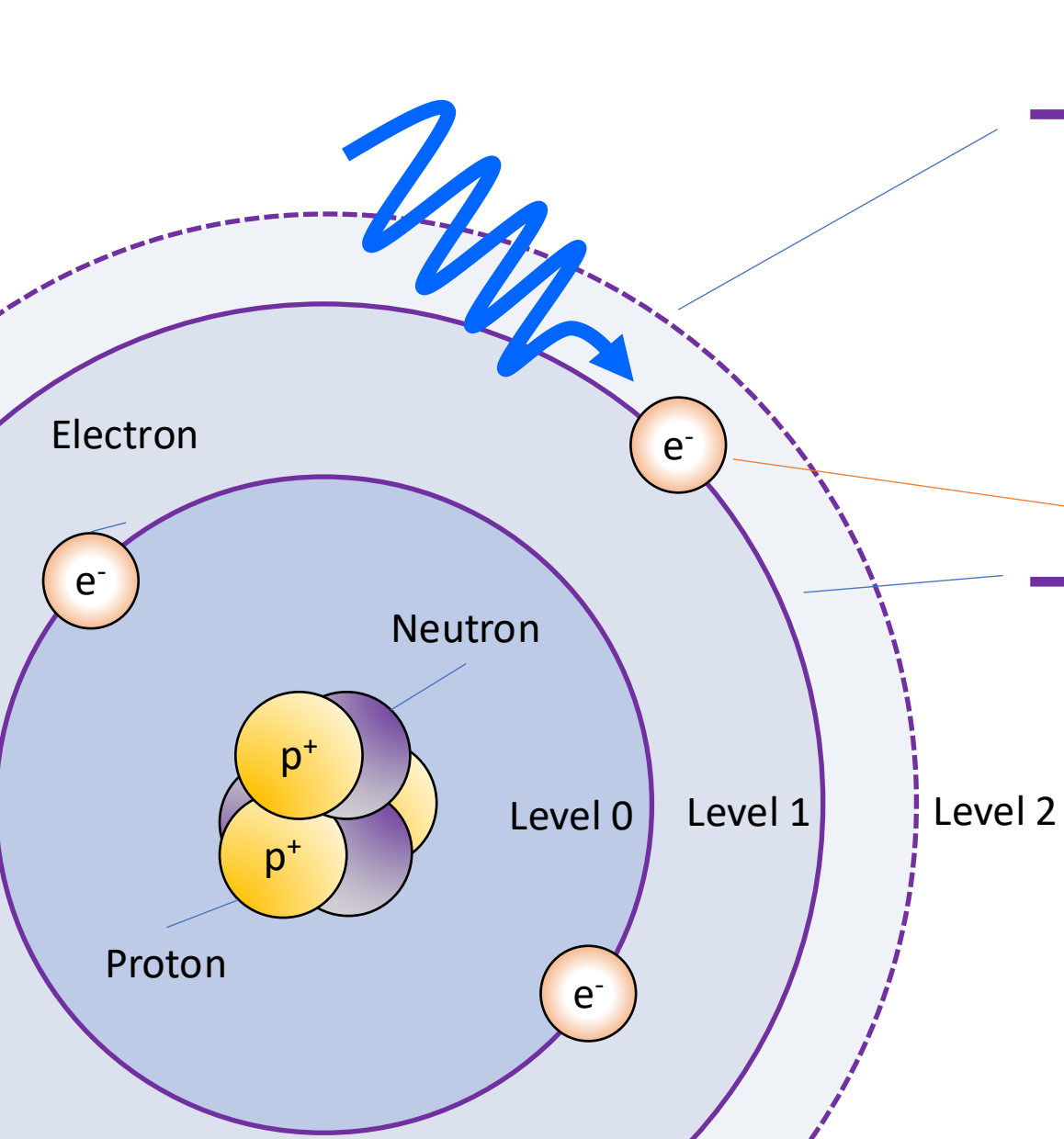
Bohr Atomic Model

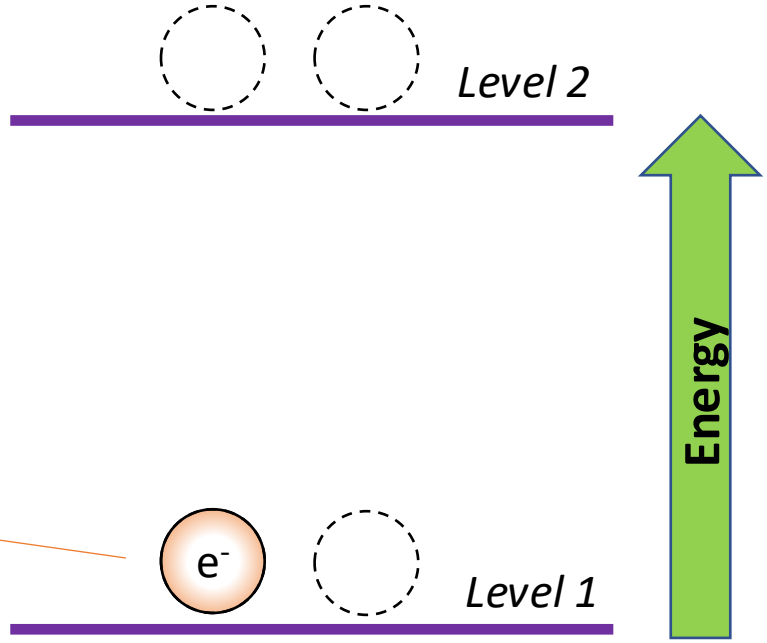
Atomic levels



Lithium Atom Model

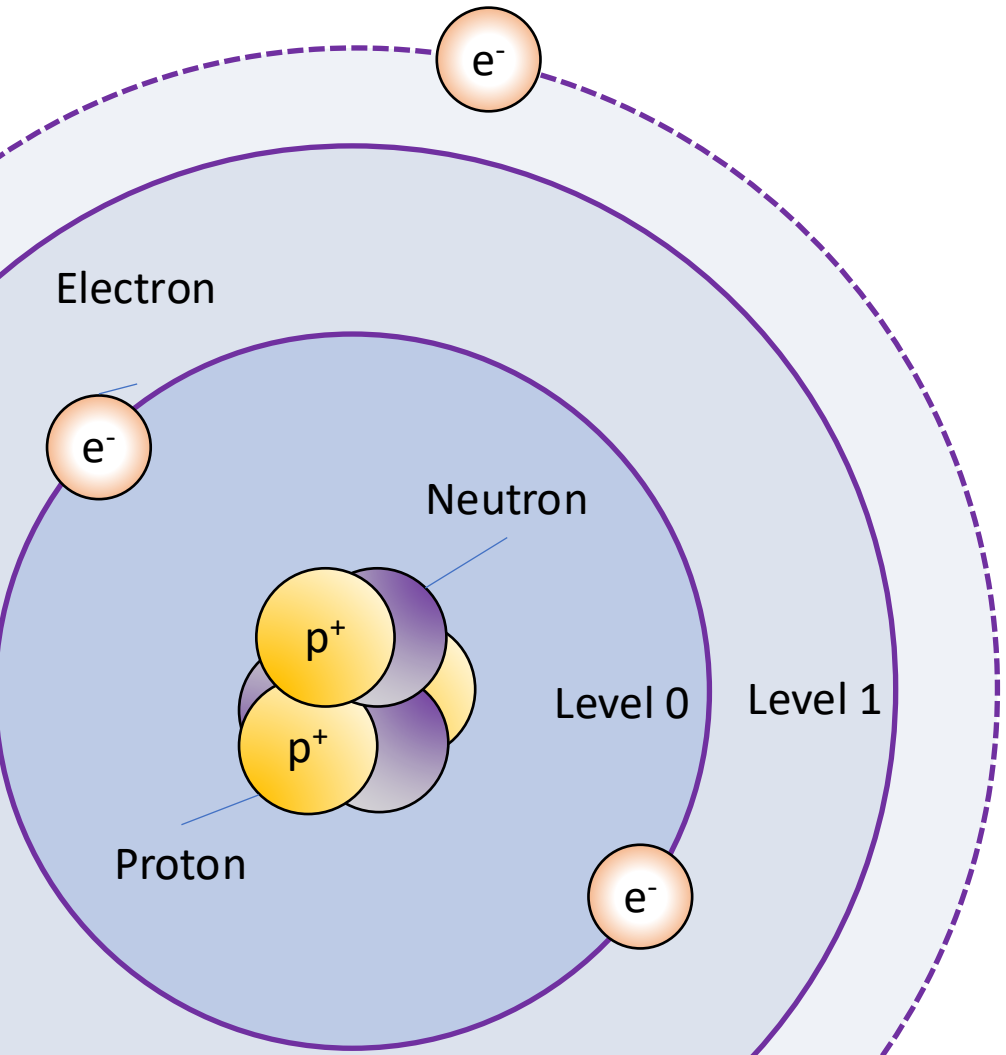
Atomic levels

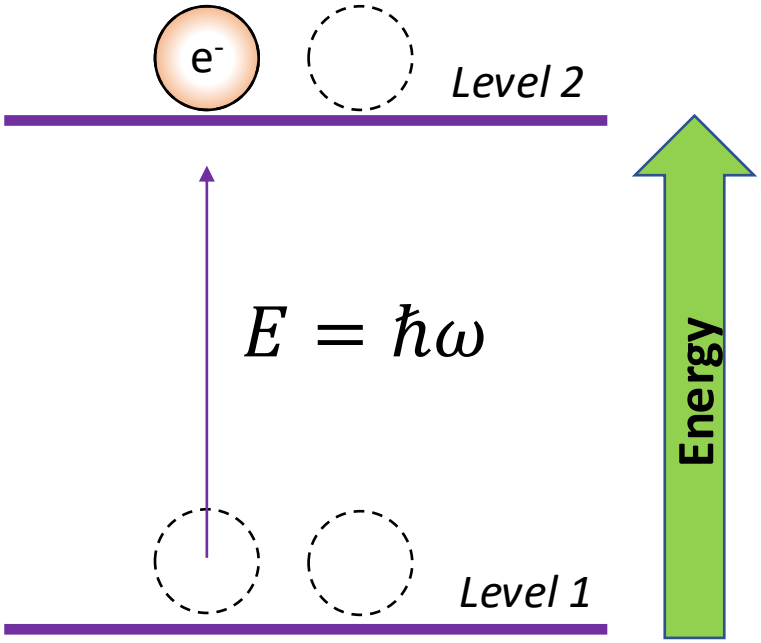


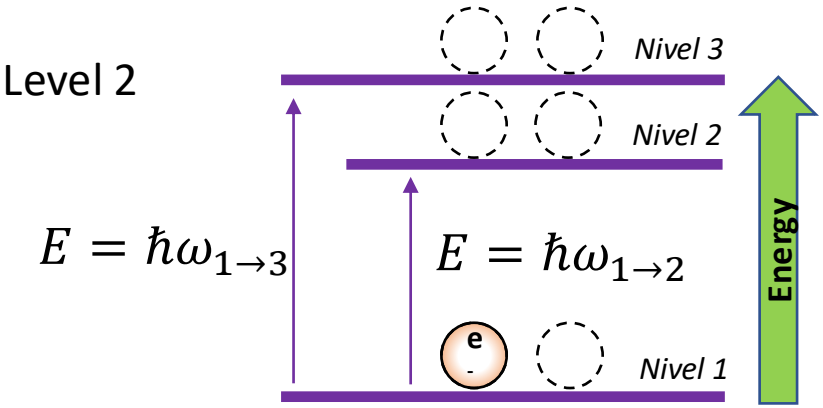


If the energy is adequate....

Atomic levels

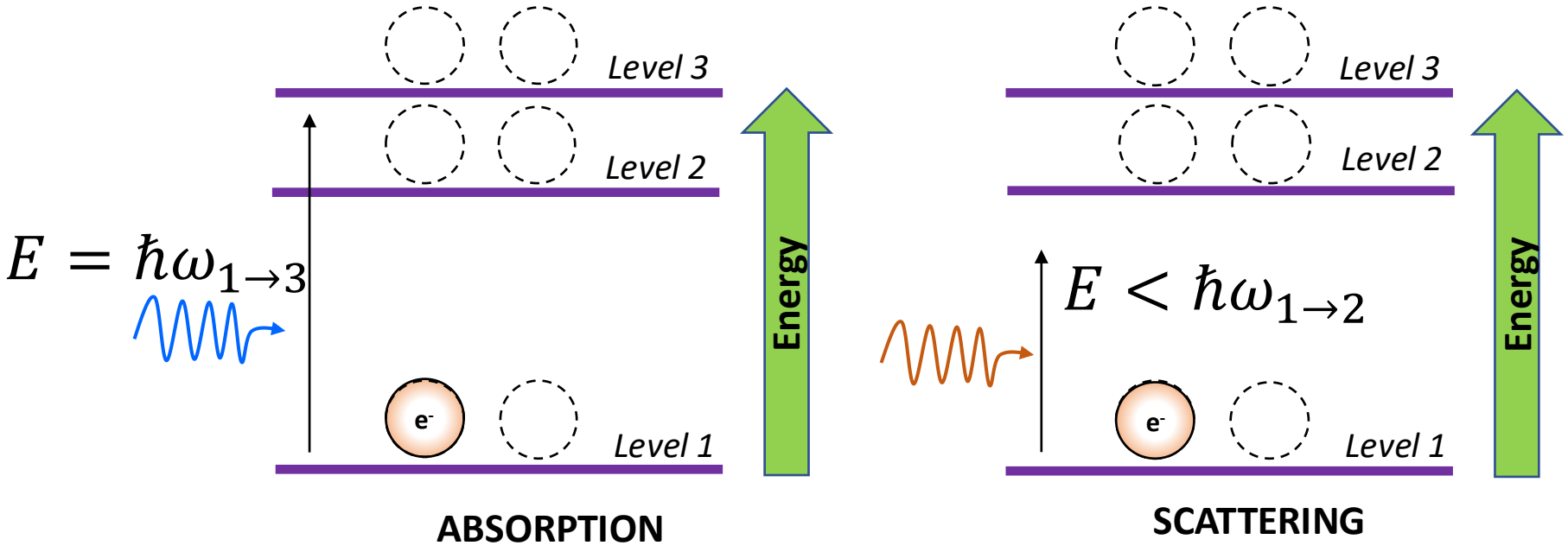






Atomic levels

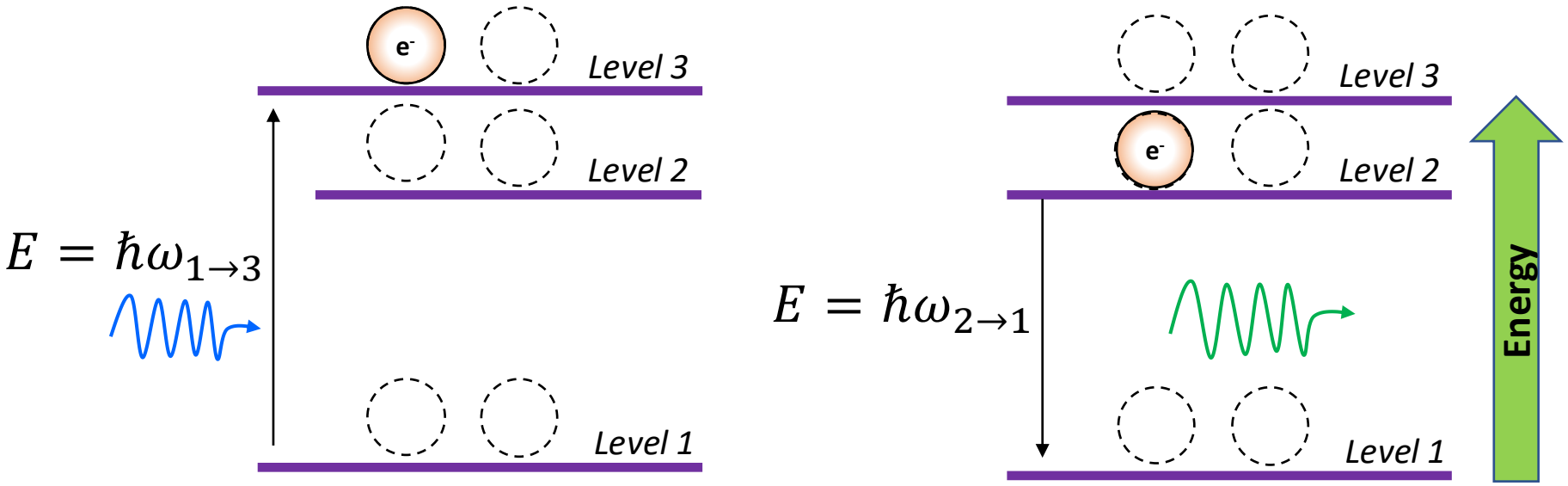
- Each molecule has its own energy levels
- Each energy level has an associated frequency
- To promote an electron at that level of excitement, the energy provided has to be $E = \hbar\omega$



In case of absorption...

There are two options:

A) Go back to relaxed state by emitting an EM wave

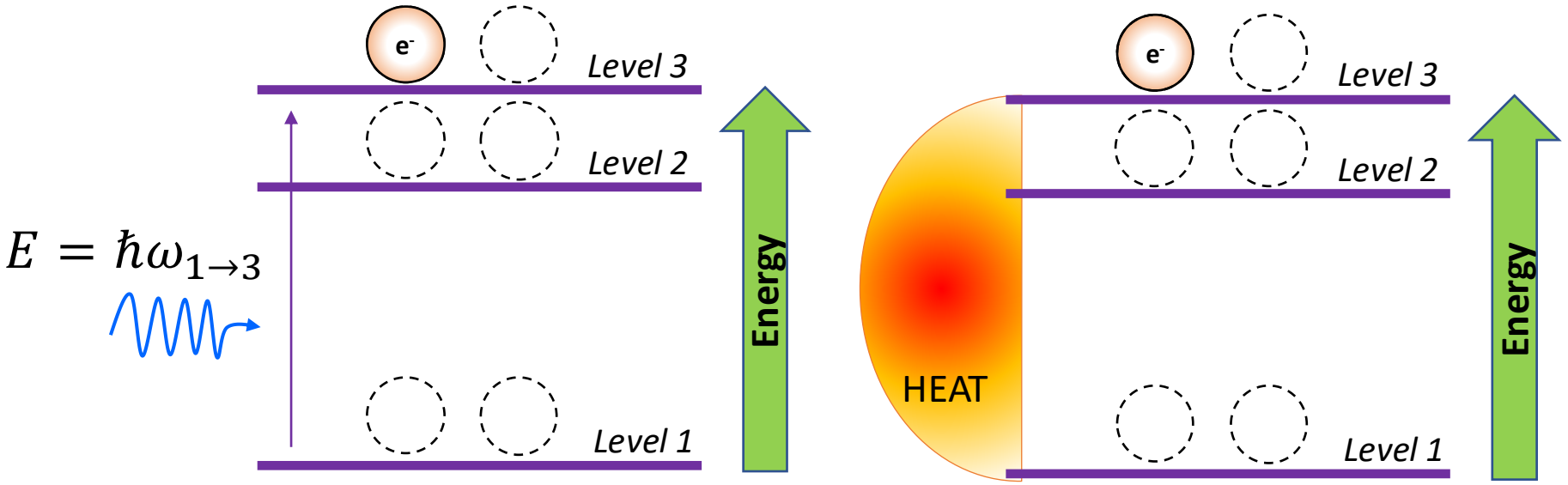


In case of absorption...

There are two options:

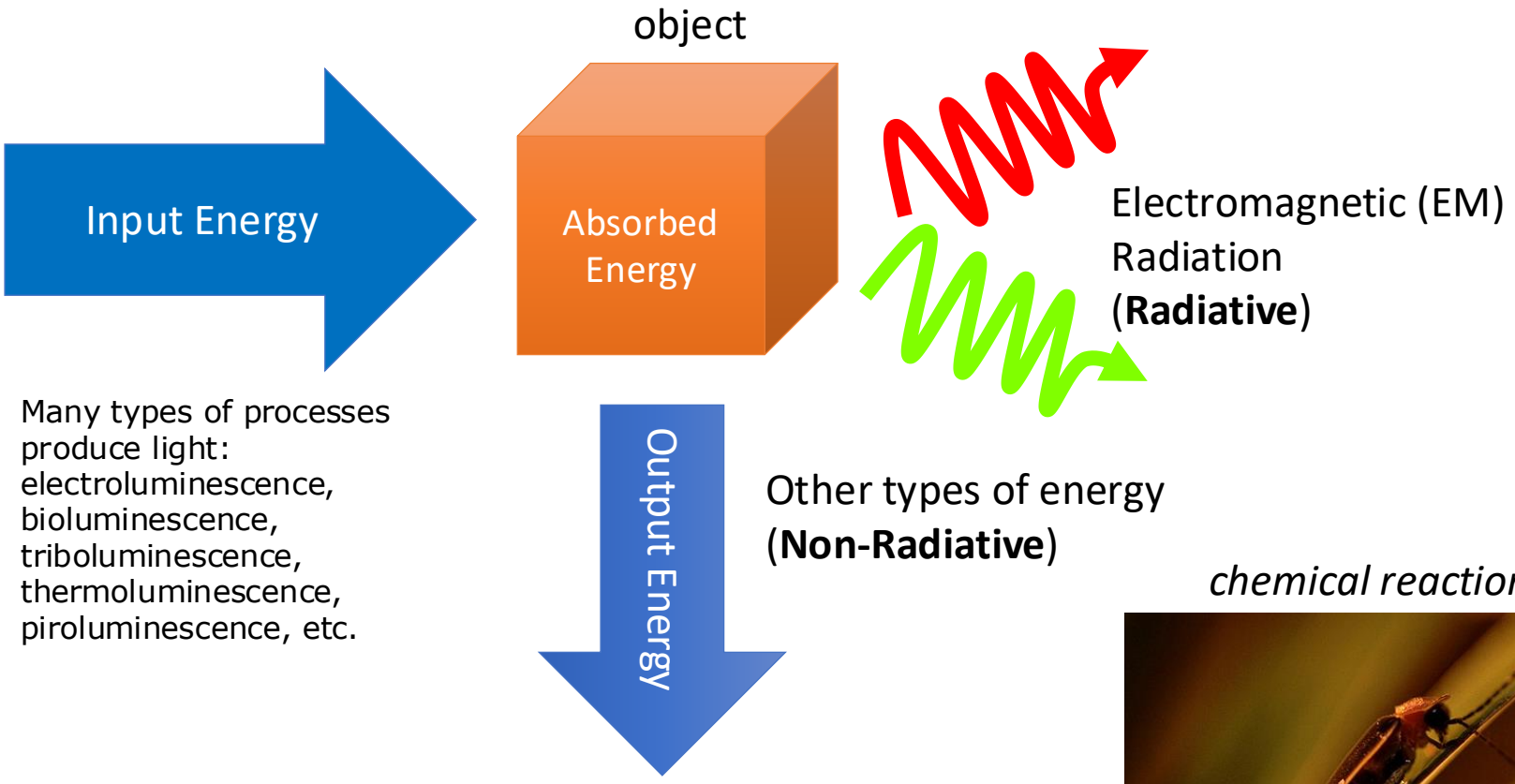
A) Go back to relaxed state by emitting an EM wave

B) Returning that energy to the medium in the form of heat



So, all color we see is due to

I) Light Emission

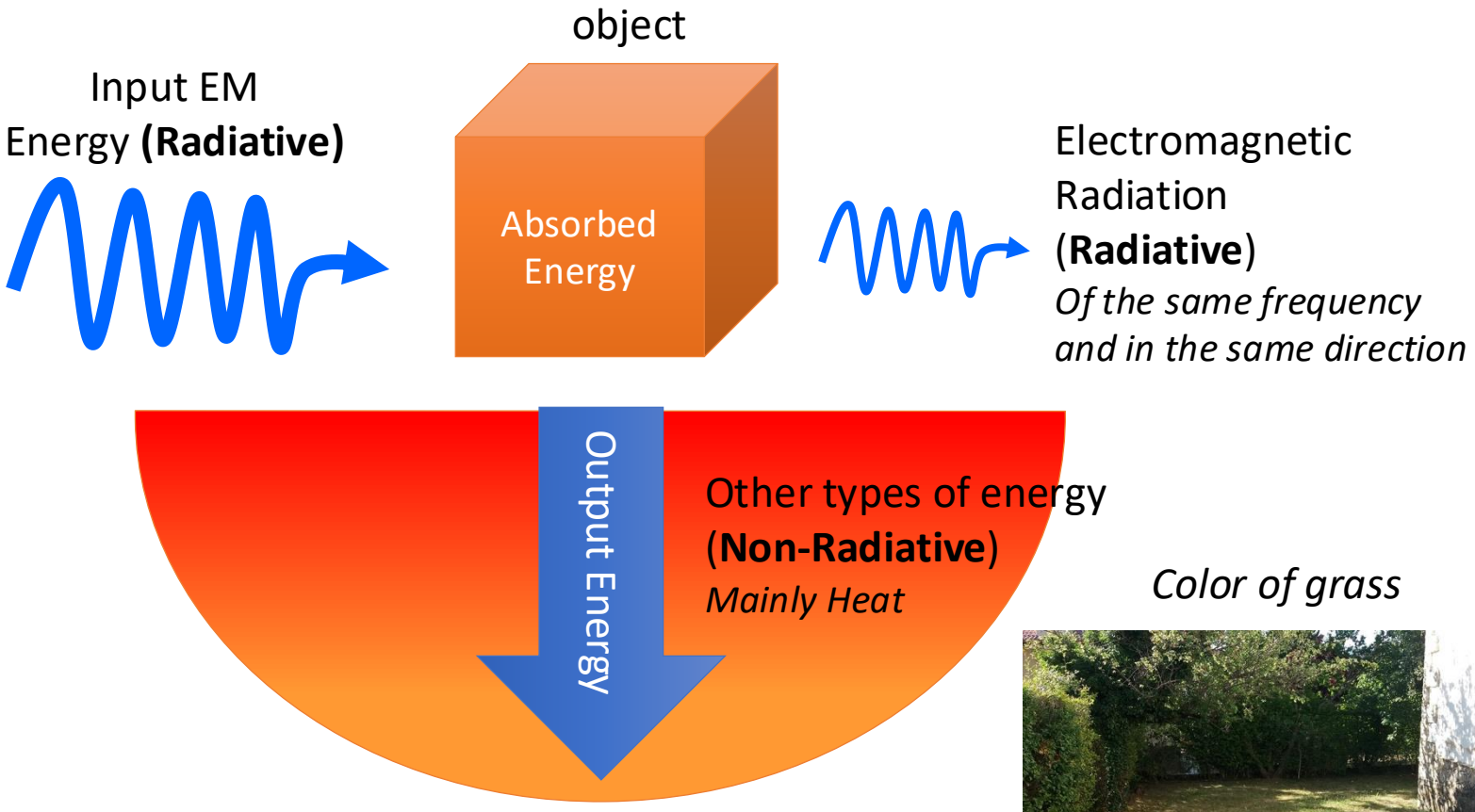


chemical reaction



and

II) Light Absorption

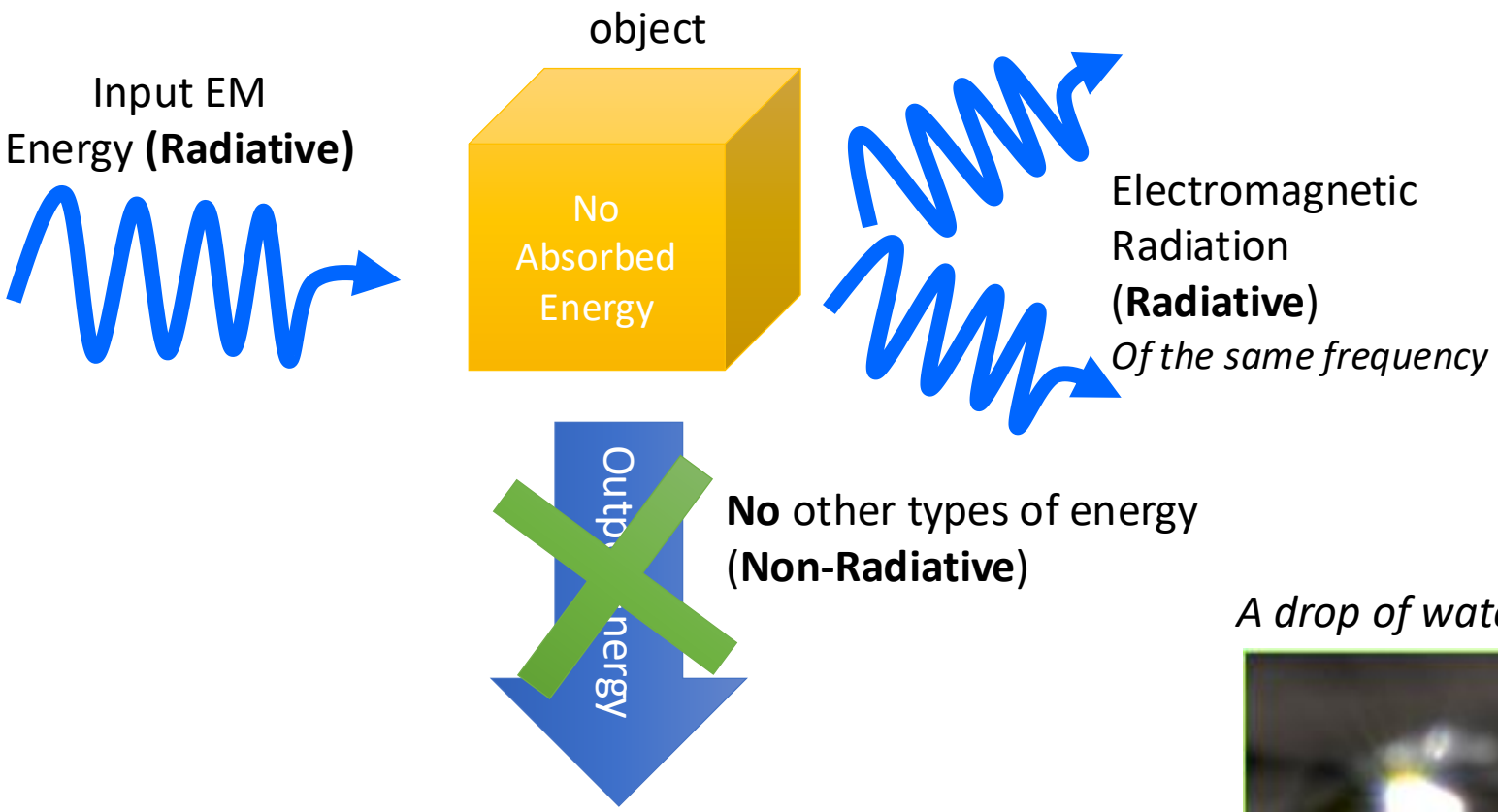


Color of grass



and finally

III) Light Scattering

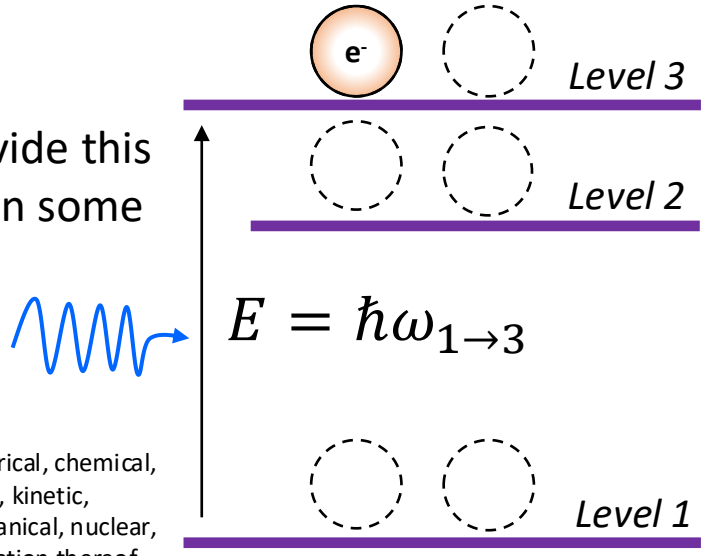


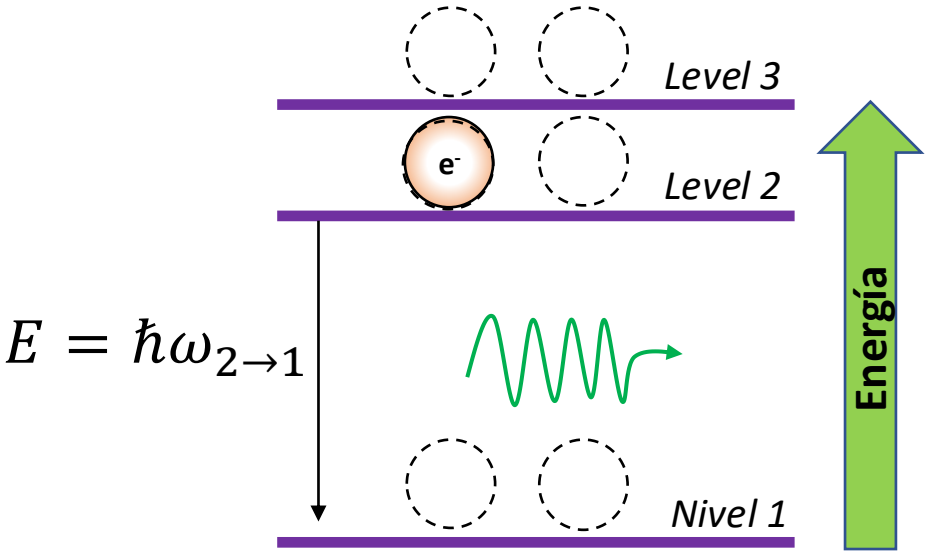
A drop of water



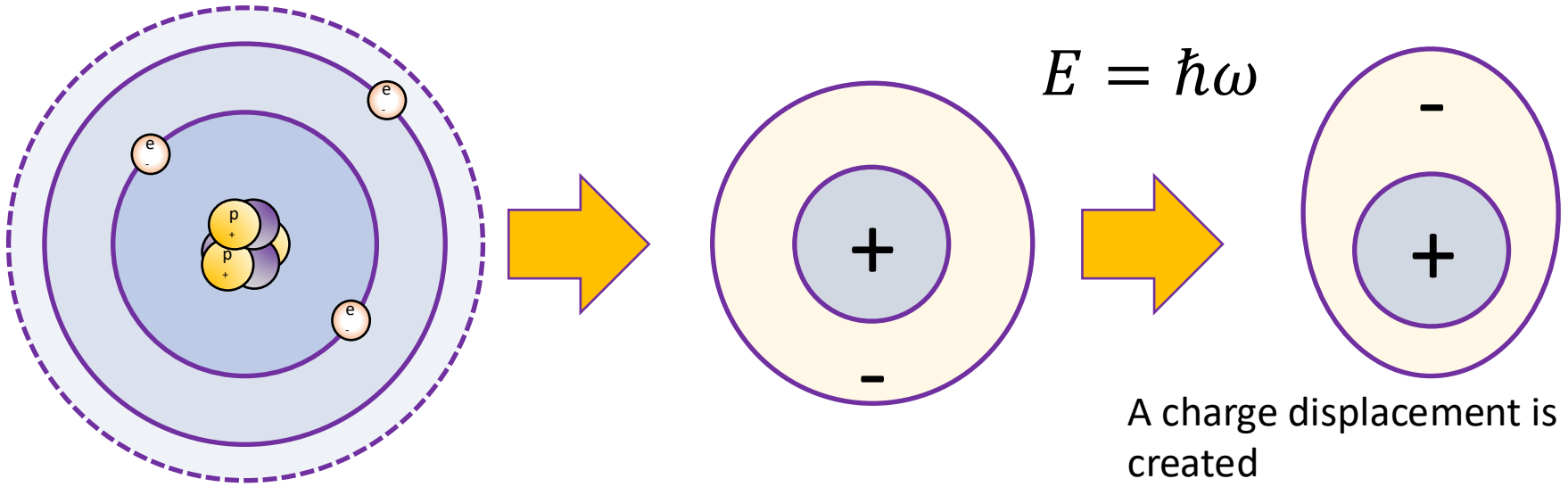
let's go to the first case: **Light emission**

We provide this energy in some way*





* Thermal, electrical, chemical, electromagnetic, kinetic, magnetic, mechanical, nuclear, and any combination thereof.



A charge displacement is created

Electronic orbitals: excited states

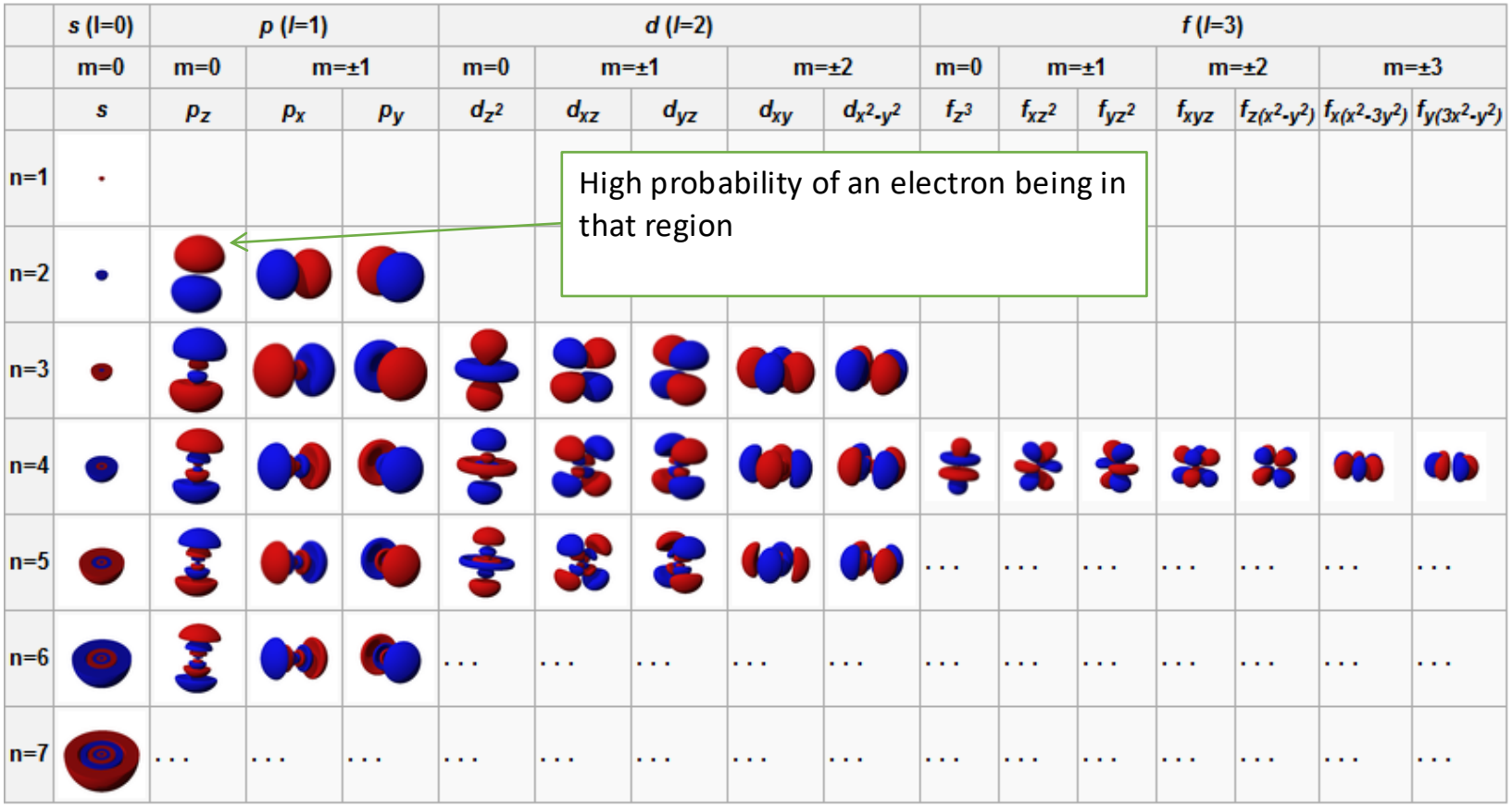
Represented by the 3 quantum numbers (n, l, m):

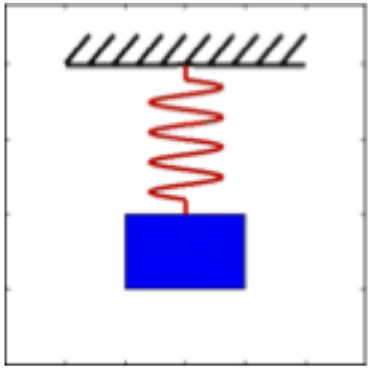
n ' main number (several orbitals can be associated with each n: these orbitals together are the electron levels)

l ' azimuth number

m ' magnetic number

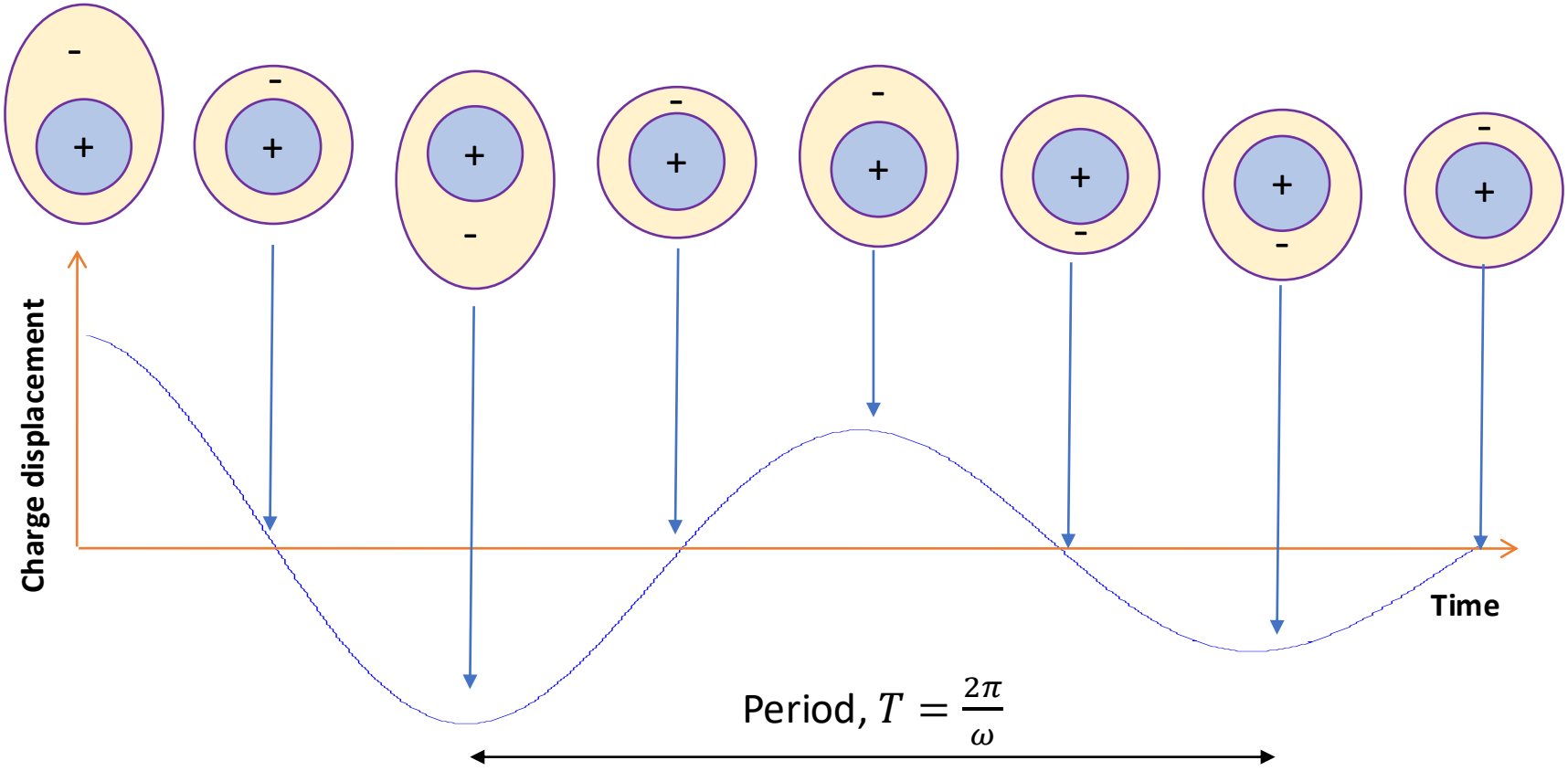
Impressive → https://ptable.com/#Orbital





Light emission

The evolution of the excited state to the stable state produces an oscillation similar to a damped spring:

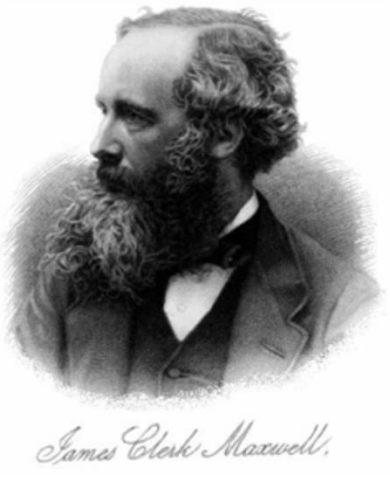


The frequency of this oscillation only depends on the difference between the energy levels:

$$E = \hbar\omega$$

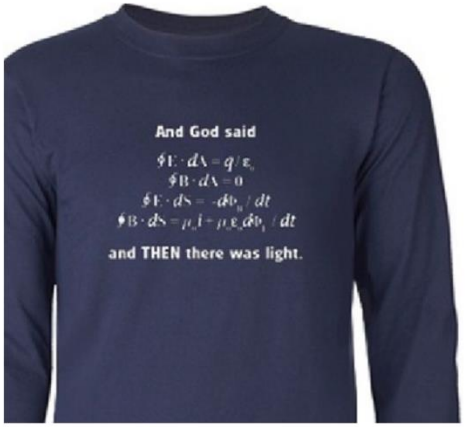
Frequency (rad/s)

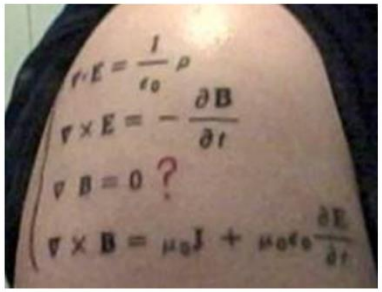
Maxwell's equations













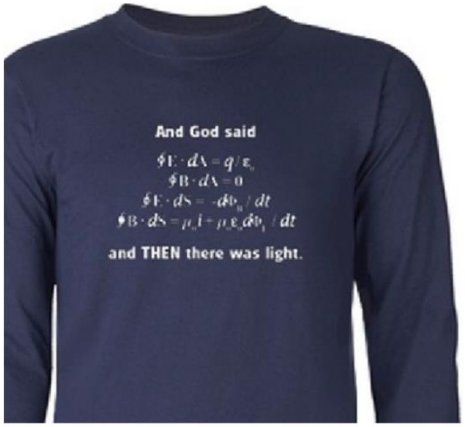
J. Maxwell	Developed the electromagnetic field theory and original equations.
O. Heaviside	Reduced Maxwell's cumbersome system to the compact vector form; eliminated many unnecessary potentials; made the equations practical.
H. Hertz	Reformulated electrodynamics, developed moving-body equations, clarified the physical interpretation, and experimentally confirmed electromagnetic waves.
J. W. Gibbs	Developed vector analysis as a mathematical language, independently of electromagnetism.

Maxwell's equations

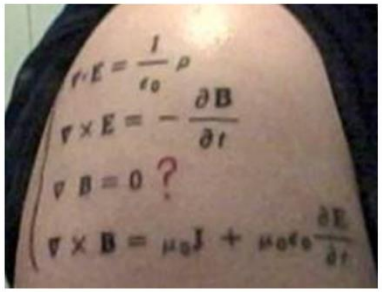








$\nabla \cdot \mathbf{B}=0$?

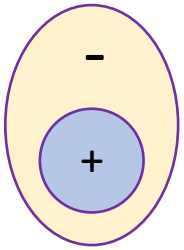


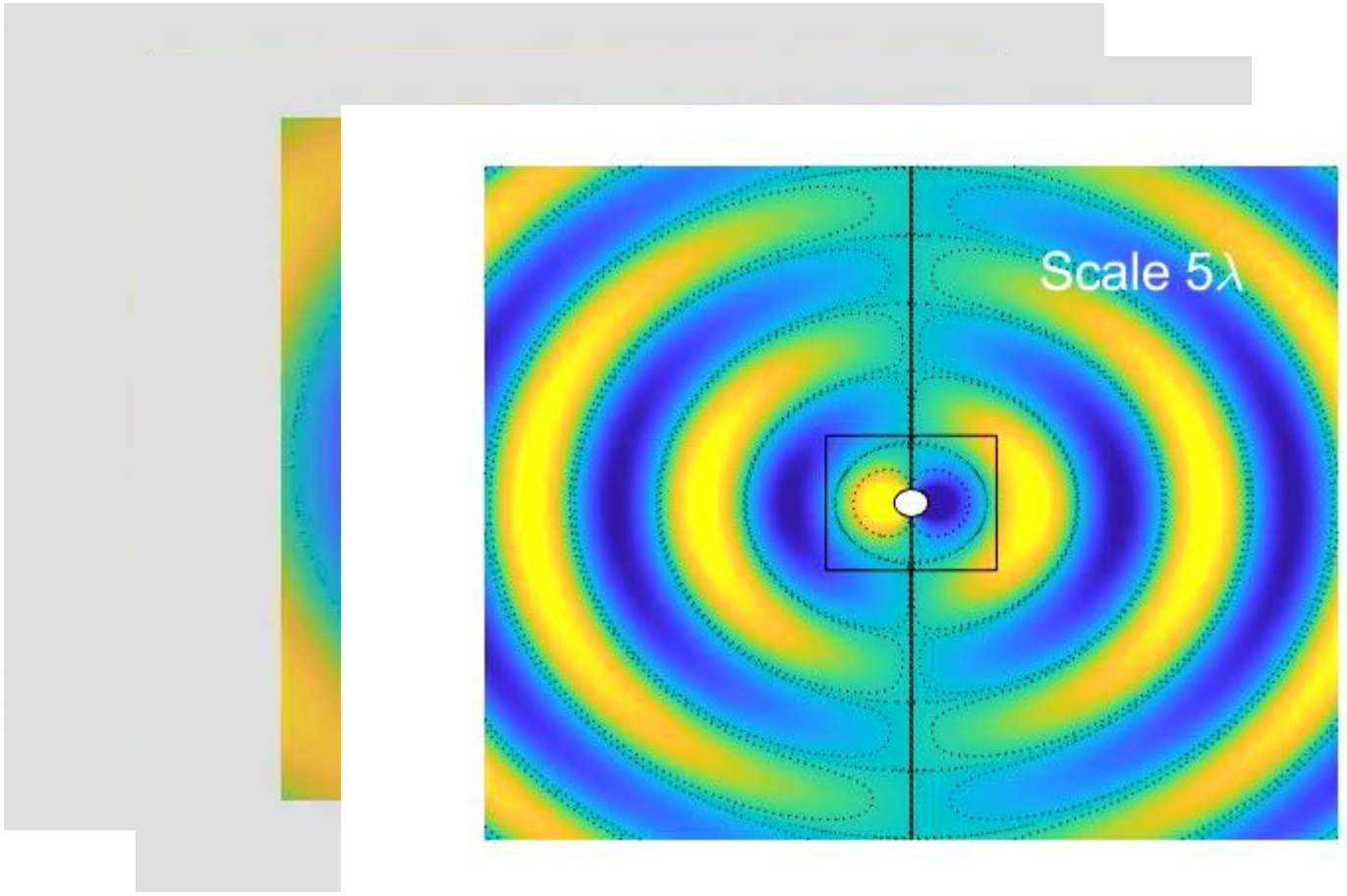


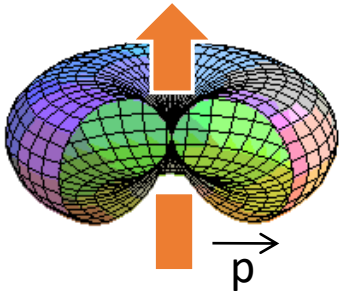
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Light emission

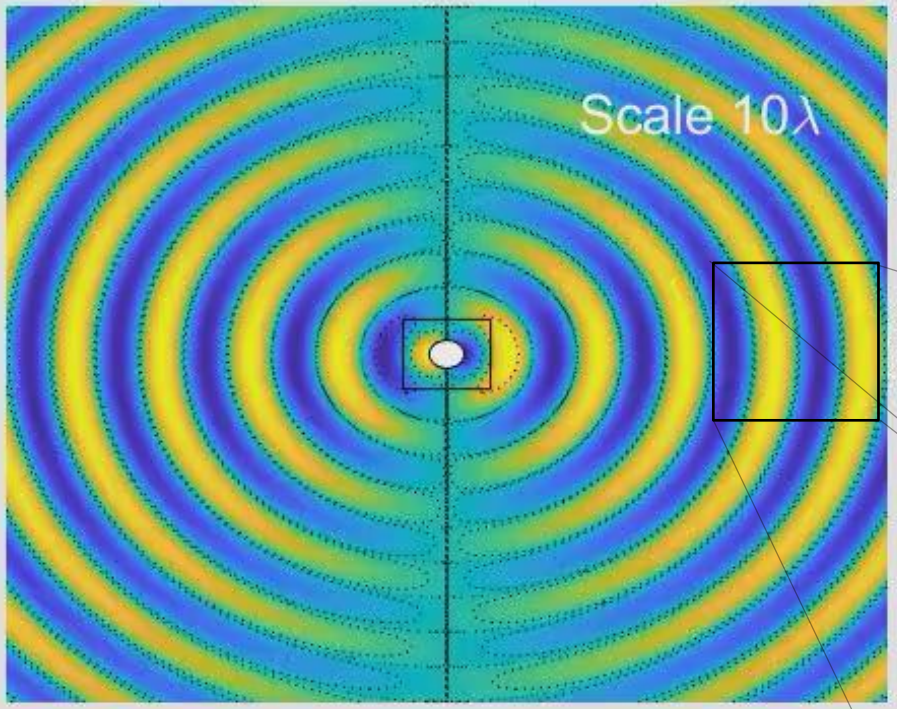
As Maxwell´s equations state, a moving charge generates an electromagnetic field. Thus we obtain radiation by a dipole that propagates at the speed of light:







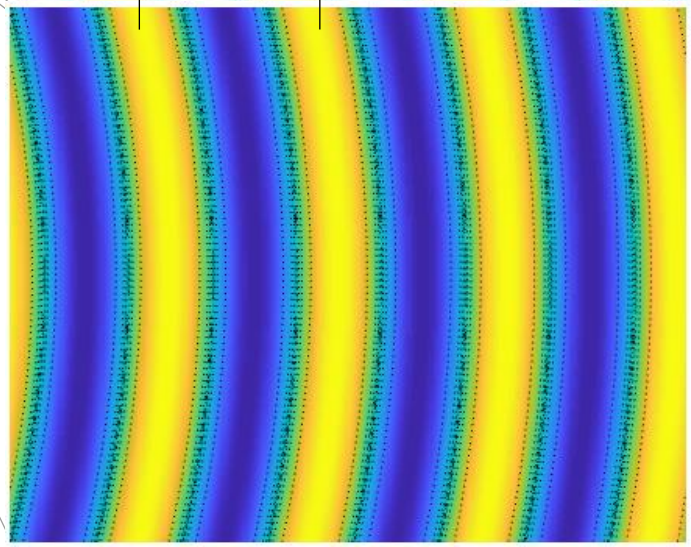
Light Emission





wikipedia

Wavelength, λ



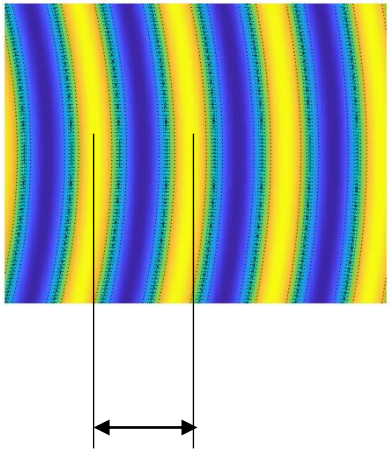
If we look far away from the source what we have is a ***plane wave***

Light emission

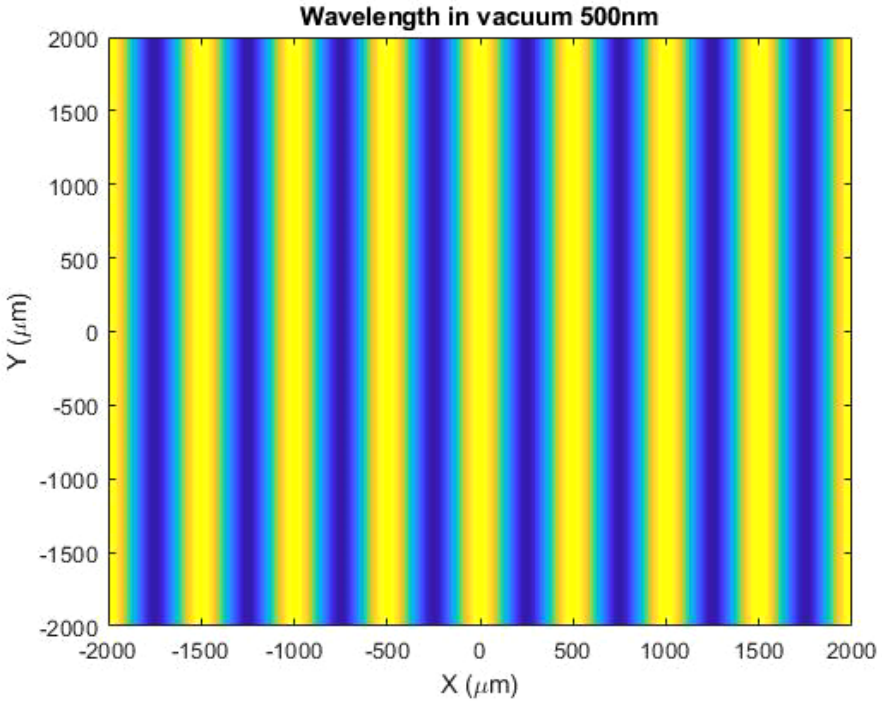
- As we've seen, an energy difference has an associated frequency,

$$E = \hbar\omega$$

- This oscillation frequency is propagated at the speed of light, c
- The spatial distance between maximums is the wavelength. λ



Wavelength, λ



Propagation of a plane wave

Light emission: important points

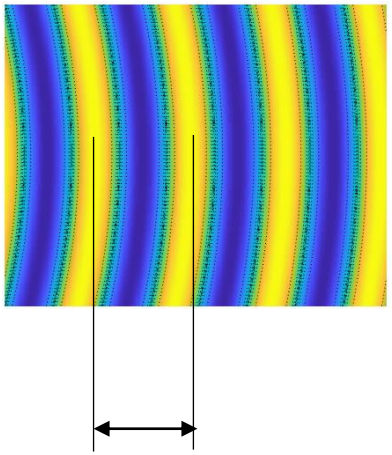
- The oscillation frequency, ω propagates at the speed of light, c of the medium

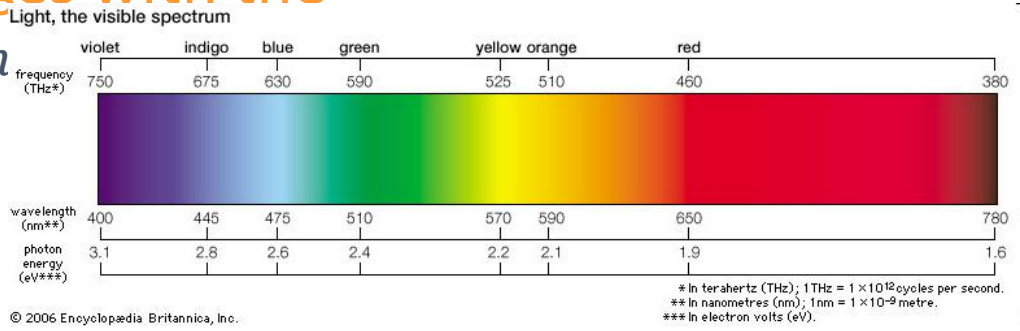
- The speed of light will be different in each medium, depending on the refractive index:
  $c_{medium} = c_0/n_{medium}$

- When changing medium the frequency of oscillation ω does not change

- The wavelength, λ, changes with the refractive index, n_{medium}

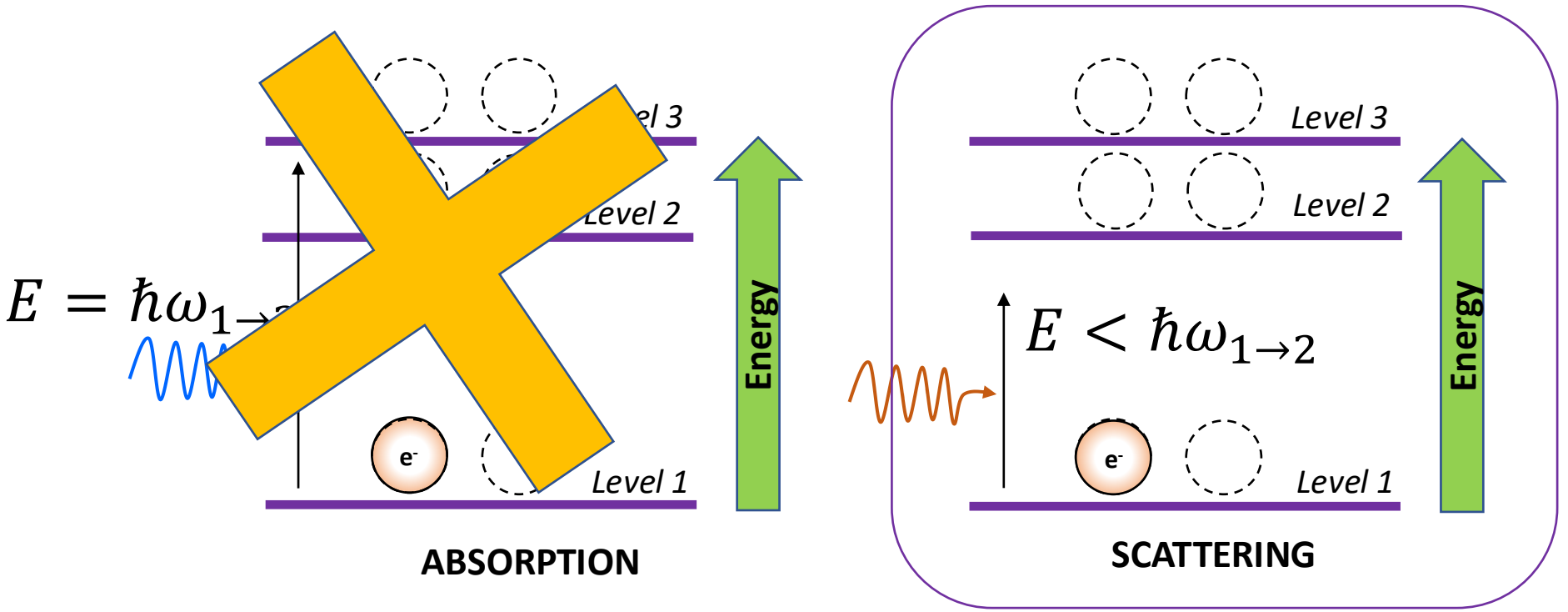
$$\lambda_m = \frac{2\pi}{\omega} c_m = \frac{2\pi}{\omega} \frac{c_0}{n_m}$$





Interaction with non-absorbing media

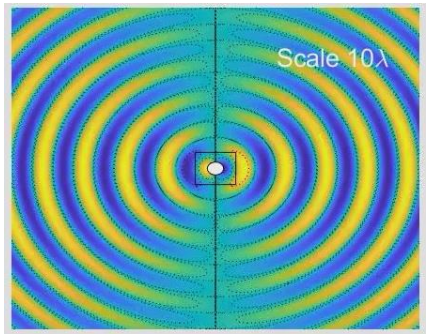
- We have already generated a plane wave through an oscillation.
- Now let's see what happens when this electromagnetic wave interacts with a medium that does not absorb that frequency...

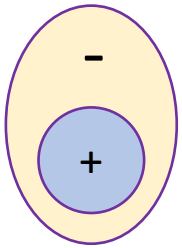


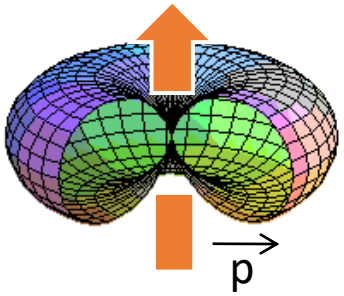
Scattering

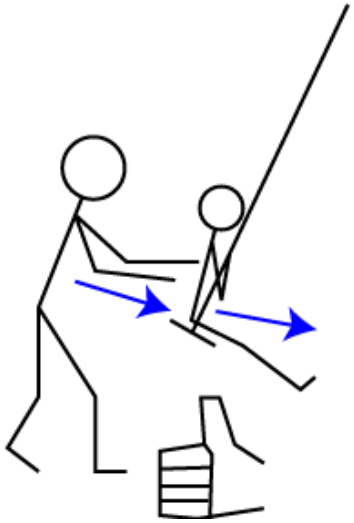
- When light is not absorbed, the electric field forces the bound charge to oscillate.
- The charge behaves as a forced oscillator driven at the frequency of the incident wave.
- The oscillating charge acts as an oscillating electric dipole.
- The total scattered field is the superposition of the radiation emitted by all these dipoles.

https://phet.colorado.edu/sims/normal-modes/normal-modes_en.html

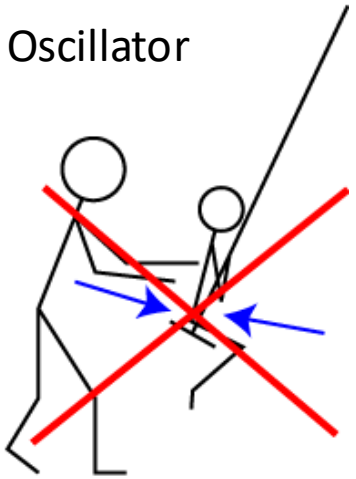








Forced Oscillator



Interaction with different non-absorbing media: *How lenses work*

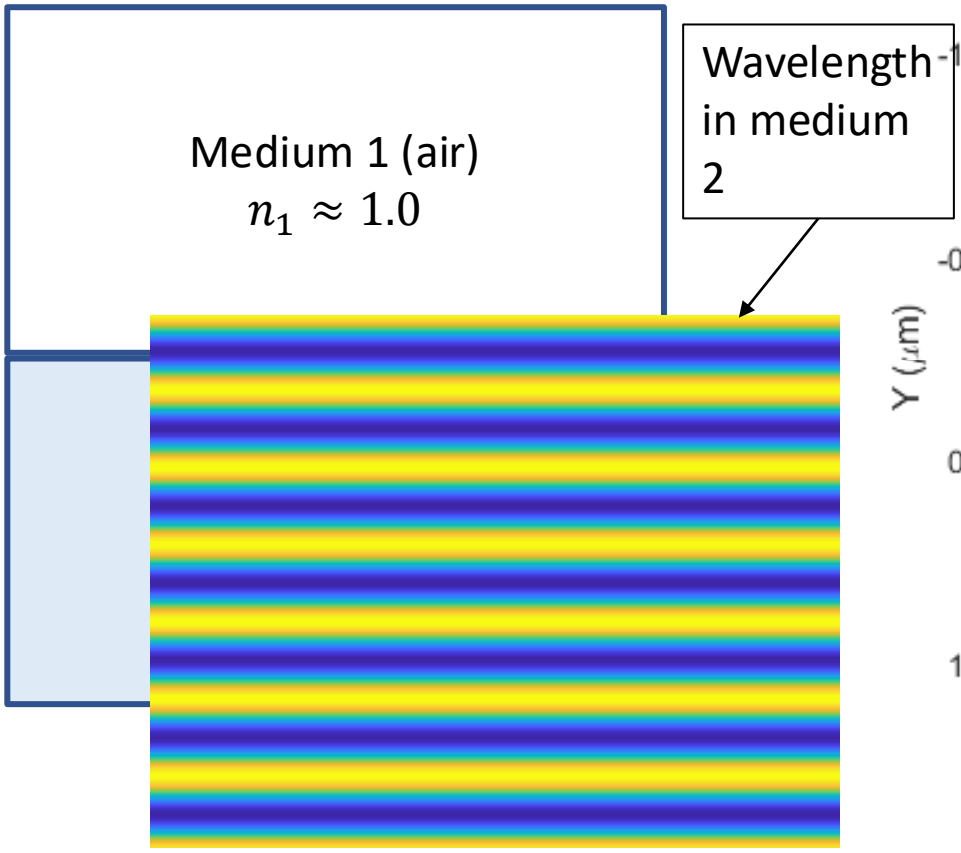
From a plane to a lens

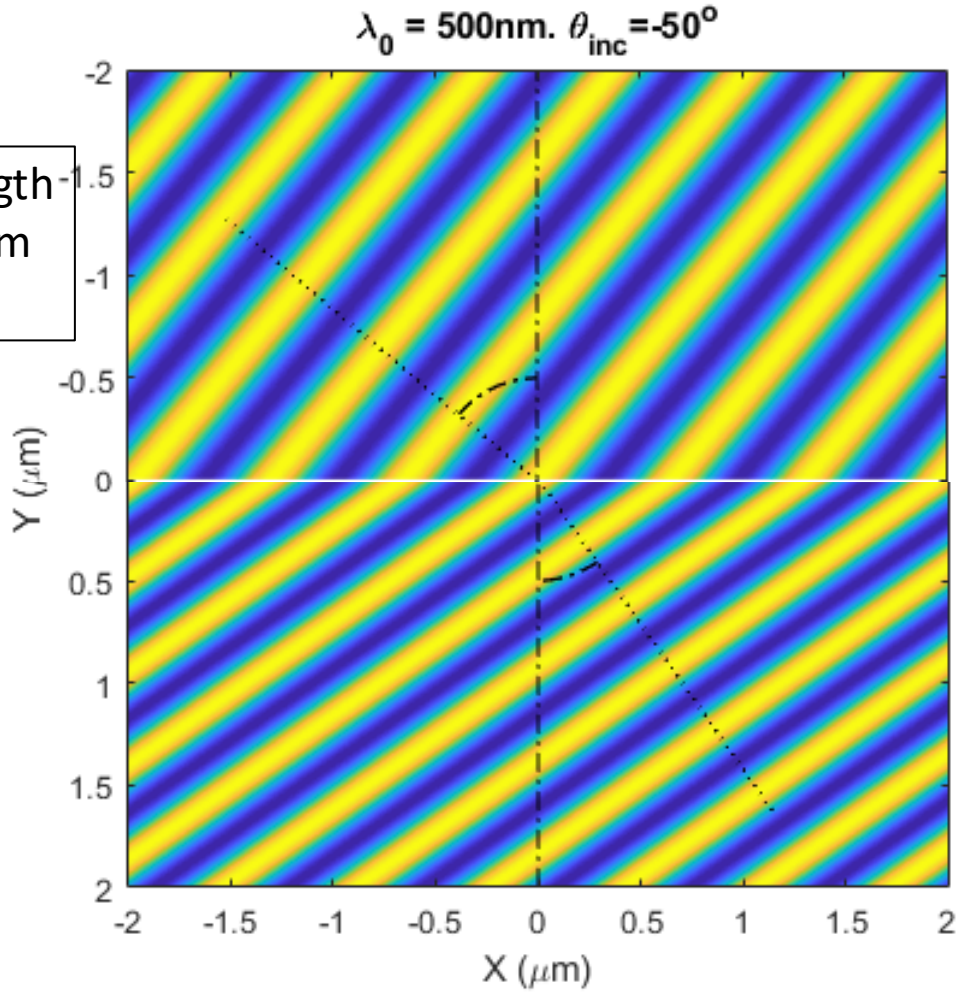
Snell's Law

Snell's law appears from the continuity of the electric field on the interface between two different media:

$$n_1 \sin\theta_{inc} = n_2 \sin\theta_r$$

$$\lambda_m = \frac{2\pi}{\omega}\frac{c_0}{n_m}$$



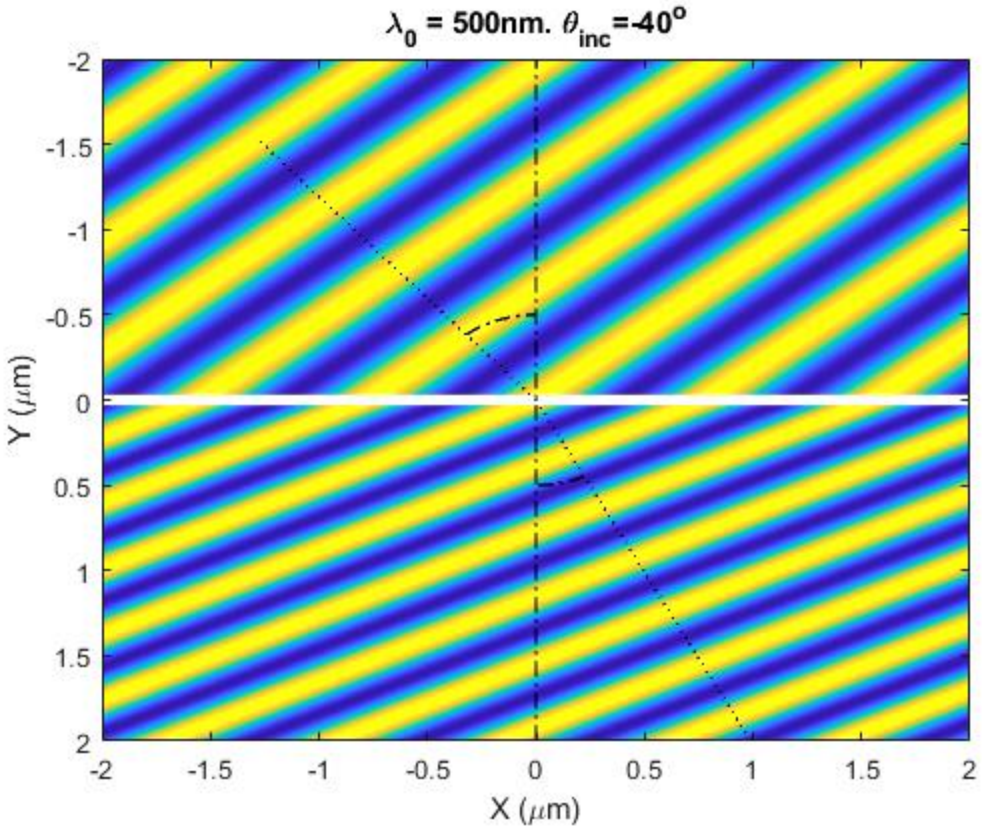


Interaction with a plane

$$n_1 \sin\theta_{inc} = n_2 \sin\theta_r$$

Medium 1 (air)
$n_1 \approx 1.0$

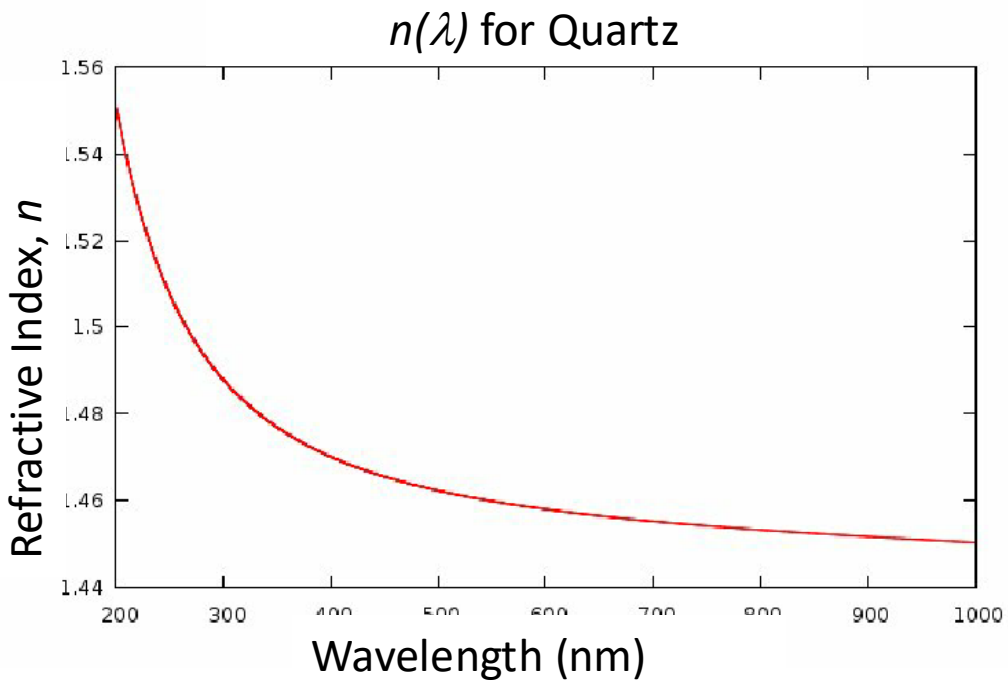
Medium 2 (water)
$n_2 \approx 1.333$



Dispersion of light

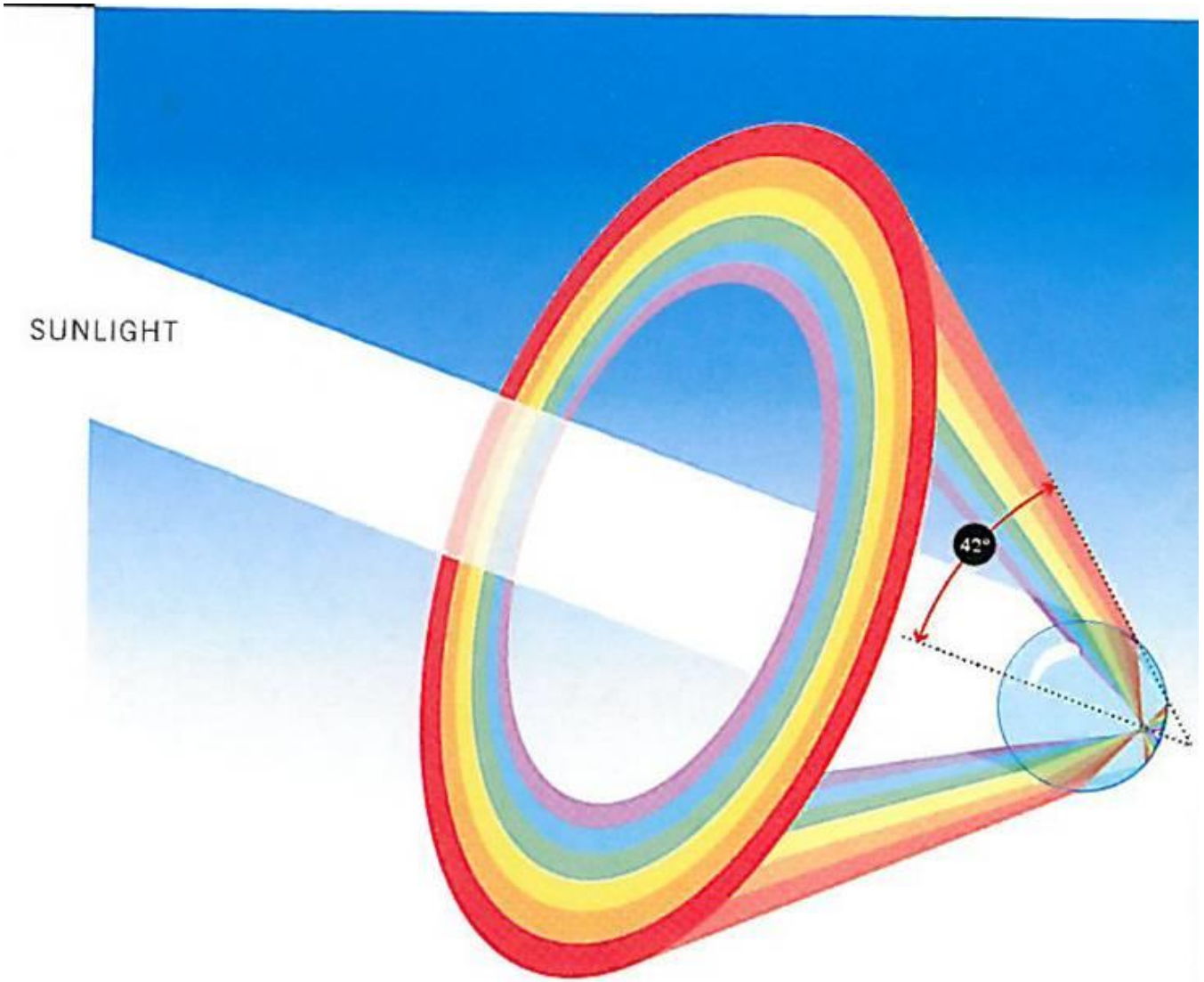
$$n_1(\omega)\sin\theta_{inc} = n_2(\omega)\sin\theta_r$$

- Although Snell's law does not depend on frequency, the refractive index does. This produces light dispersion by a prism, a drop of water, etc.





The Rainbow

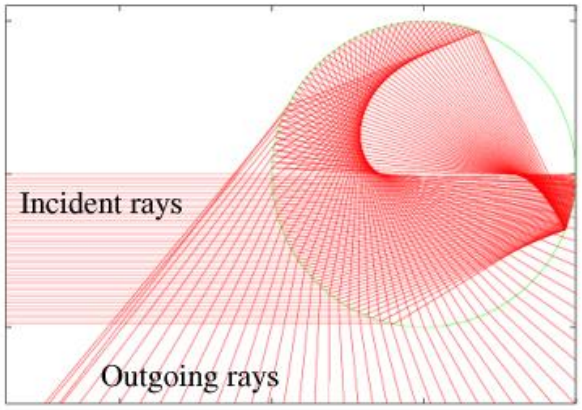


Color and Light in Nature by David K. Lynch and William Livingston

The Rainbow

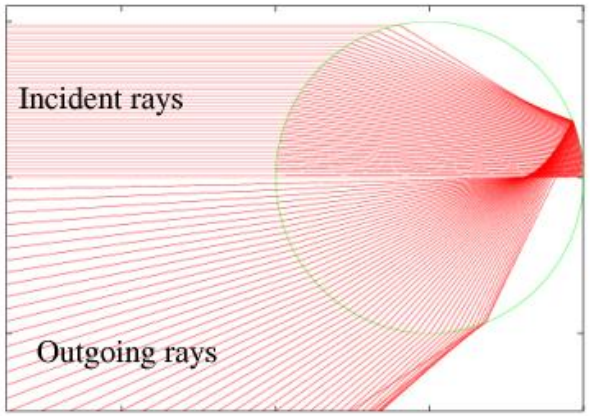


Secondary Rainbow



Alexander's Band *

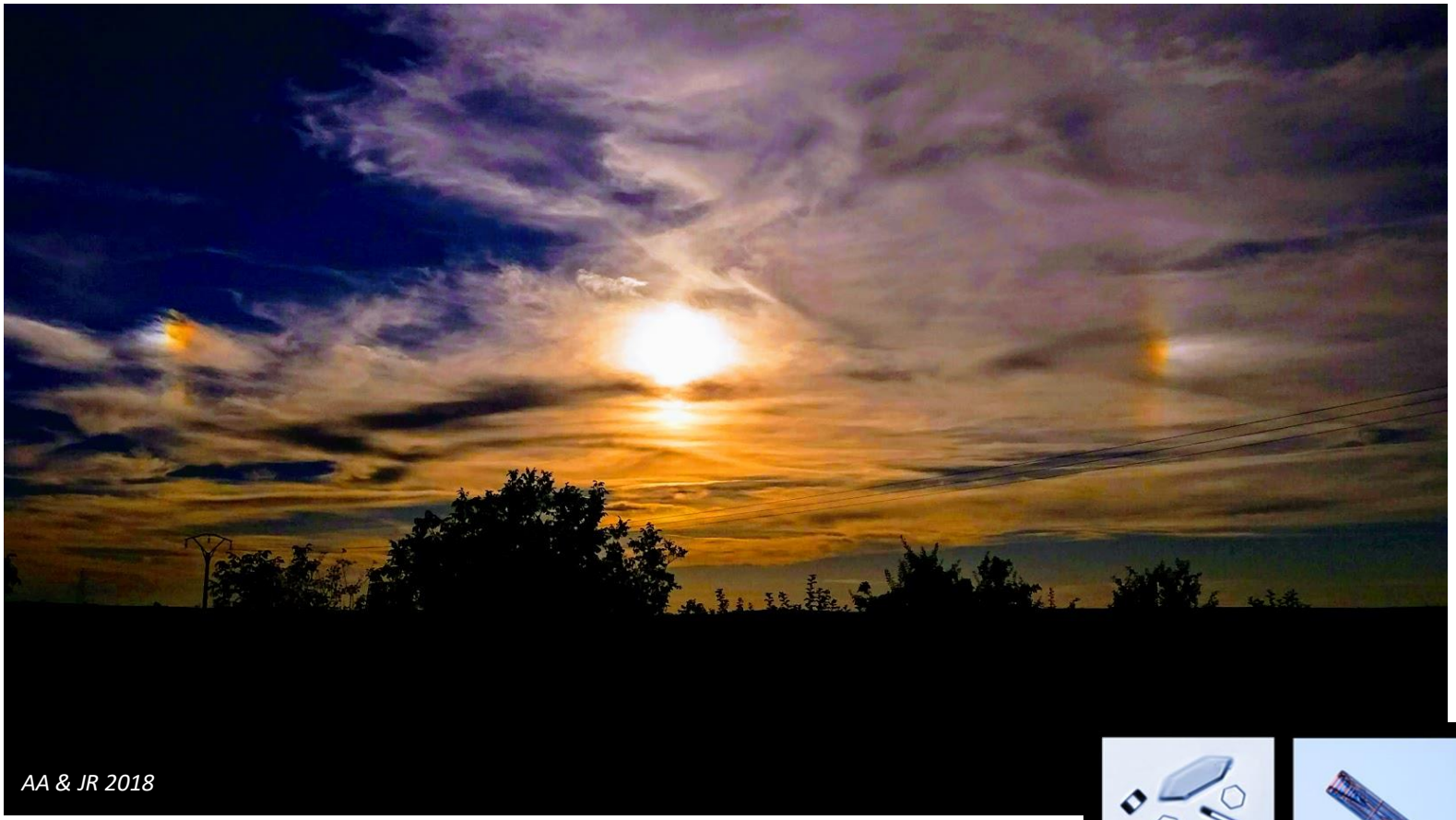
Primary rainbow



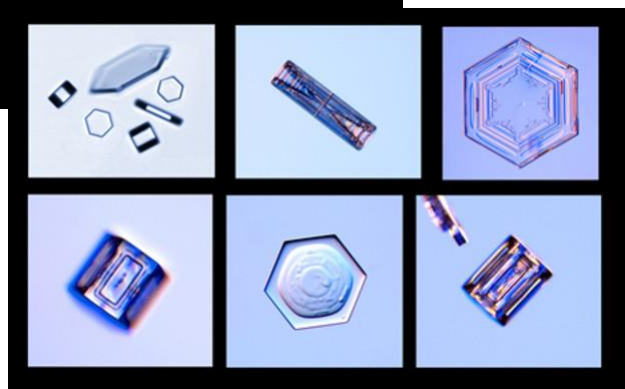
* Alexander of Aphrodisias (lived and taught in Athens, 200AD)

AA & JR 2011

“Sun dogs” (parhelion)



AA & JR 2018

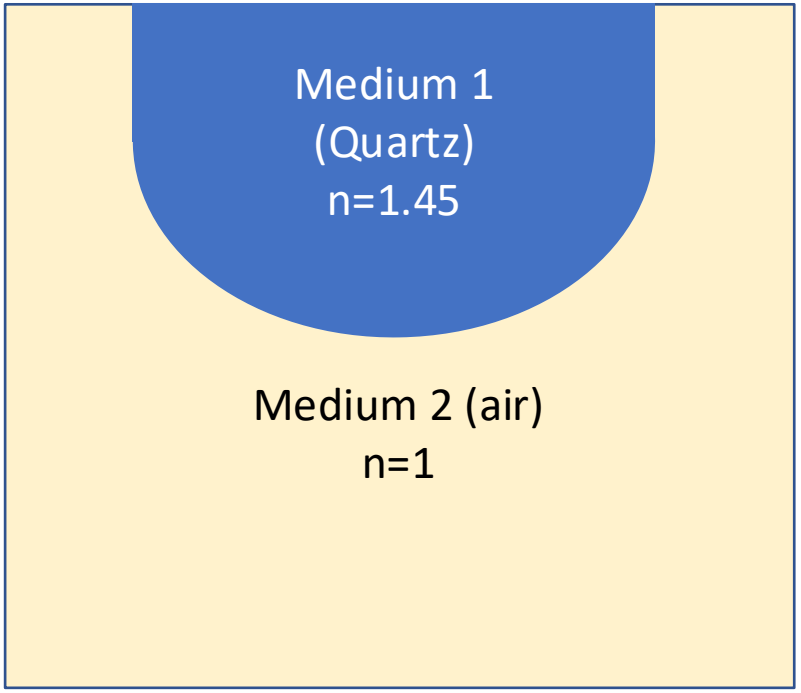


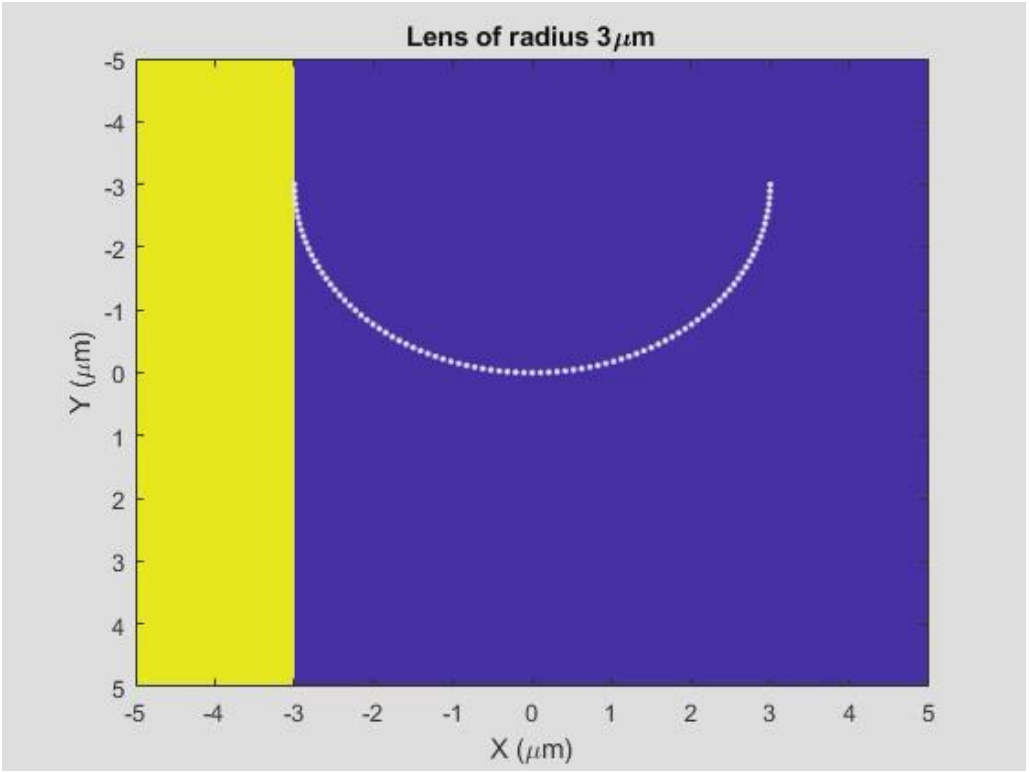
Agencia Sinc

Sun dogs are commonly caused by refraction and light scattering of hexagonal plate-shaped ice crystals, either suspended in high, cold cirrus clouds or cirroestratus clouds.

Interaction with a curved medium

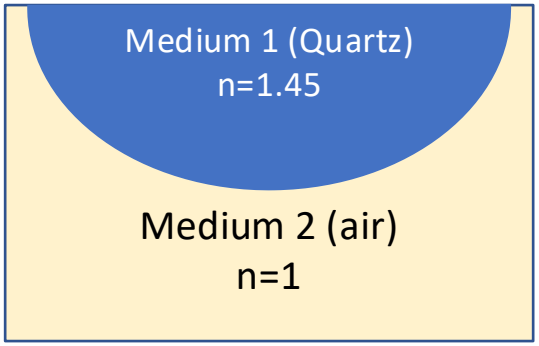
- We can consider each point on the surface as locally flat and add all of them together:

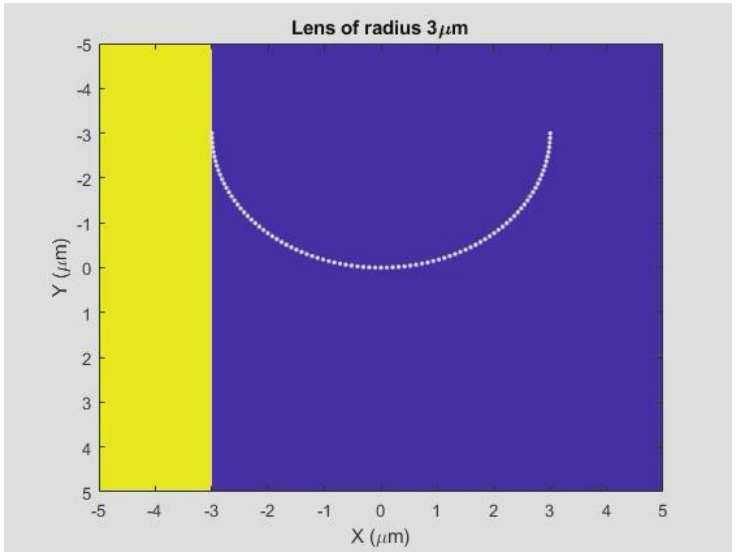


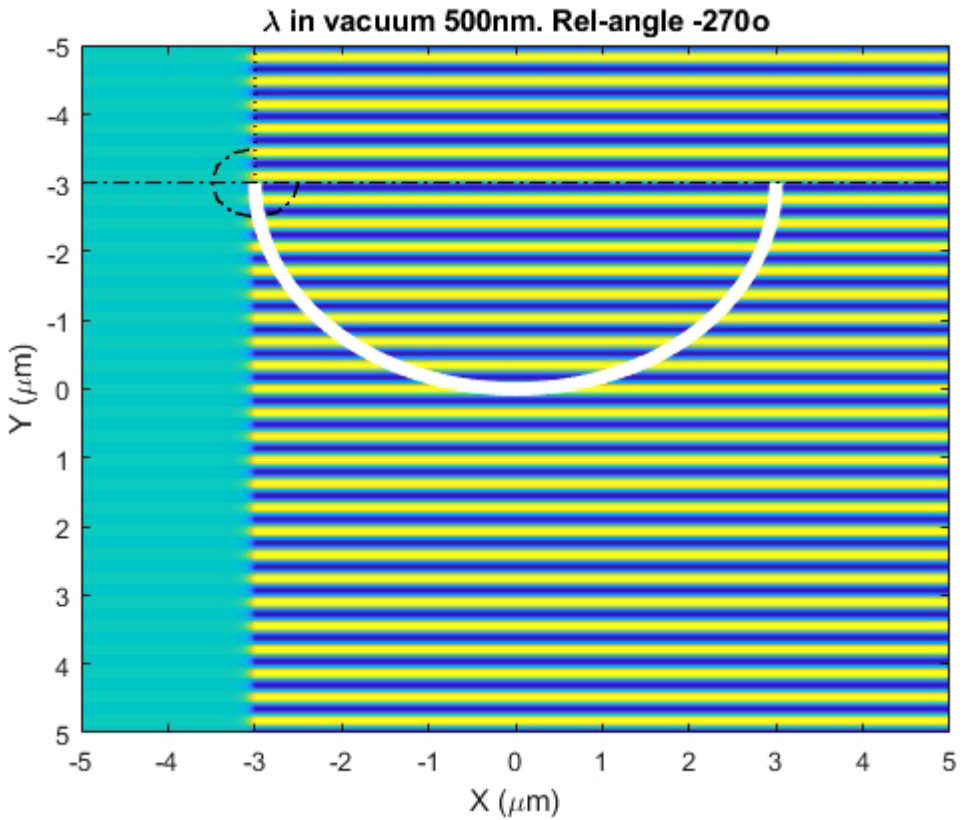


Interaction with a curved surface

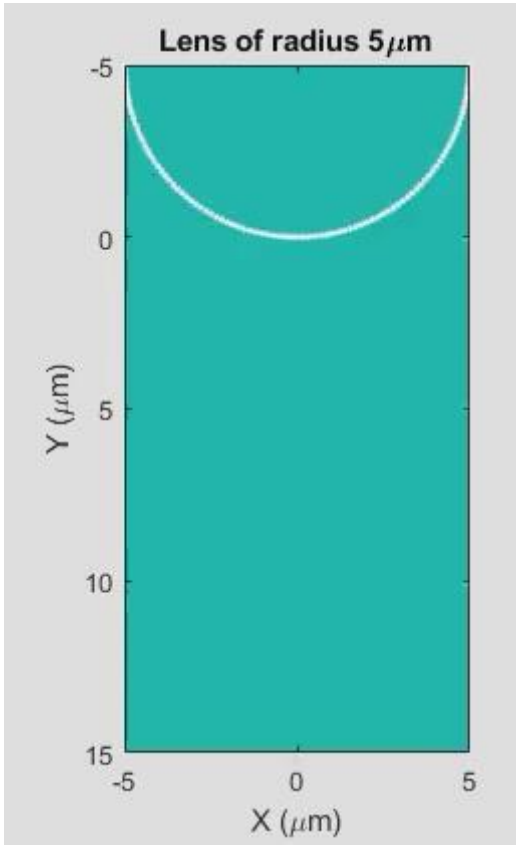
- We can consider each point on the surface as locally flat



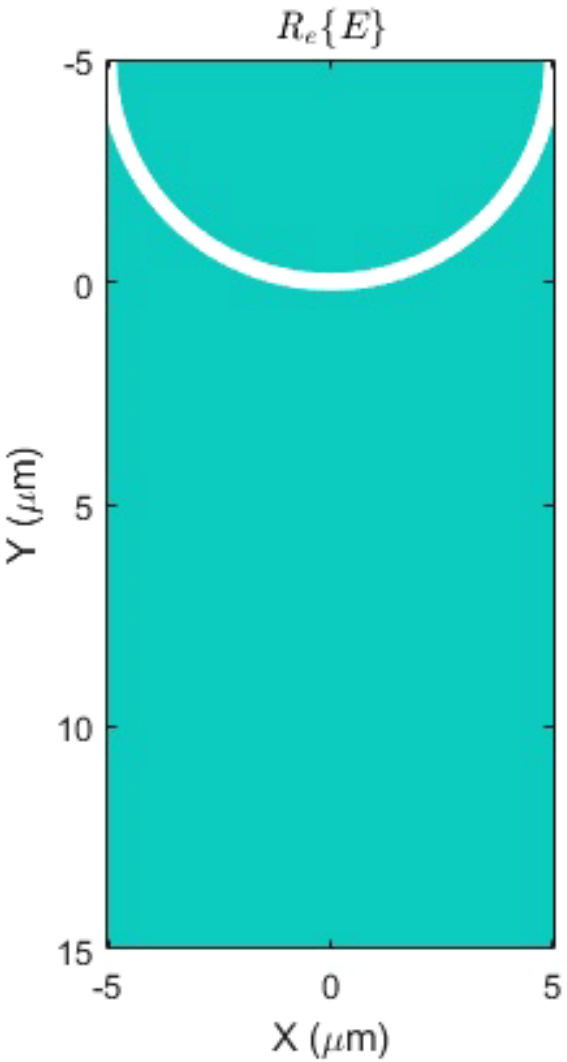




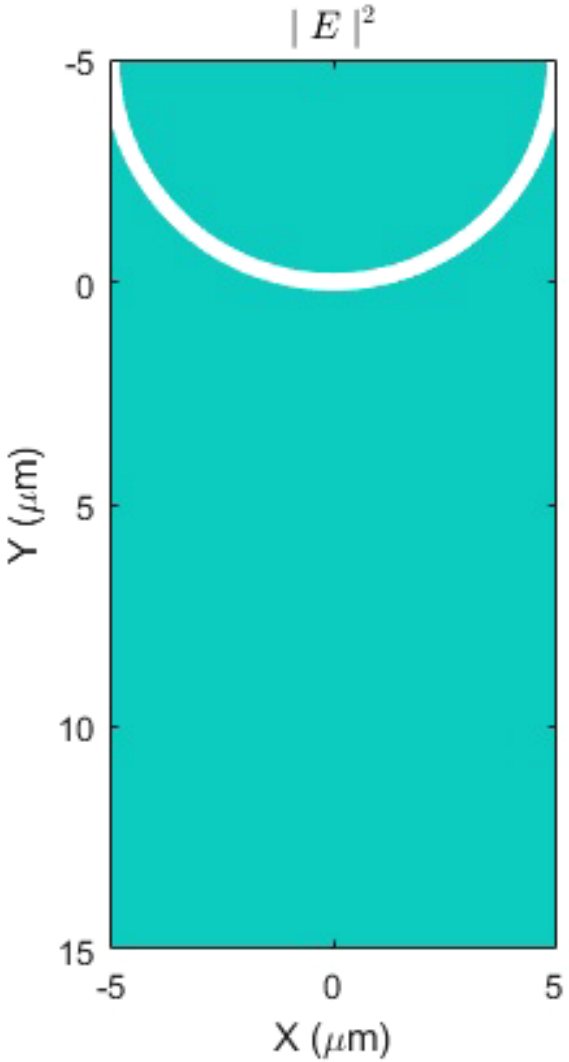
Interaction with a lens



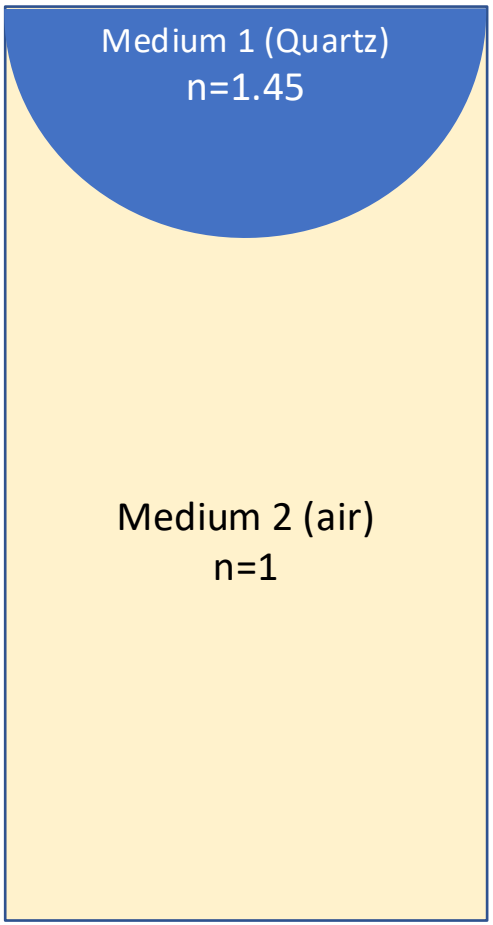
Electric field

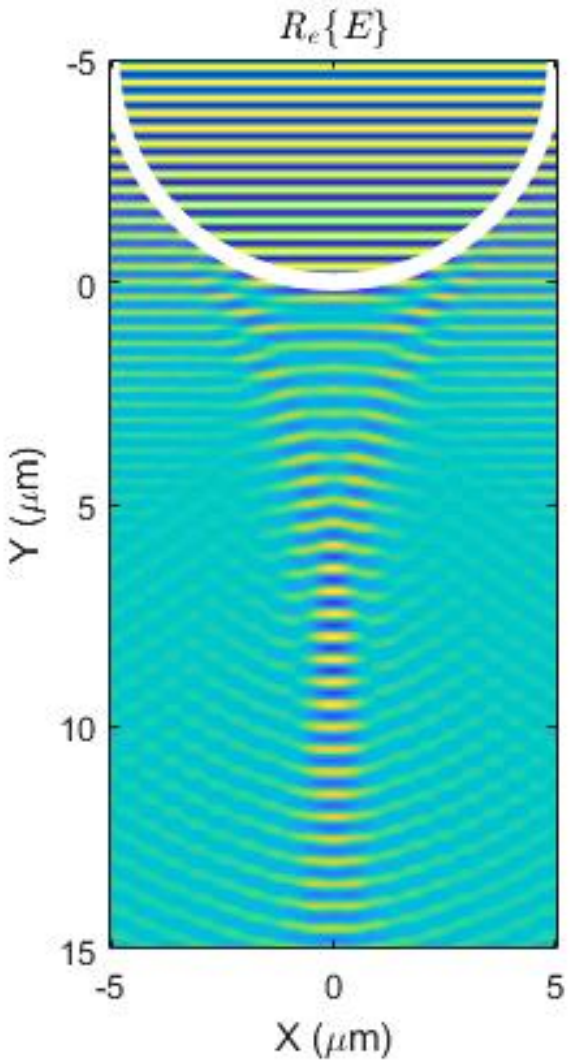


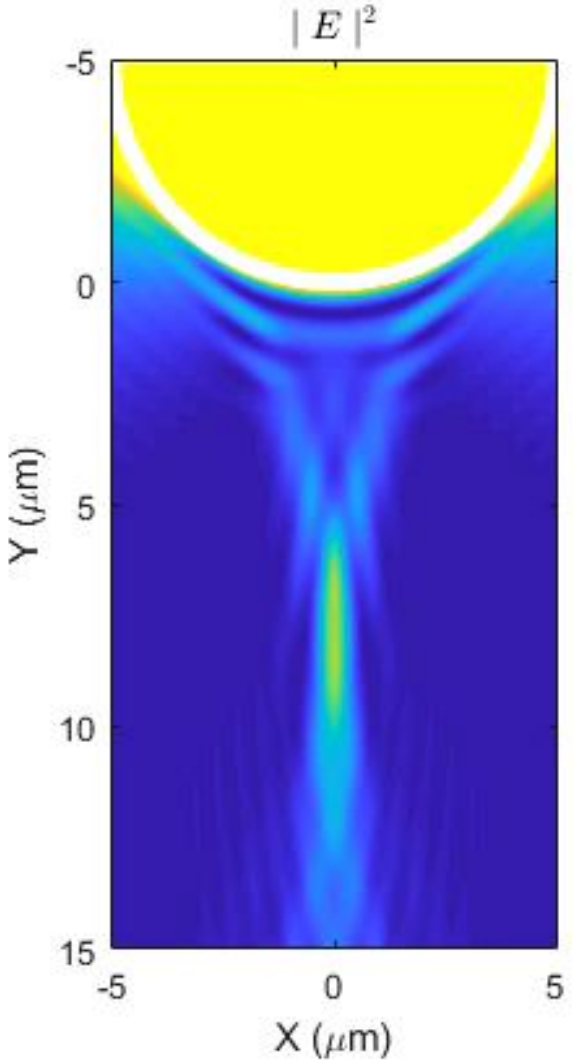
Intensity



Interaction with a lens

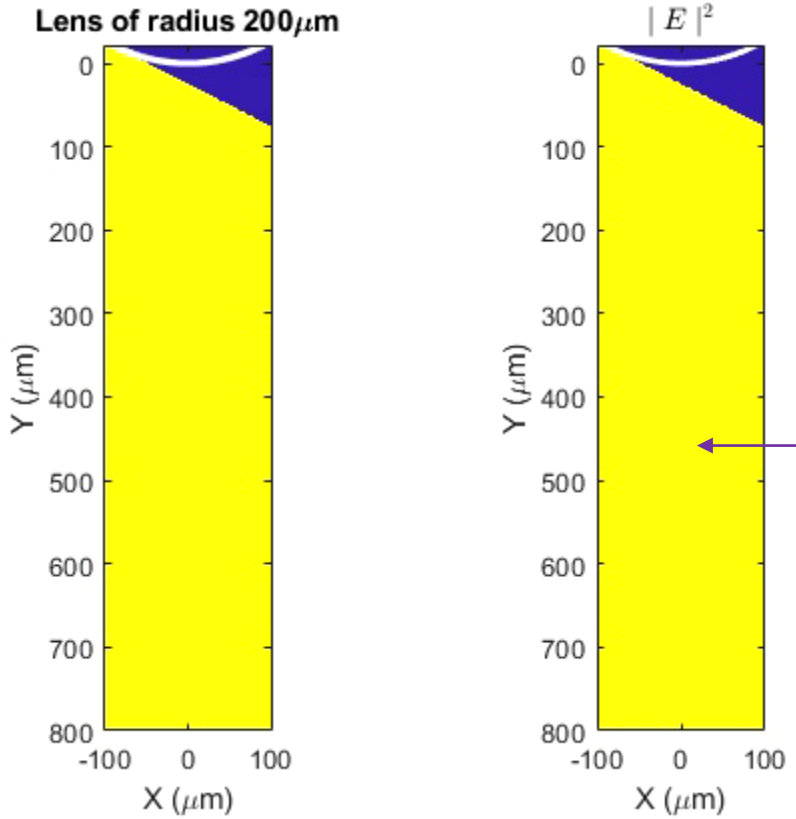






Larger lenses - focal length

$$\frac{1}{f} \approx (n-1)\frac{1}{R}$$ *Lens equation for quasi-flat lenses* $$f \approx \frac{R}{(n-1)}$$



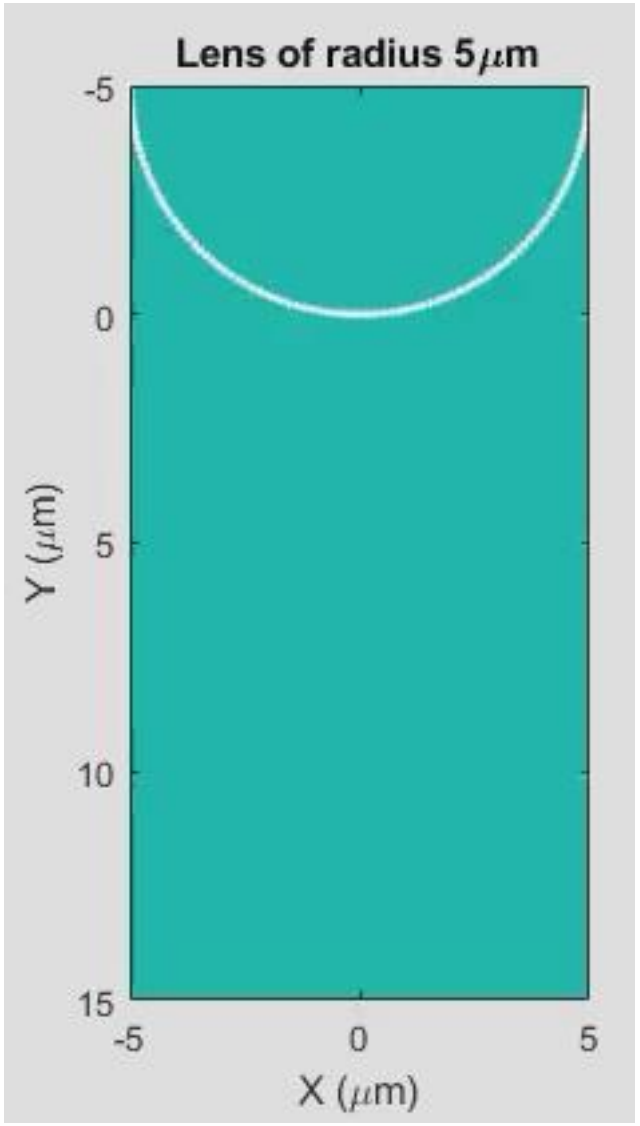
In this case *n*=1.45, R=200μm:

$$f \approx \frac{200\mu m}{0.45} = 444\mu m$$

Let's recap... where do we want to go with this?

So far we have seen that...

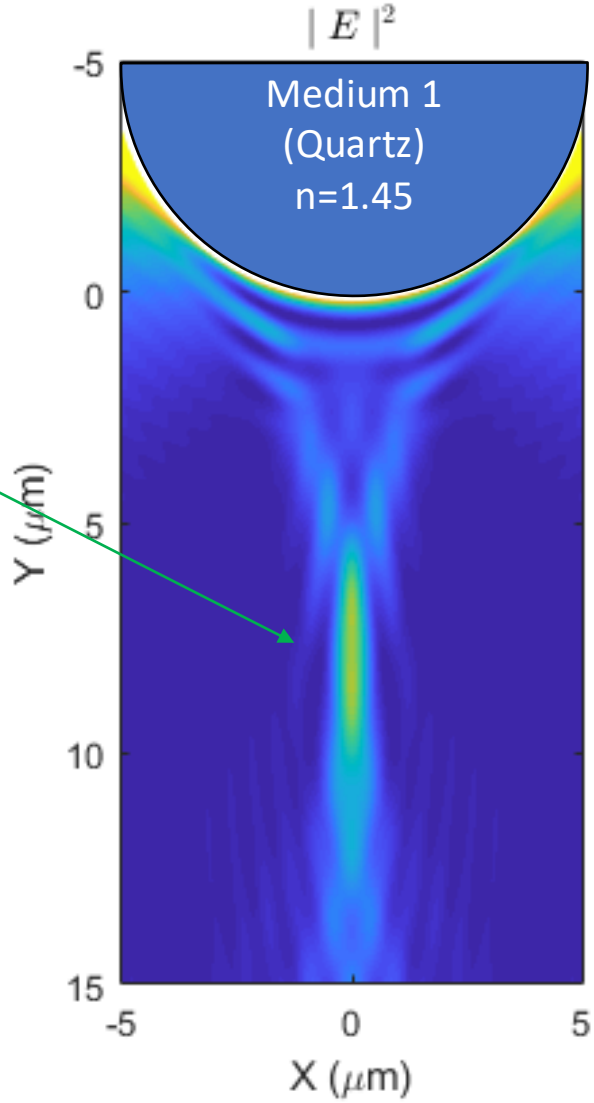
- What a lens does (and therefore a microscope lens), we can simulate by adding plane waves.



Let's recap... where do we want to go with this?

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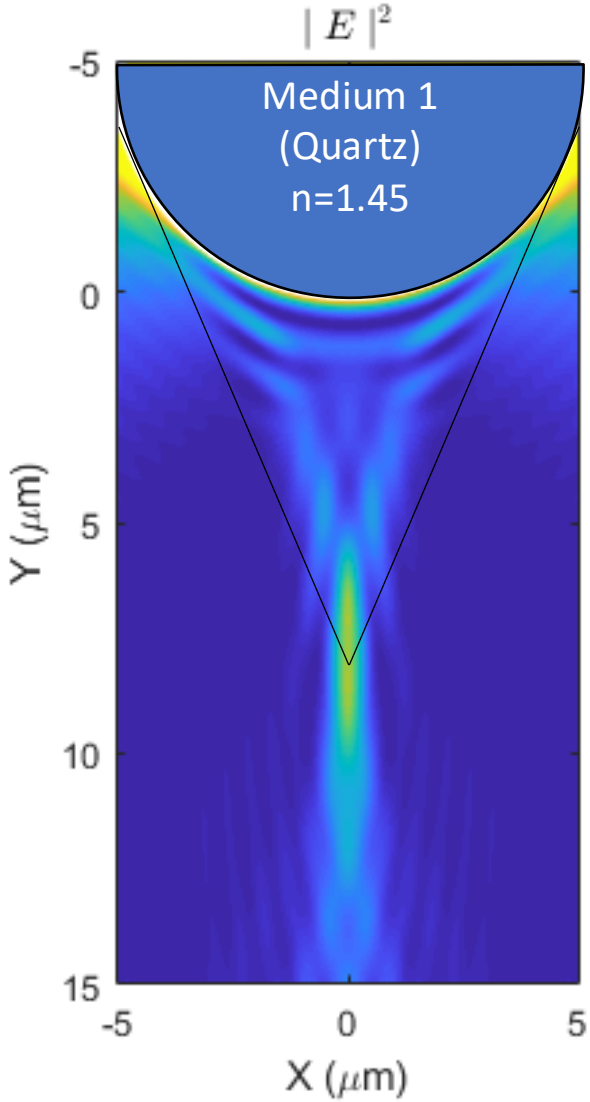
- What a lens does (and therefore a microscope lens), we can simulate by adding plane waves.
- When adding all the plane waves there is an area where they converge, which is the focal of the lens.



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So far we have seen that...

- What a lens does (and therefore a microscope lens), we can simulate by adding plane waves.
- When adding all the plane waves there is an area where they converge, which is the focal of the lens.
- With this in mind, the larger the lens, the more angles of incidence of these flat waves I can consider.



Let's recap... where do we want to go with this?

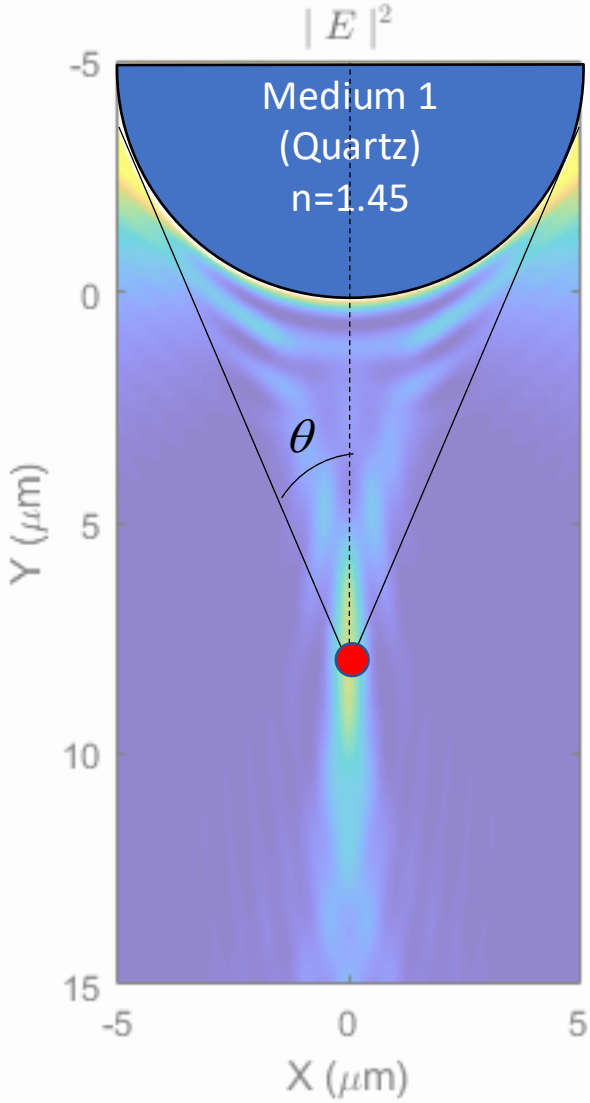
So far we have seen that…

- What a lens does (and therefore a microscope lens), we can simulate by adding plane waves.
- When adding all the flat waves there is an area where they converge, which is the focal of the lens.
- With this in mind, the larger the lens, the more angles of incidence of these flat waves I can consider.

This is the definition of **numerical aperture:**

$$NA = n_0 \sin\theta$$

Refractive index of the medium (air in this case)



The numerical aperture

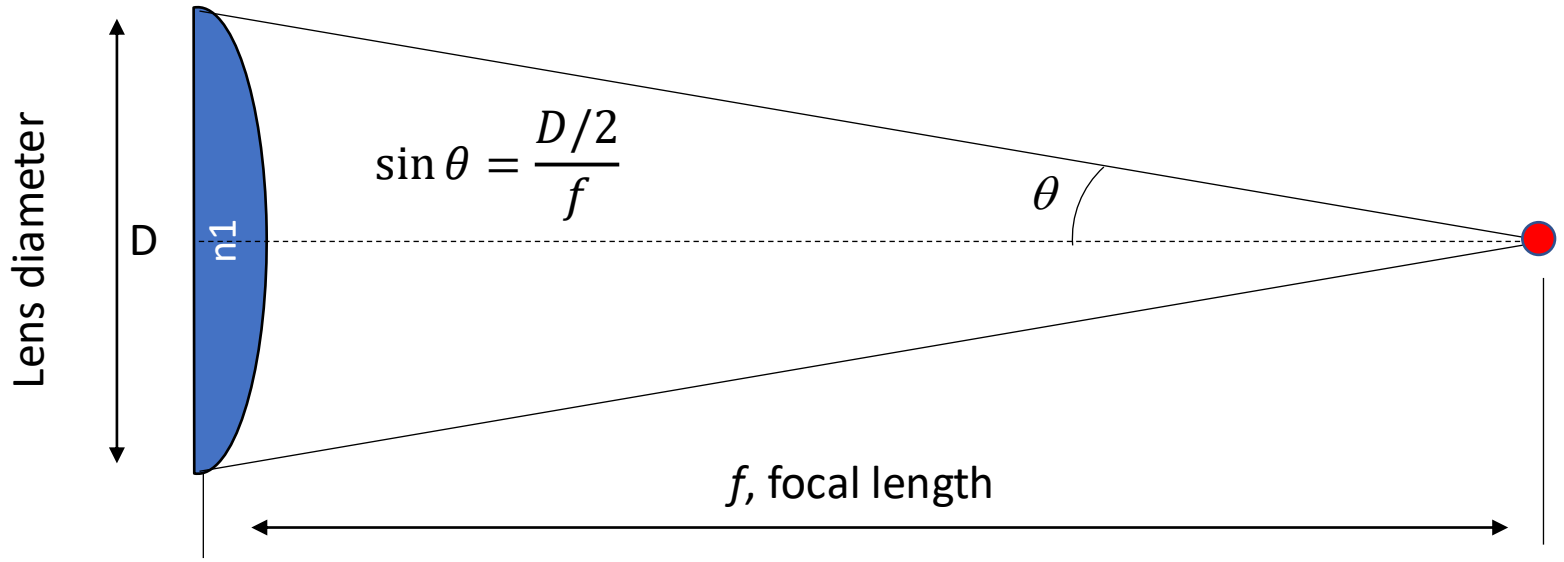
We have seen that…

- This is the definition of numerical aperture:

$$NA = n_0 \sin\theta$$

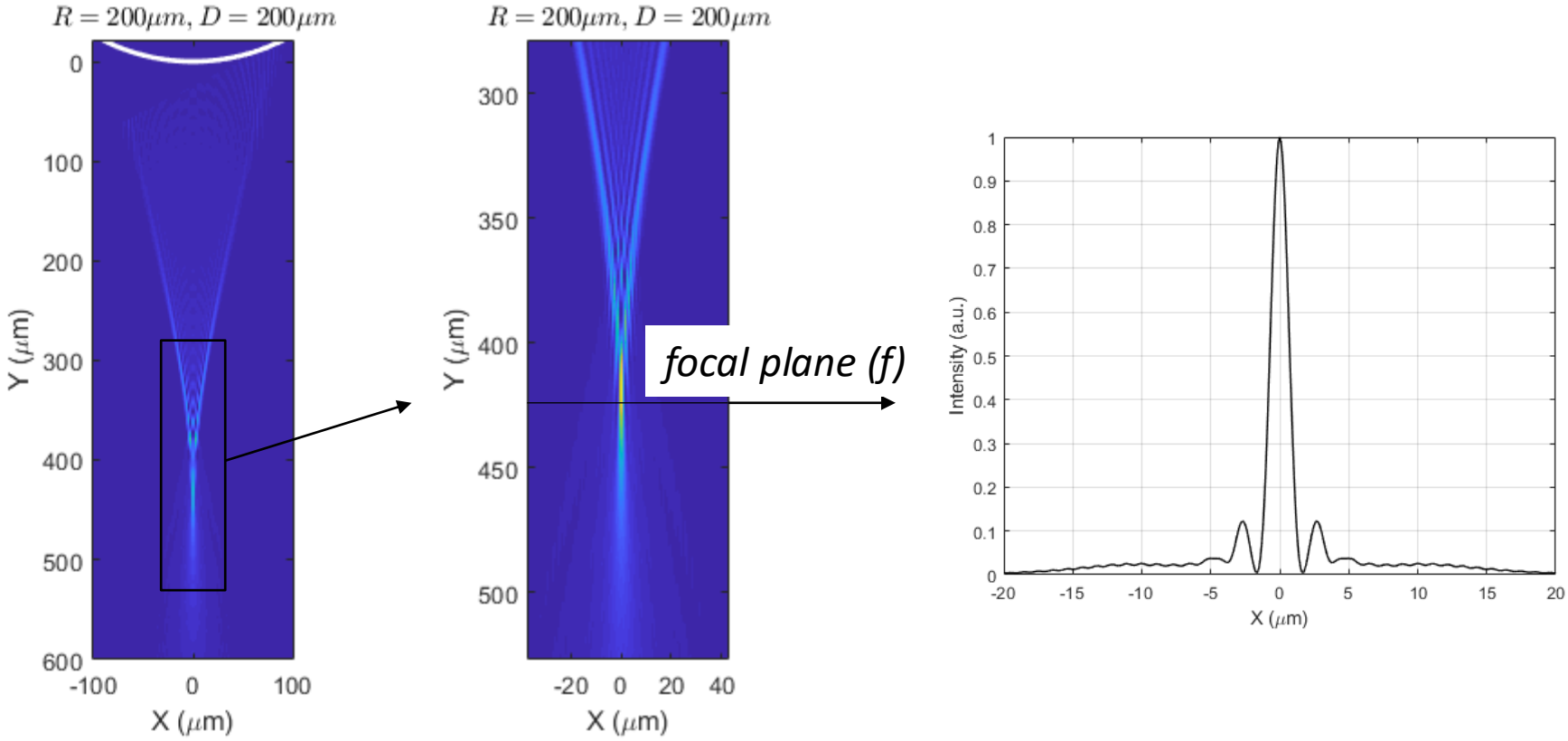
- For typical lenses of a microscope, we can assume that the distances are very large (compared to the wavelength, λ),

$$NA \approx n_0 \frac{D}{2f}$$

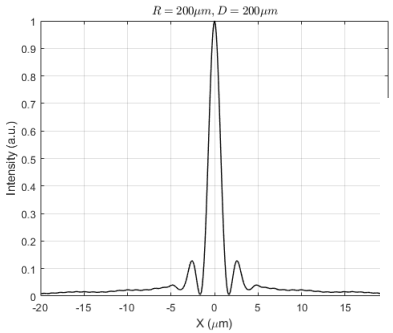


And why is the NA important?

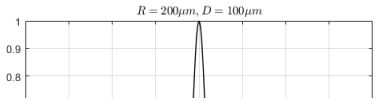
- The more different angles we can concentrate on the focal point, the higher the resolution. Let's prove it...

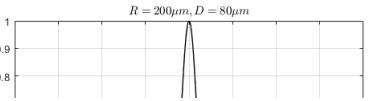


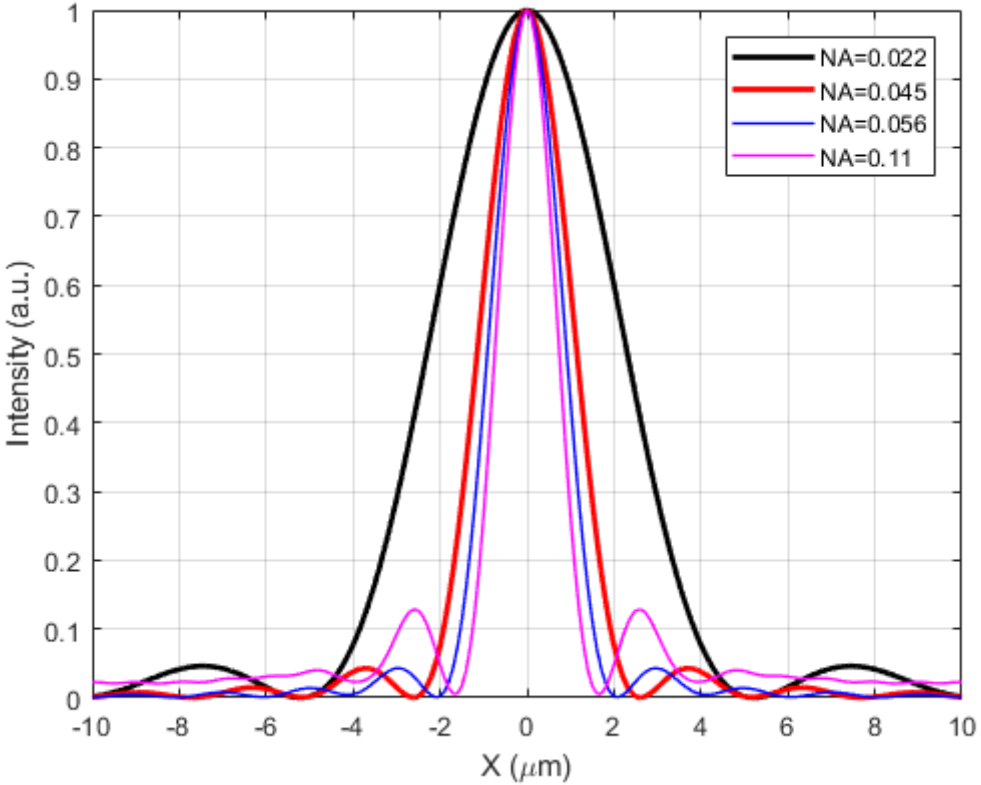
Resolution vs NA

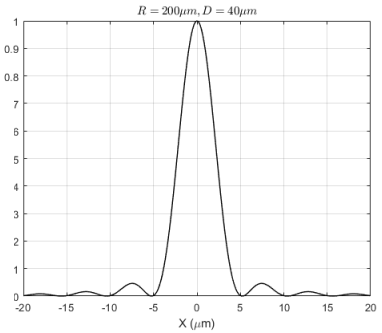


NA=0.11

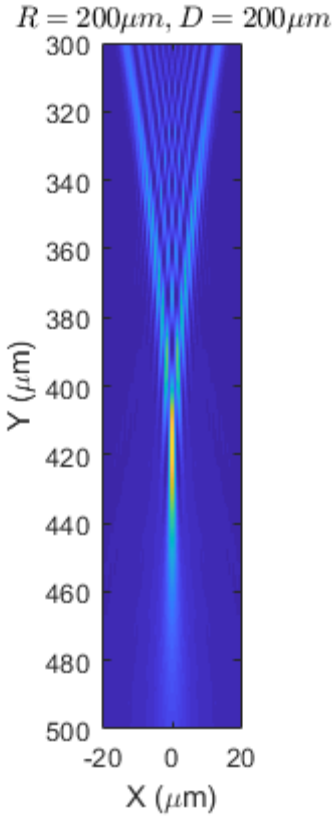


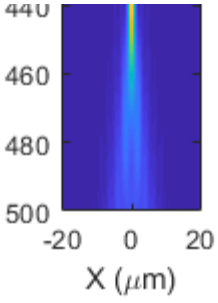


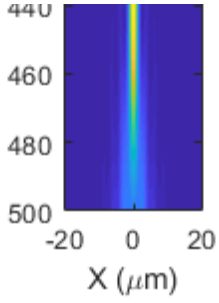


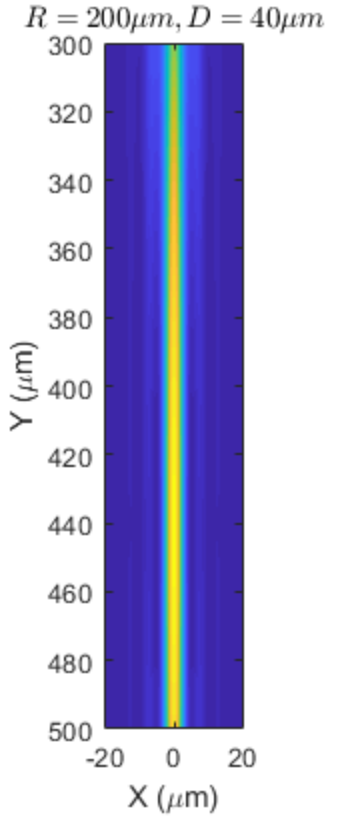


NA=0.022



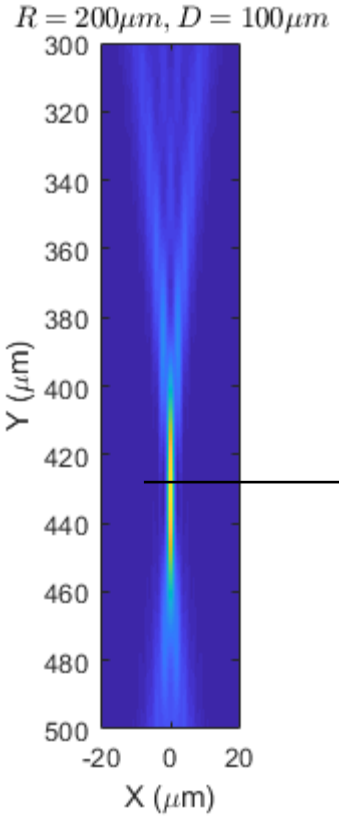




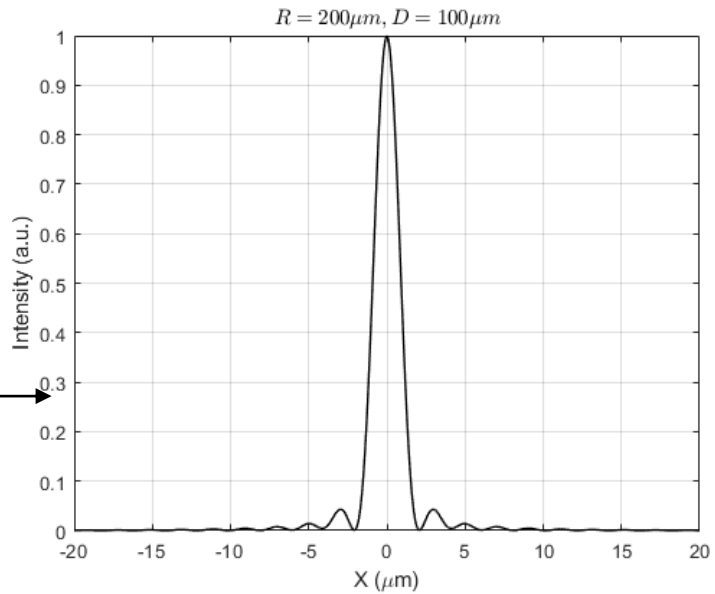


The point spread function (PSF)

- Each of these images is the equivalent of the "point spread function“ (PSF). The resulting image will be the convolution of the sample with this "point spread function".

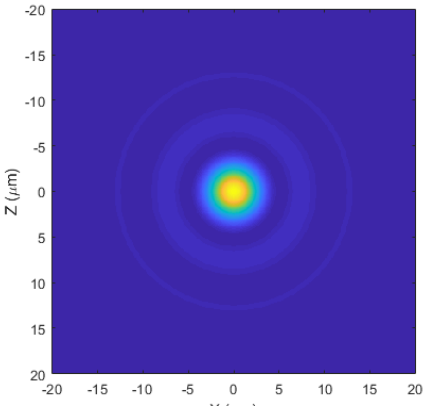


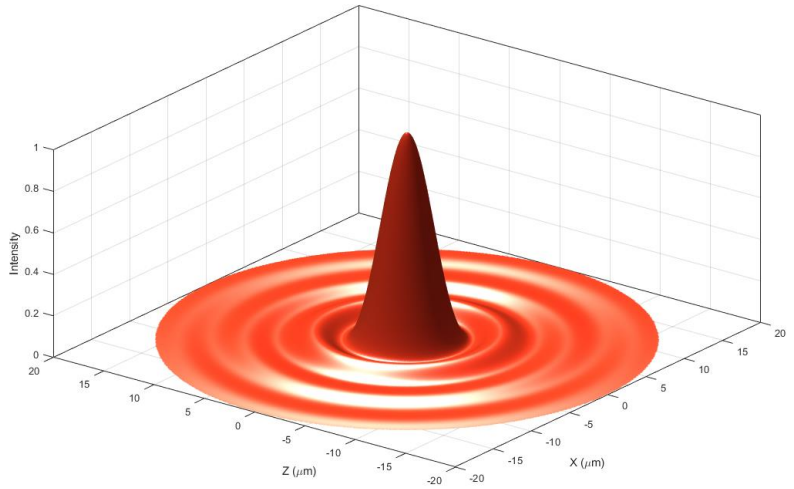
PSF 1D



PSF 2D

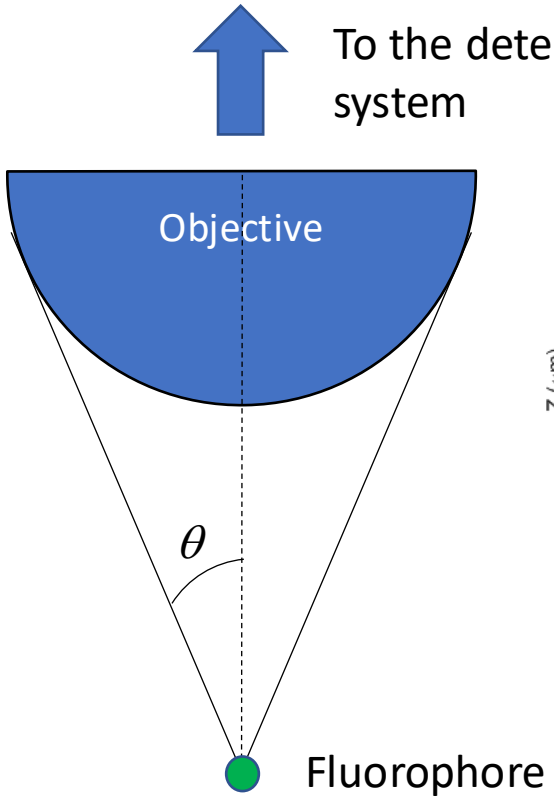


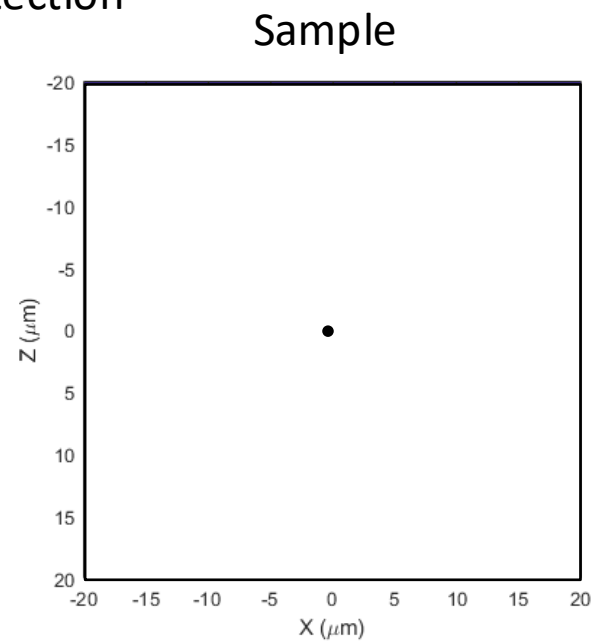


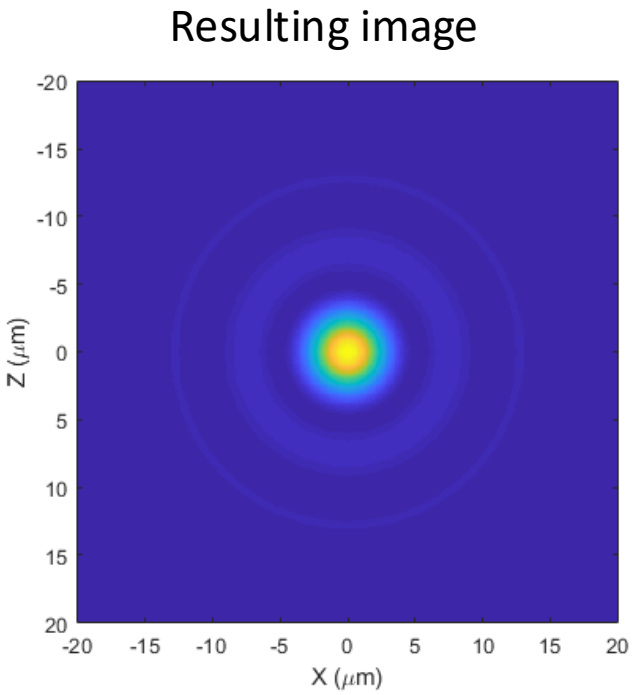


And what does this have to do with resolution?

- Put another way: if we used the smallest fluorophore possible and placed it right in the center of the focal point, the image we would get is this*:





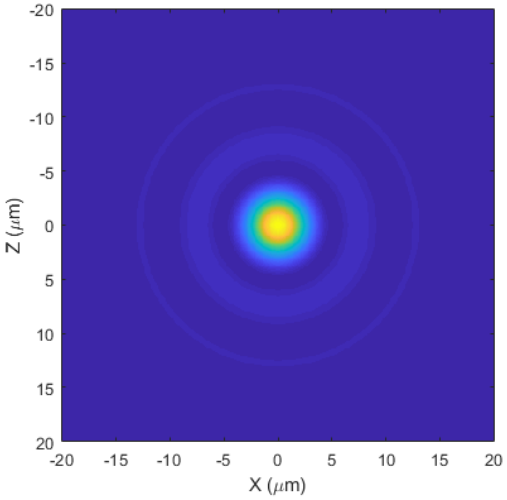


*in this lens system, of course

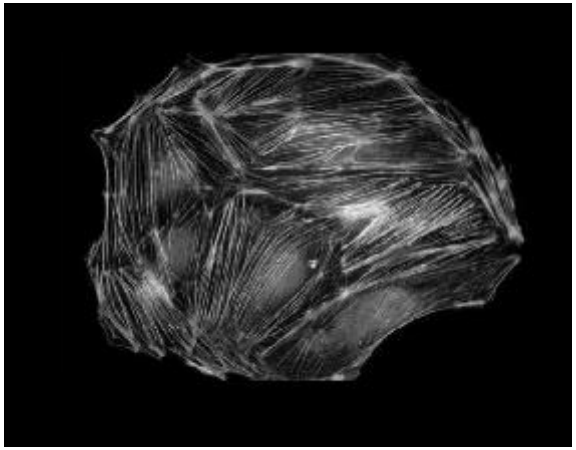
And what does this have to do with resolution?

- A more complex sample would appear blurry:

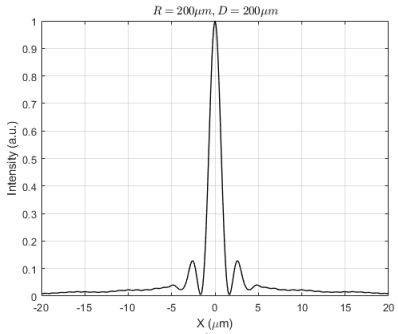
PSF



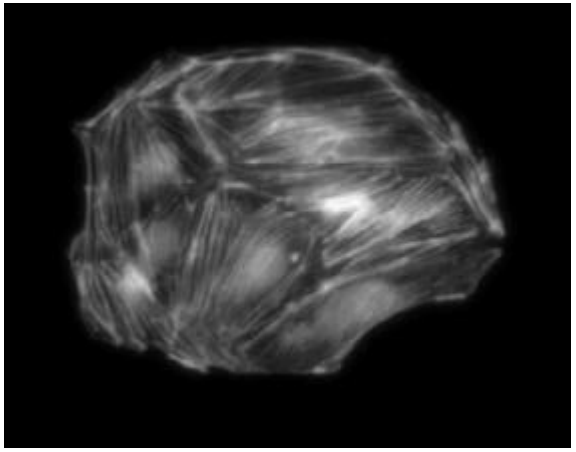
Sample



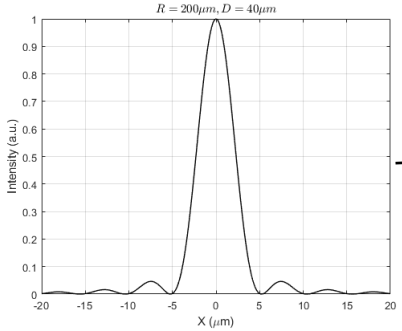
PSF Narrow

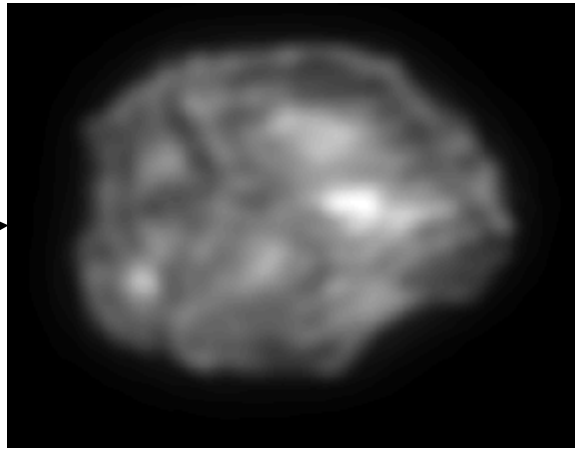


Resulting image



PSF Wide

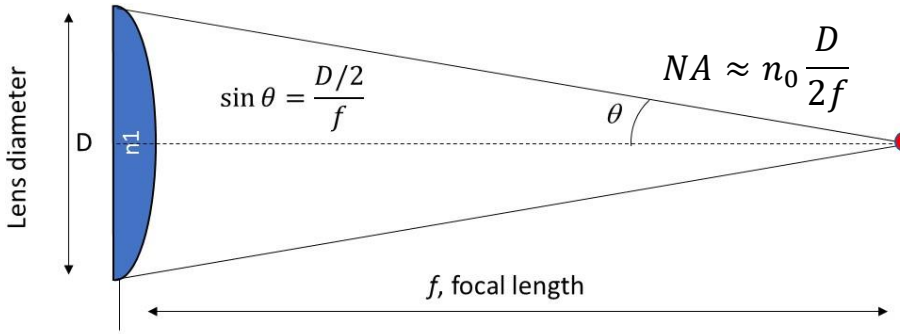


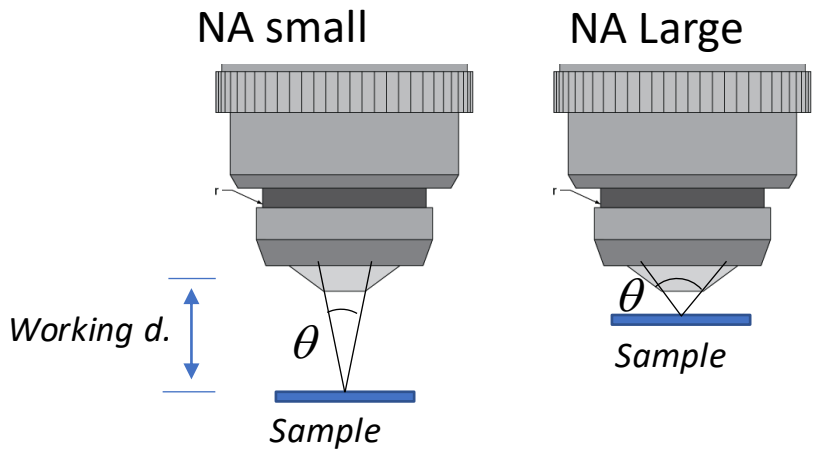


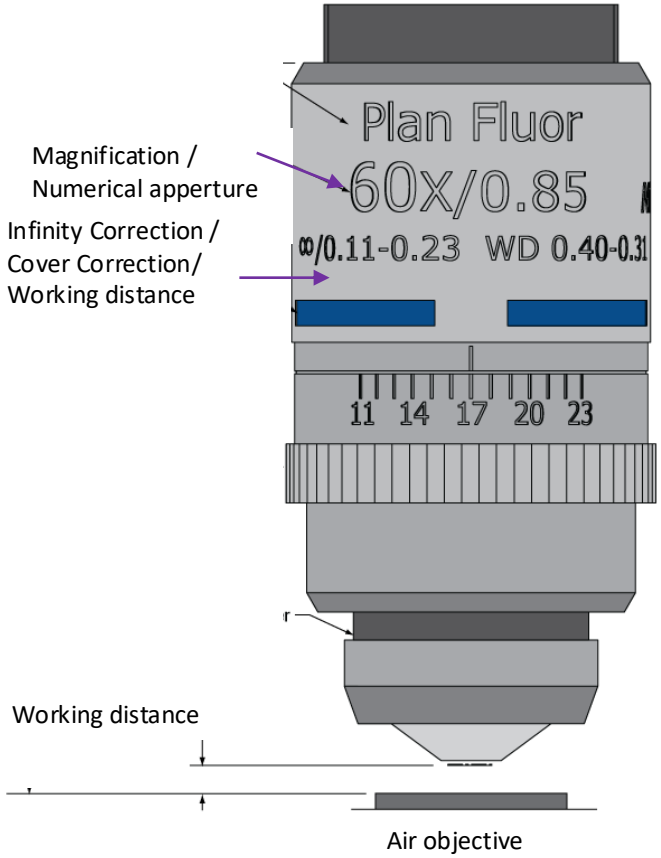
Resolution vs magnification

And magnification, does it increase the resolution?

The answer is no... the only thing that increases the resolution is the NA. What happens is that highly magnifying objectives usually have a high NA and a short working distance (*f* small)



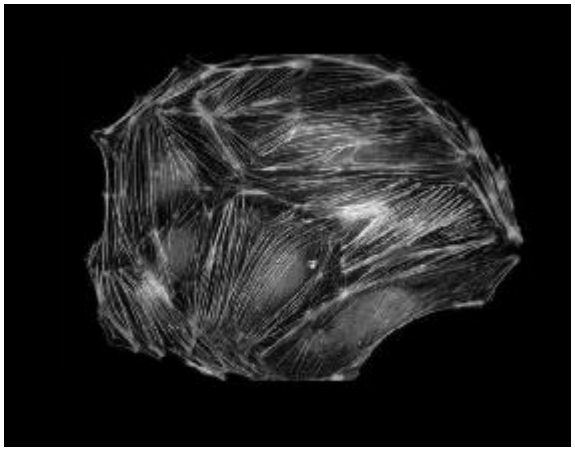




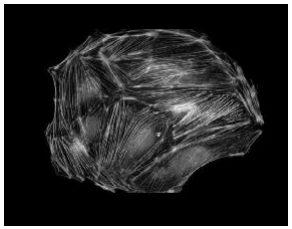
Resolution vs magnification

NA large (~1)

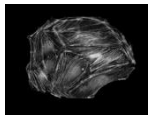
100X

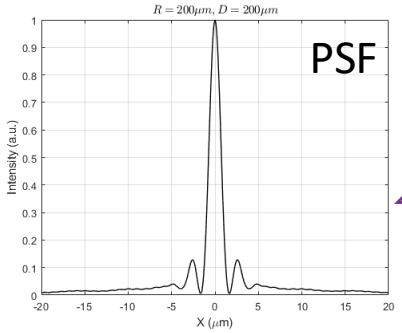


50X



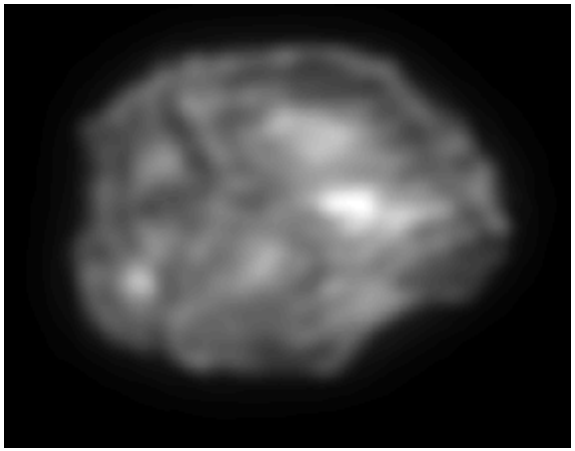
25X



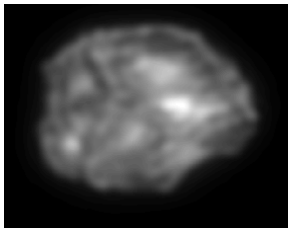


NA small (~0.1)

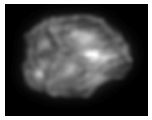
100X

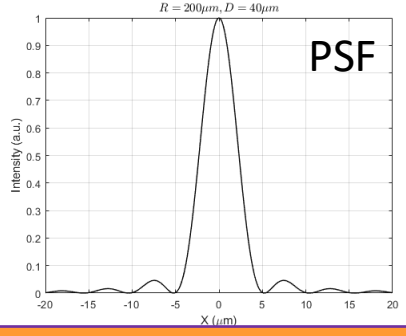


50X



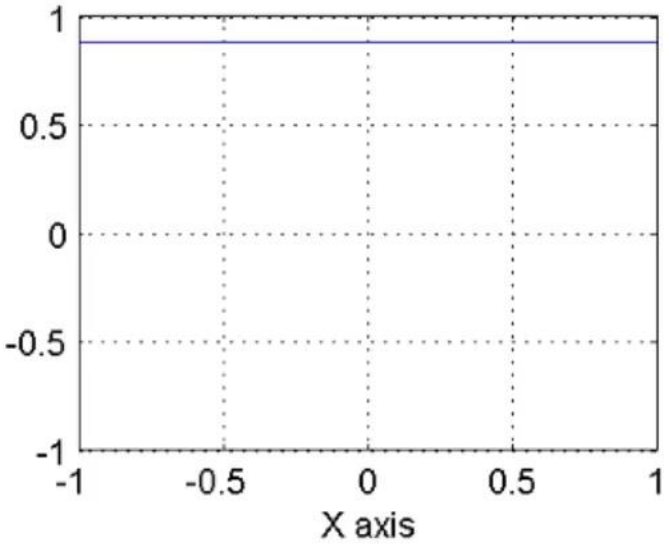
25X

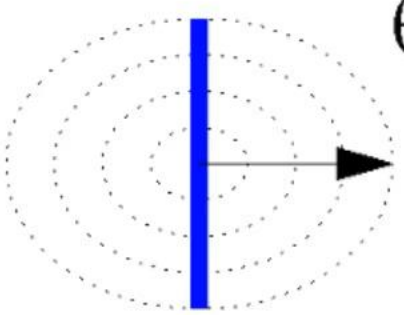




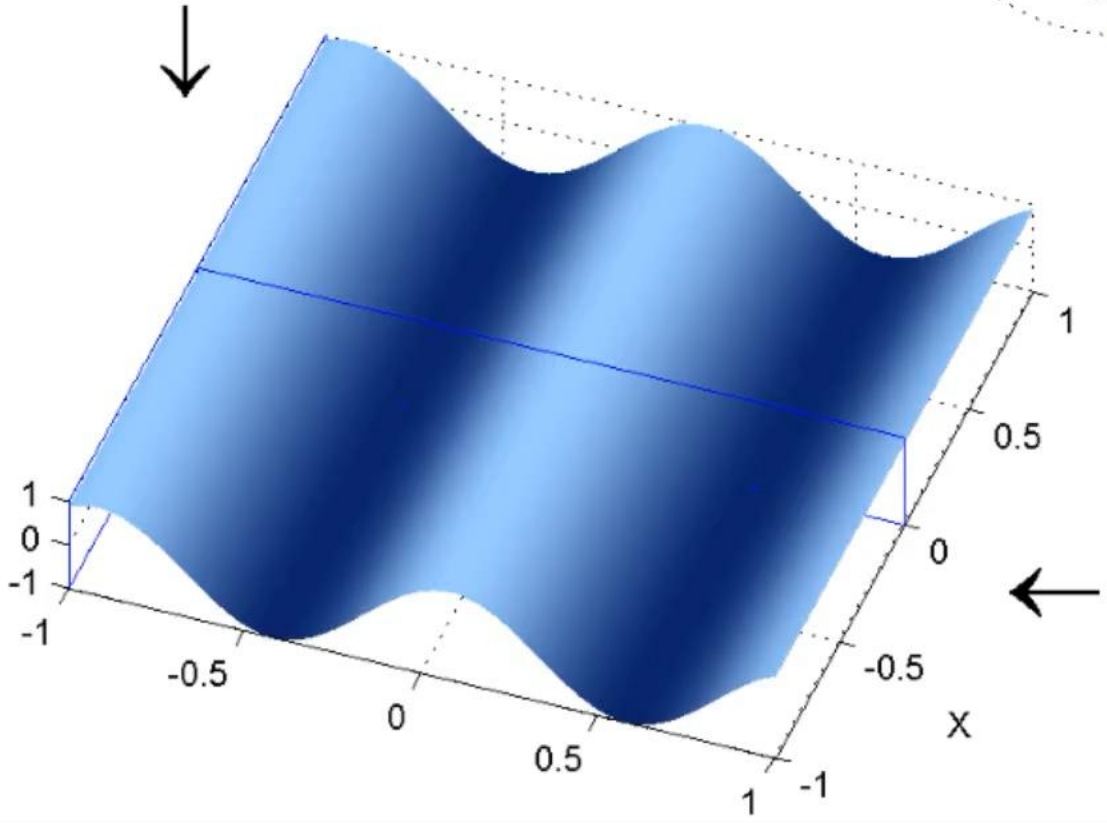
Resolution

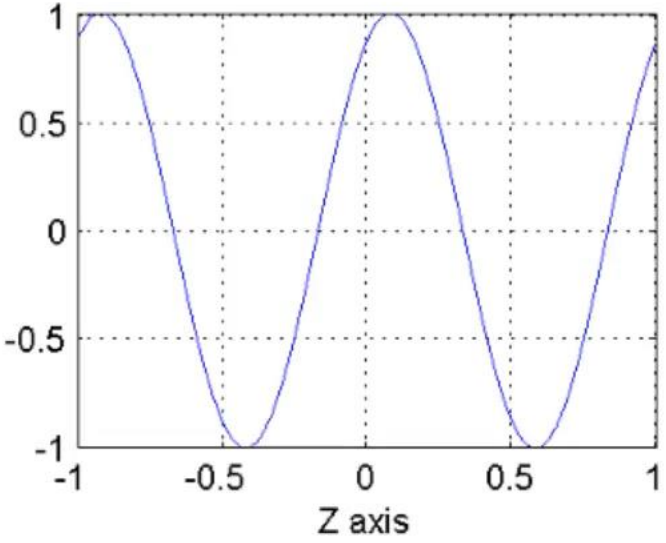
Magnification


1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
X axis

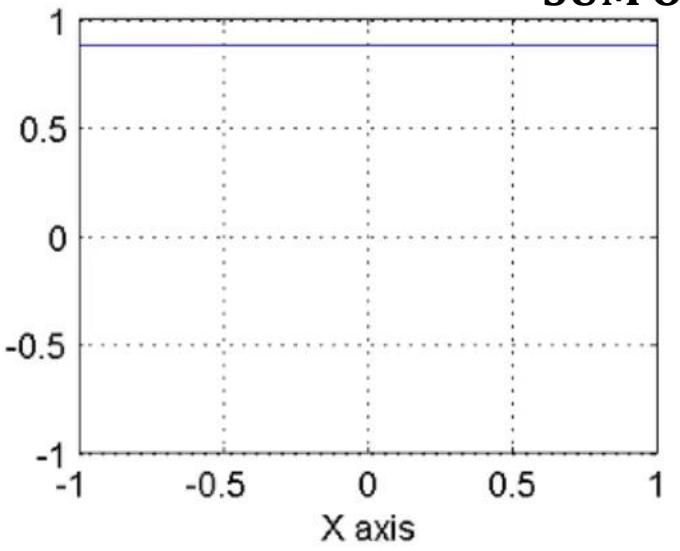


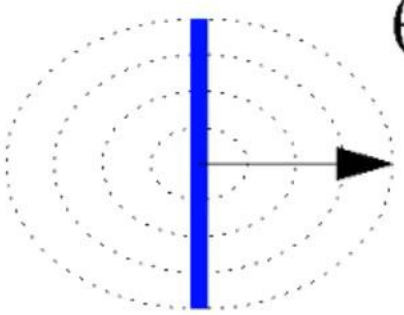
θ=0 deg


1
0
-1
-1
-0.5
0
0.5
1
1
0.5
0
-0.5
-1
X

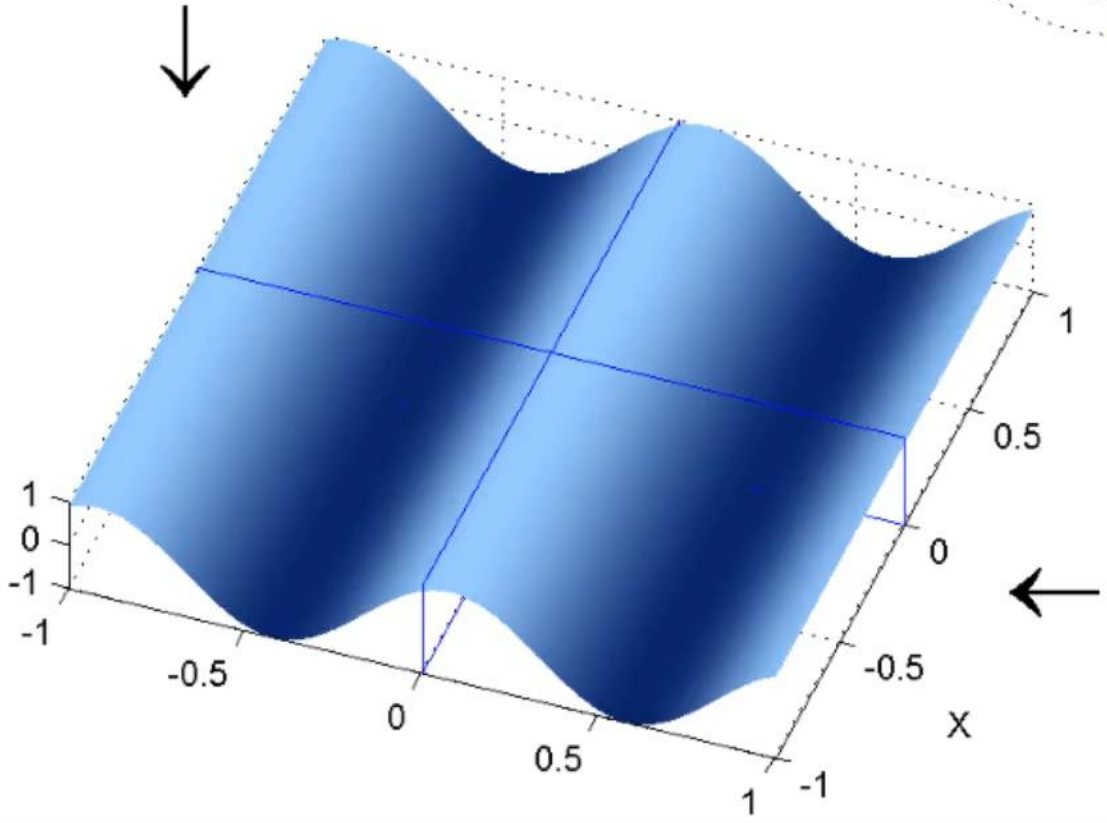

1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
Z axis

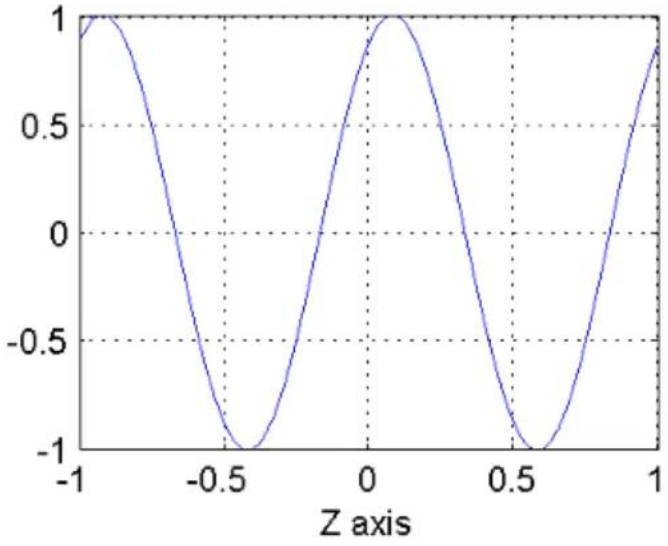
SUM OF POSSIBLE WAVE-VECTORS

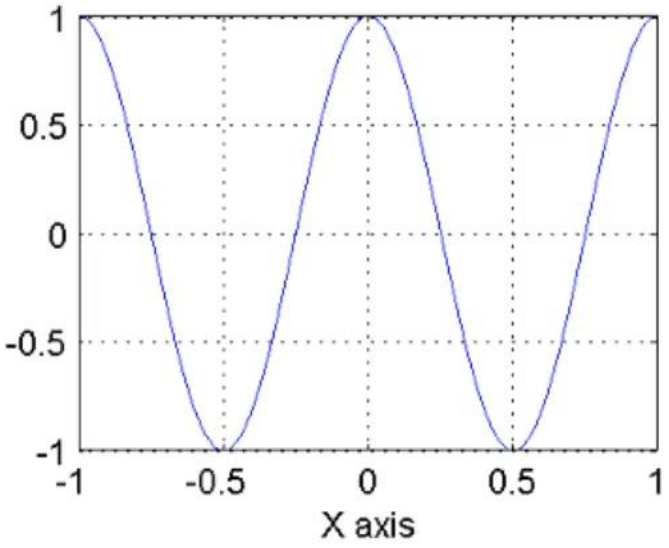


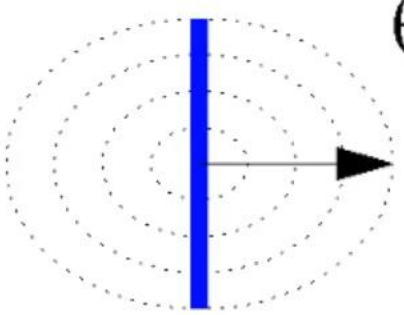


θ=0 deg

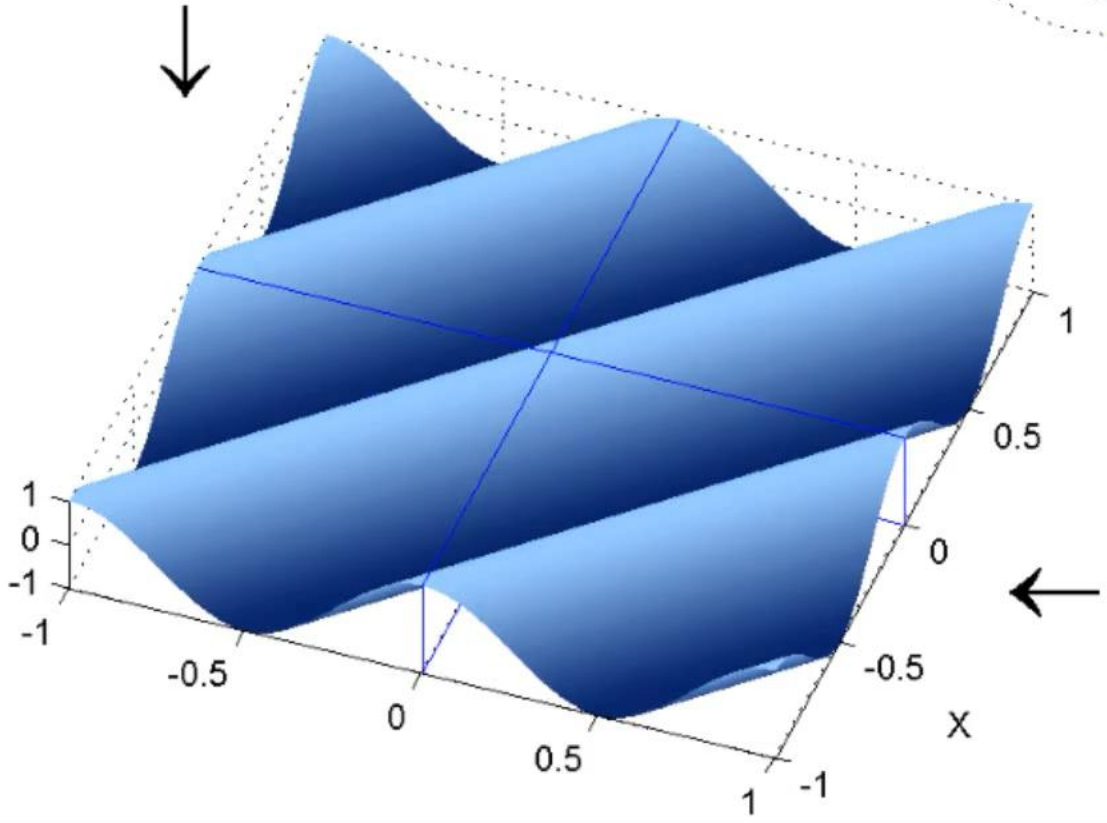


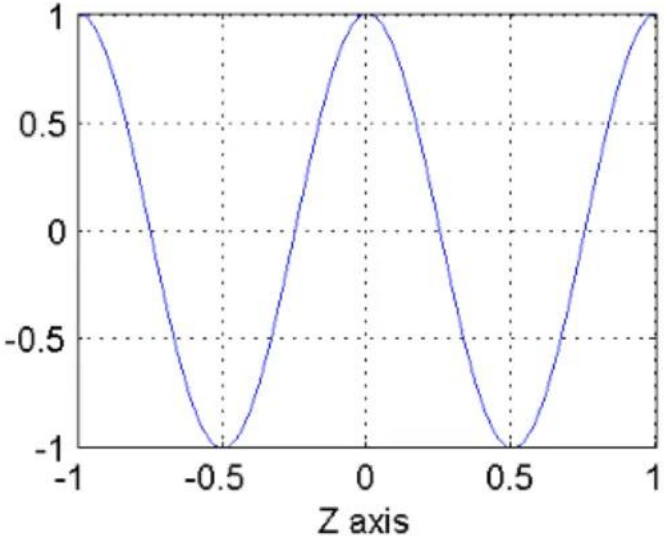



1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
X axis

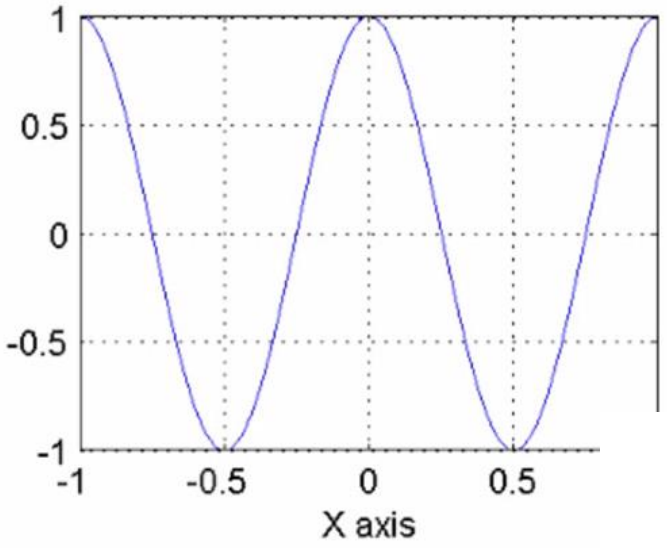


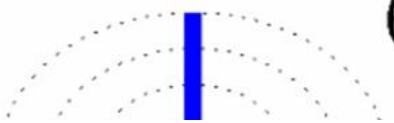
θ=0 deg


1
0
-1
-1
-0.5
0
0.5
1
1
0.5
0
-0.5
-1
X

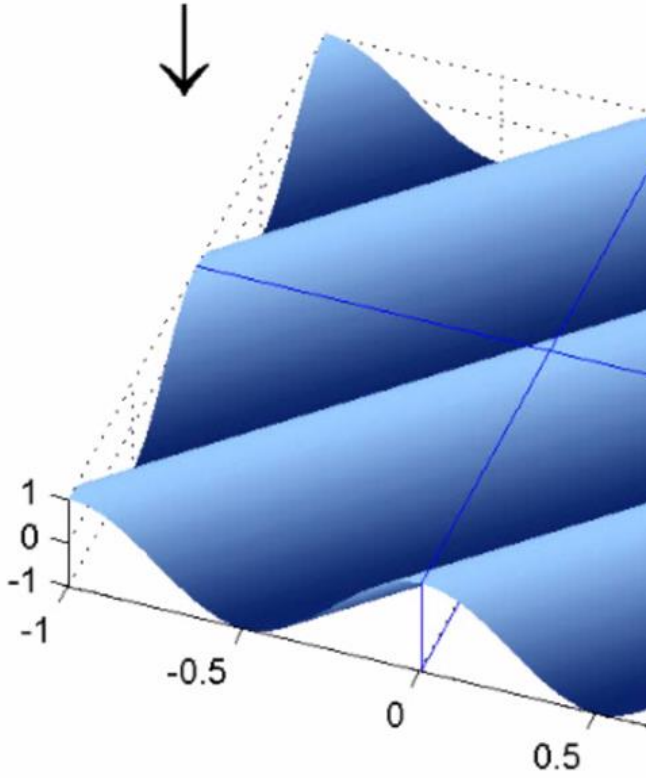

1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
Z axis

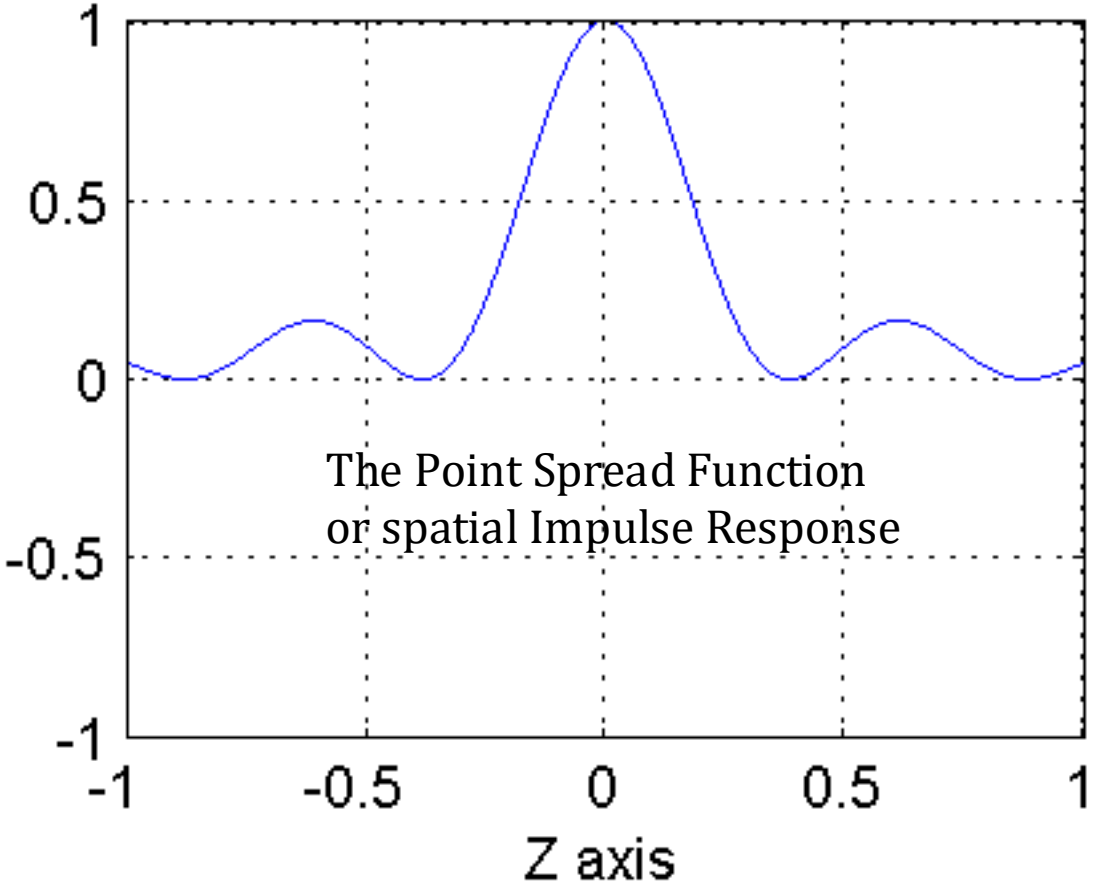
SUM OF INTENSITIES



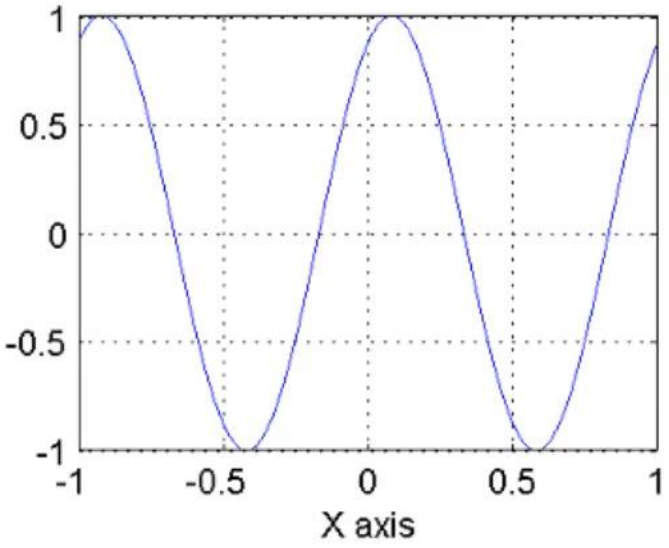


θ=0 deg



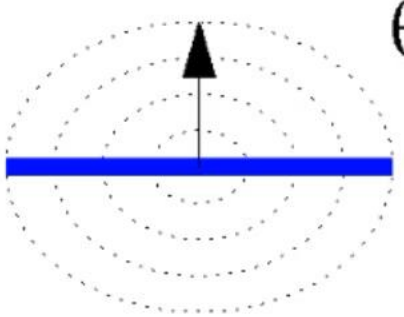


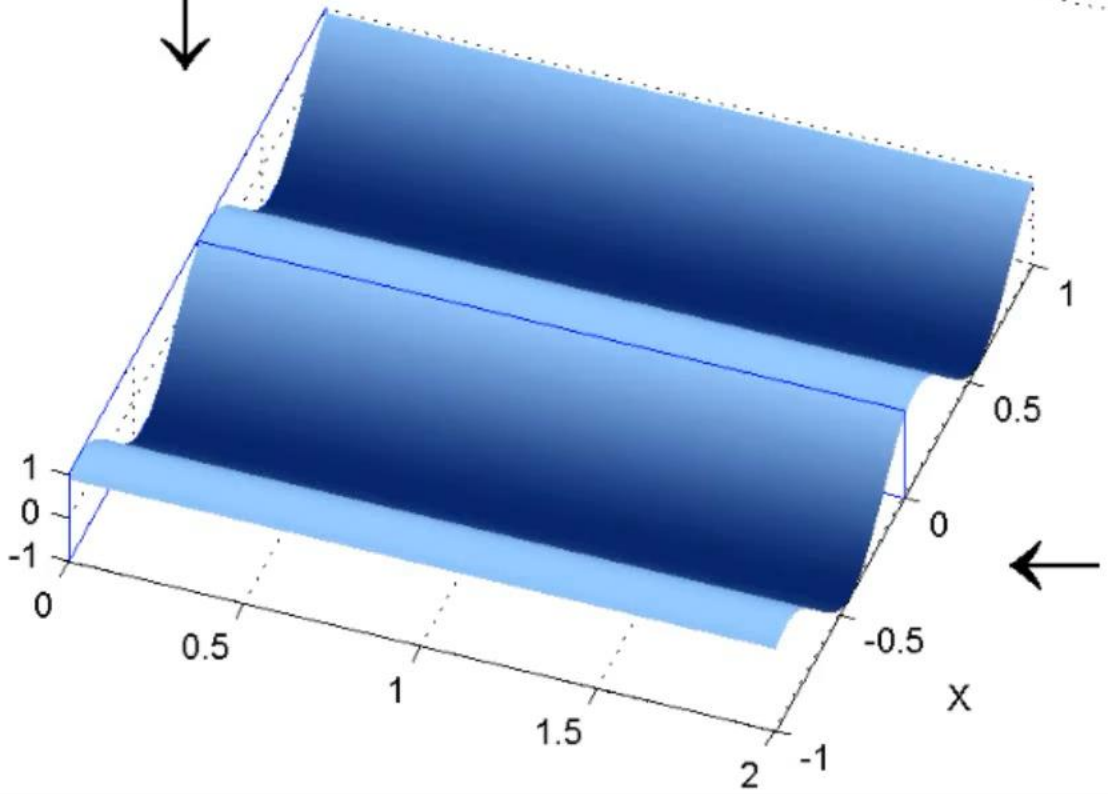
The Point Spread Function
or spatial Impulse Response

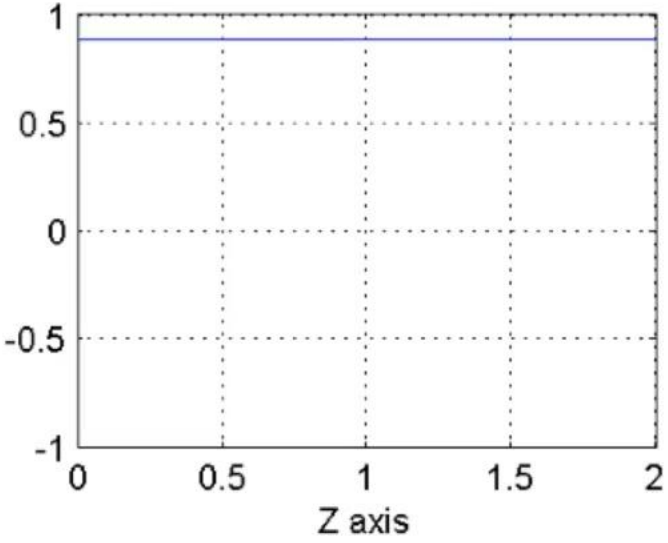


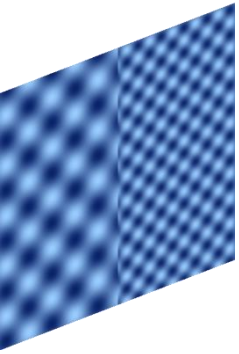
k1=1 k0

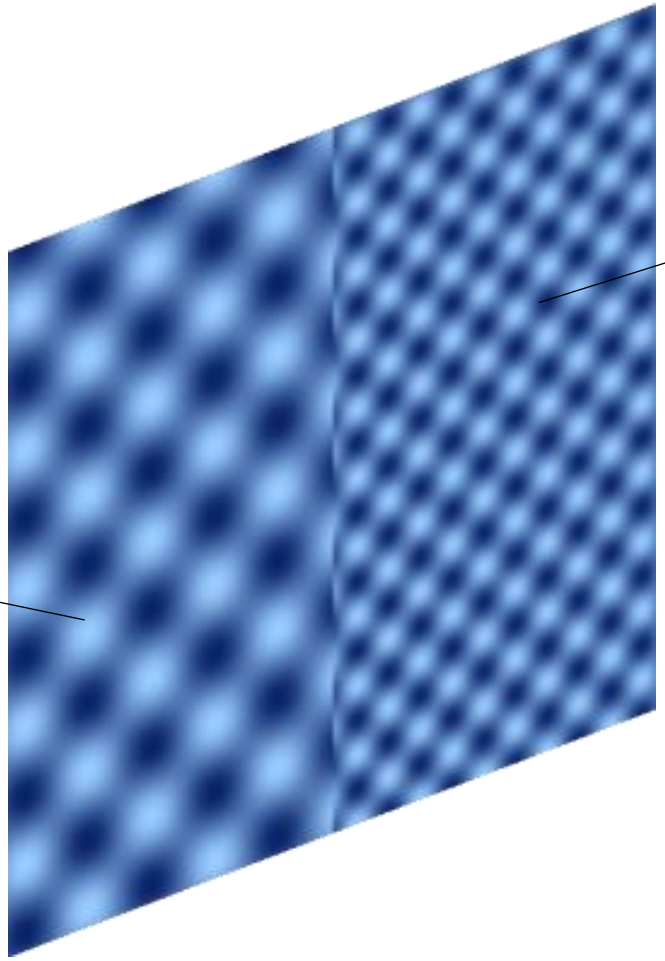
θ=90 deg





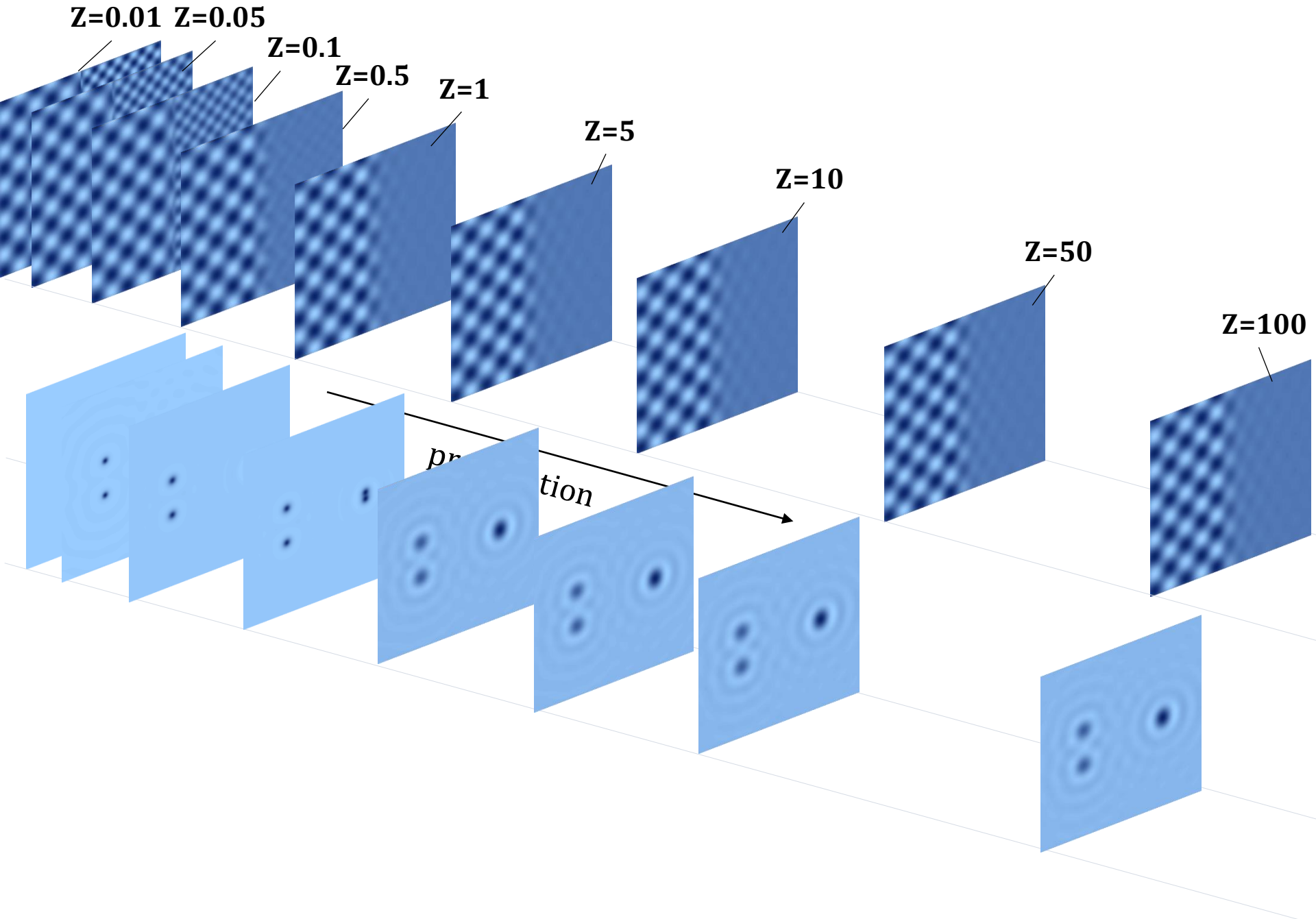


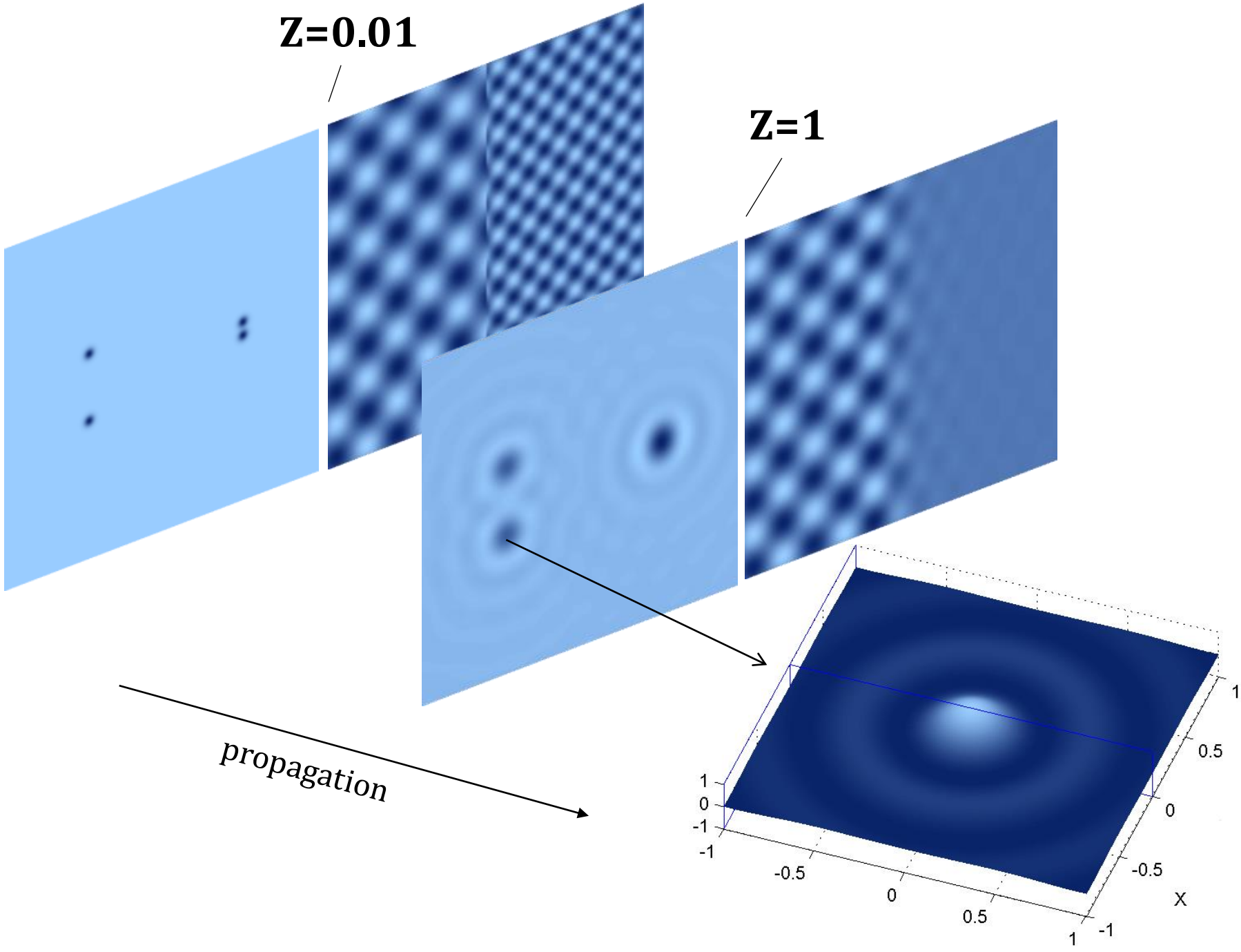


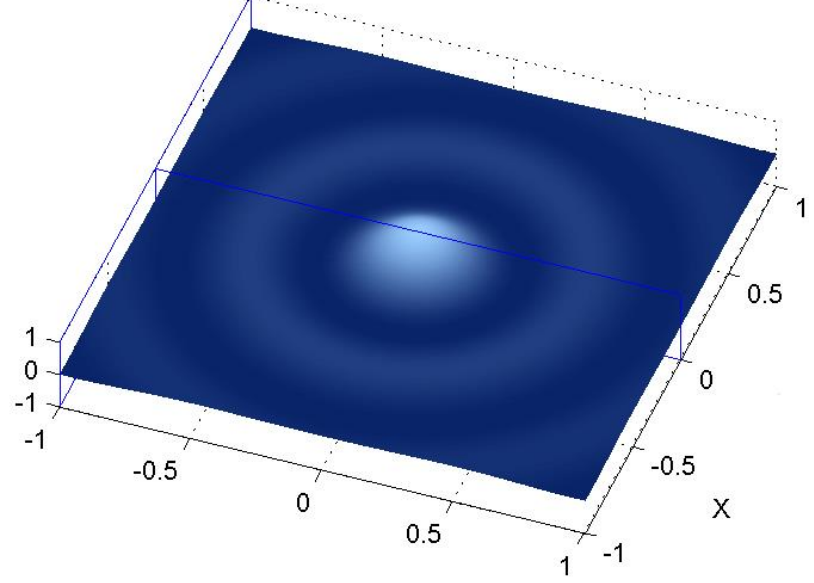


$2k_0$ frequency

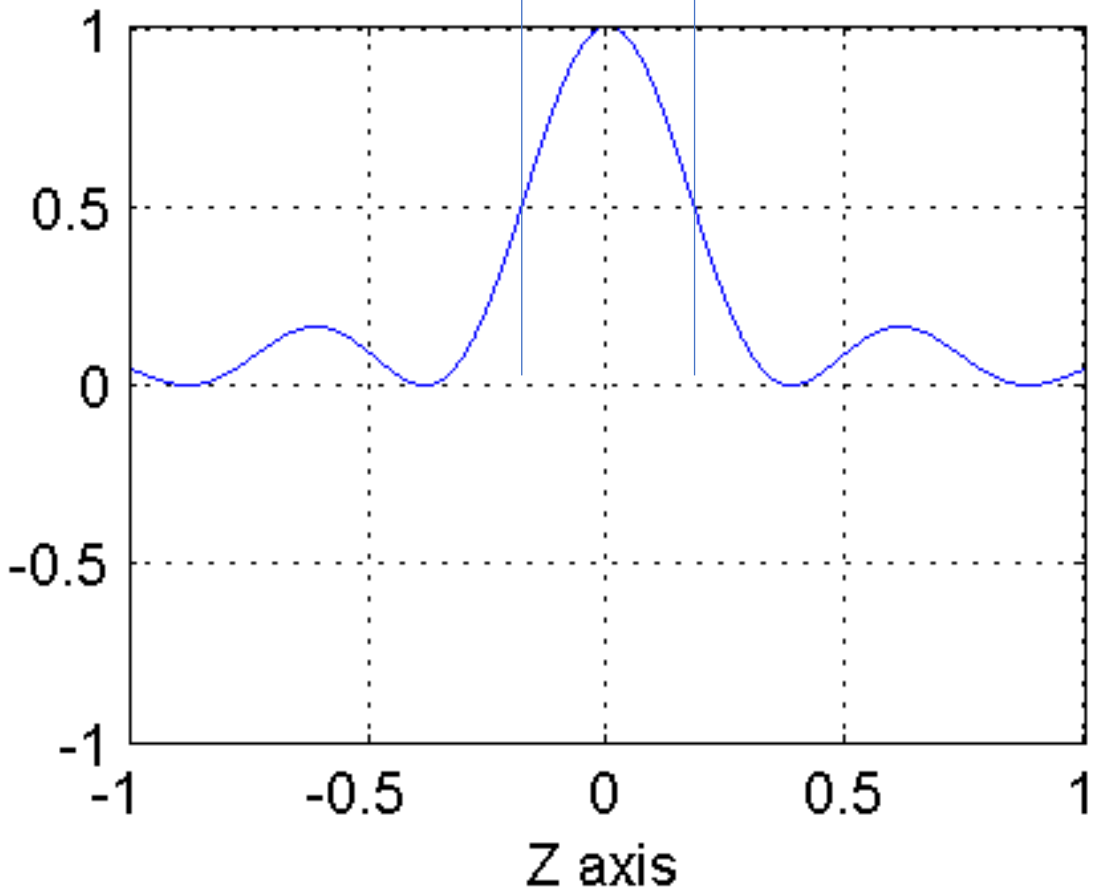
k_0 frequency


Z=0.01
Z=0.05
Z=0.1
Z=0.5
Z=1
Z=5
Z=10
Z=50
Z=100
tion


Z=0.01
Z=1
propagation
1
0
-1
-1
-0.5
0
0.5
1
1
0.5
0
-0.5
-1
X

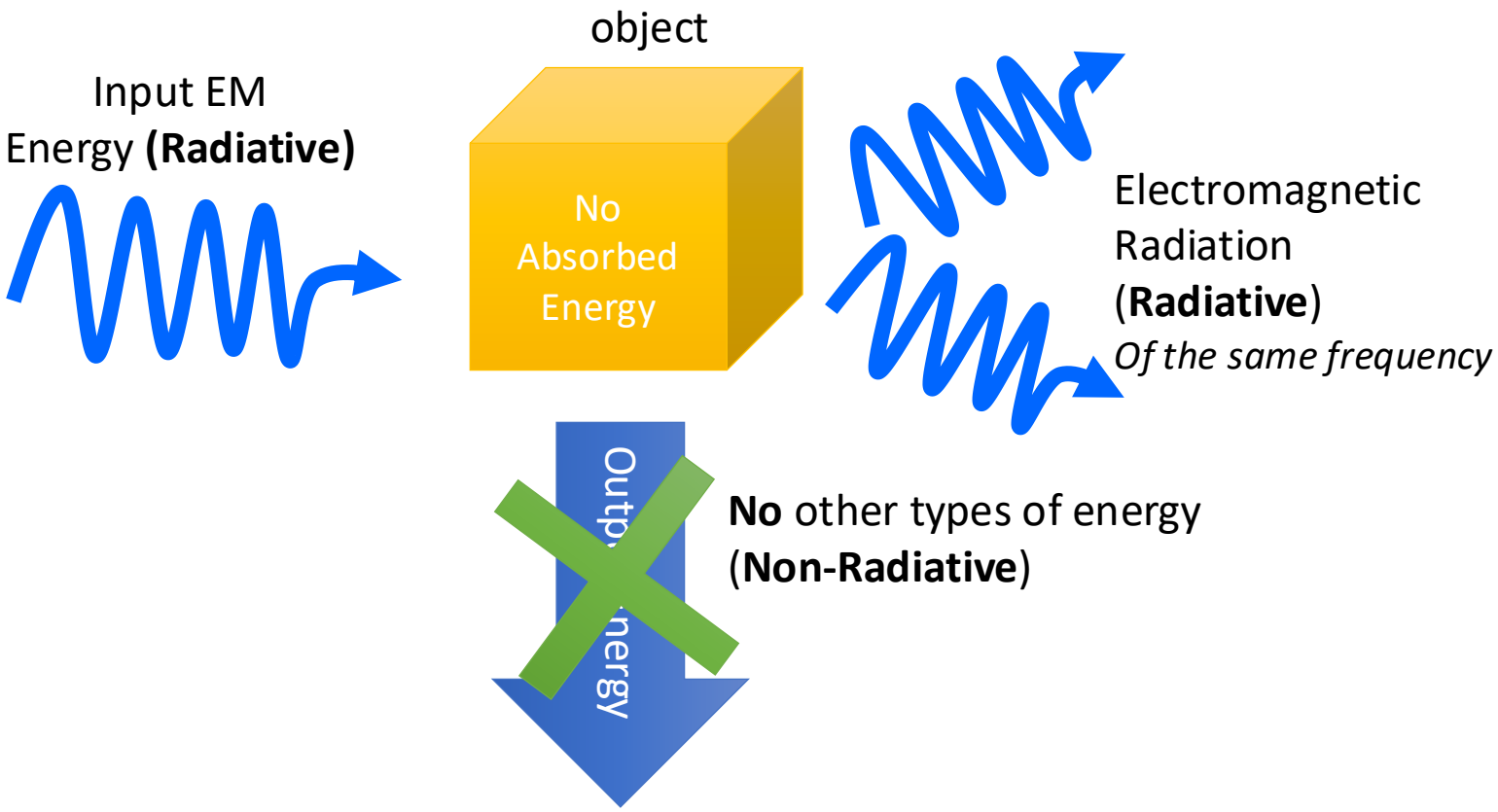


This width is ~ ½ wavelength



What happens if there is scattering?

Light Scattering



Milk Experiment







Scattering Experiment I – milk

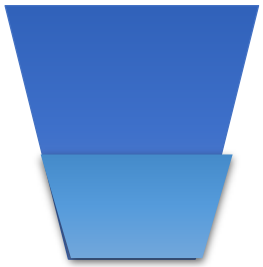
For 1cm depth of:



100% Milk



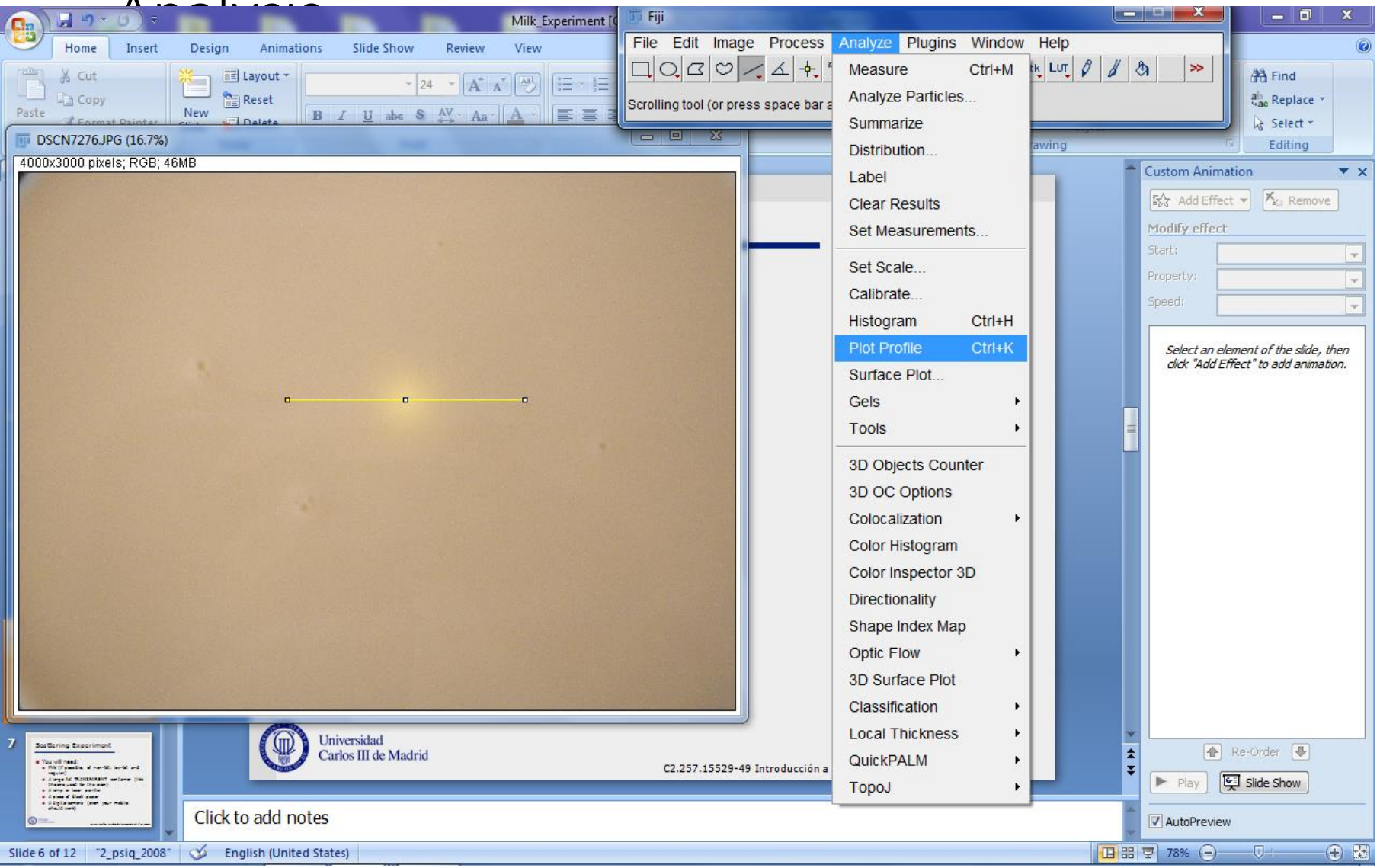
66% Milk



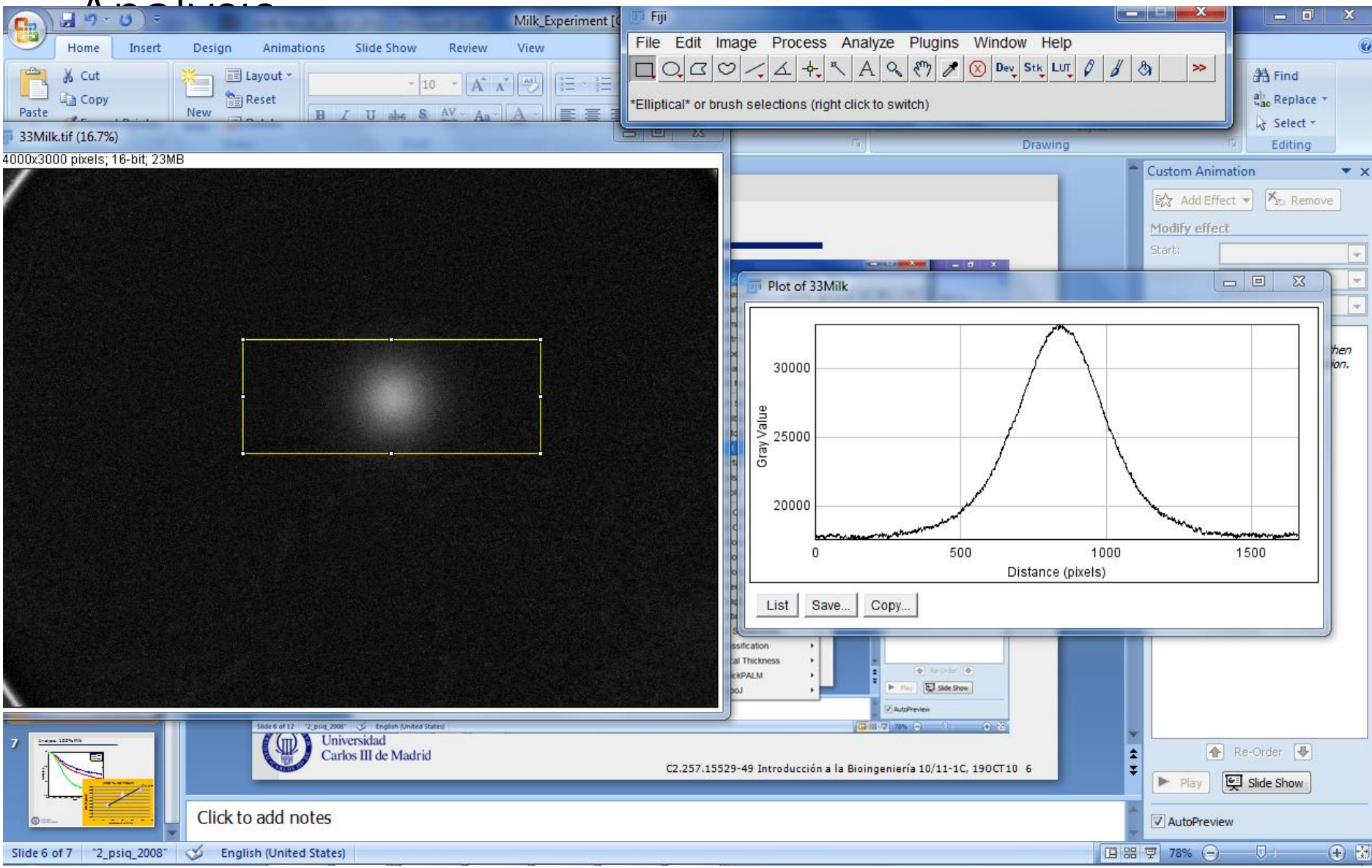
33% Milk



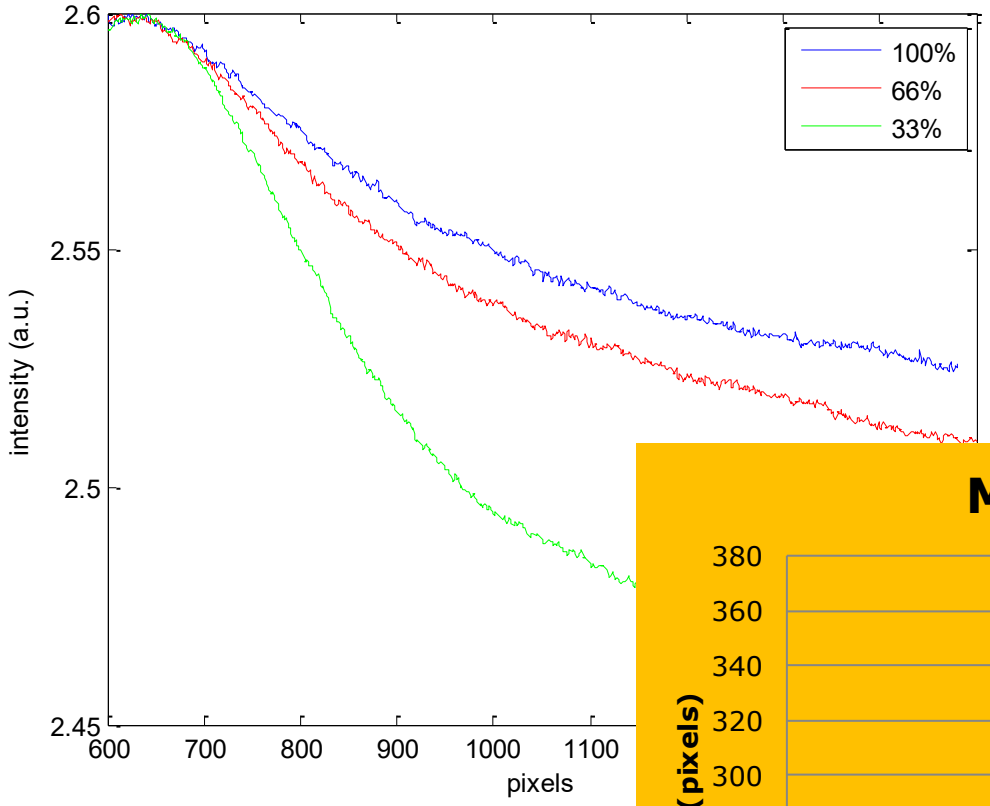
Analysis

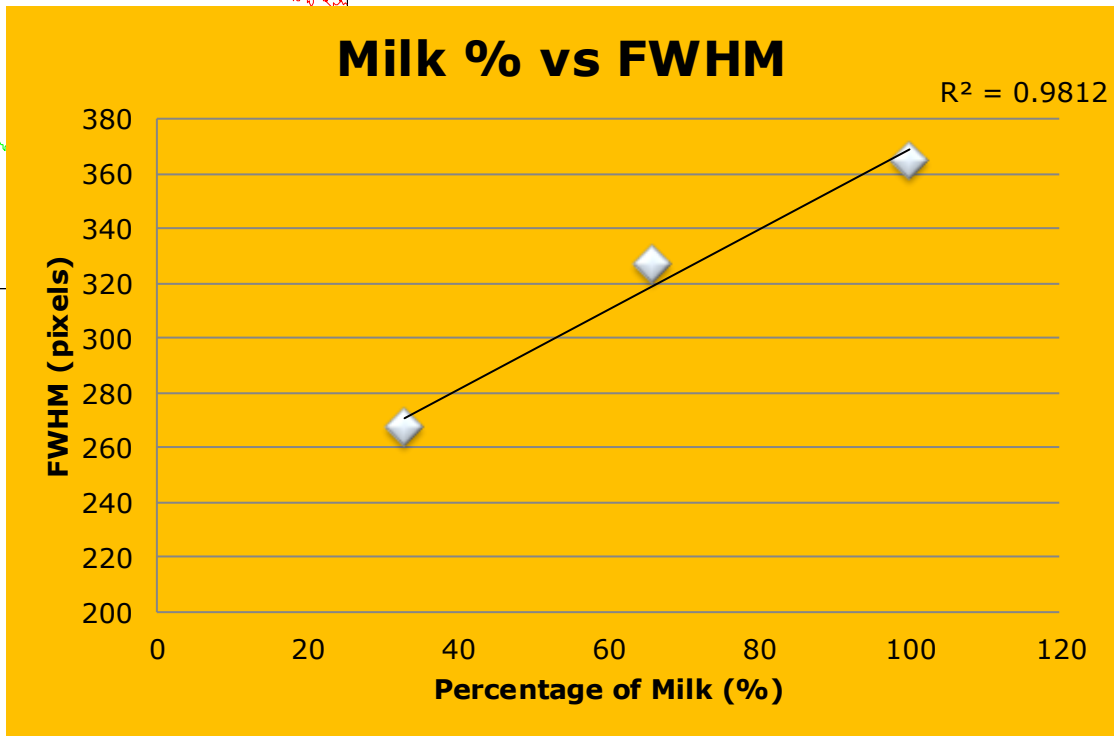

Fiji
File Edit Image Process Analyze Plugins Window Help
Measure Ctrl+M
Analyze Particles...
Summarize
Distribution...
Label
Clear Results
Set Measurements...
Set Scale...
Calibrate...
Histogram Ctrl+H
Plot Profile Ctrl+K
Surface Plot...
Gels
Tools
3D Objects Counter
3D OC Options
Colocalization
Color Histogram
Color Inspector 3D
Directionality
Shape Index Map
Optic Flow
3D Surface Plot
Classification
Local Thickness
QuickPALM
TopoJ
DSCN7276.JPG (16.7%)
4000x3000 pixels; RGB; 46MB
Custom Animation
Add Effect
Remove
Modify effect
Start:
Property:
Speed:
Select an element of the slide, then click "Add Effect" to add animation.
Re-Order
Play
Slide Show
AutoPreview
Click to add notes
Slide 6 of 12
"2_psiq_2008"
English (United States)
78%

Analysis


Milk_Experiment
Home
Insert
Design
Animations
Slide Show
Review
View
Fiji
File Edit Image Process Analyze Plugins Window Help
Elliptical or brush selections (right click to switch)
33Milk.tif (16.7%)
4000x3000 pixels; 16-bit; 23MB
Plot of 33Milk
Gray Value
30000
25000
20000
0
500
1000
1500
Distance (pixels)
List
Save...
Copy...
Custom Animation
Add Effect
Remove
Modify effect
Start:
Drawing
Editing
Find
Replace
Select
Universidad Carlos III de Madrid
C2.257.15529-49 Introducción a la Bioingeniería 10/11-1C, 19OCT10 6
Click to add notes
Slide 6 of 7
"2_psiq_2008"
English (United States)
78%
Re-Order
Play
Slide Show
AutoPreview

Analysis: 100% Milk

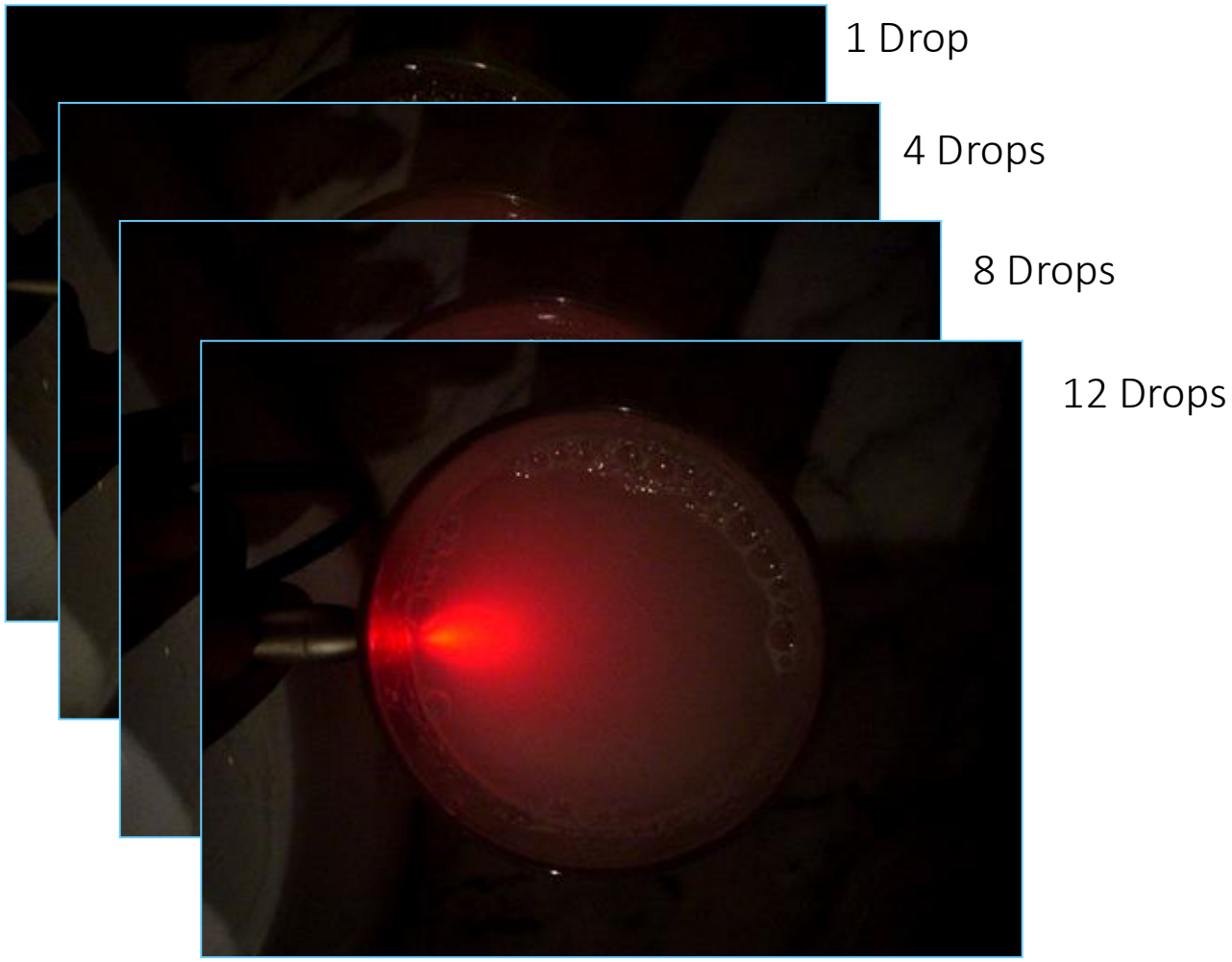




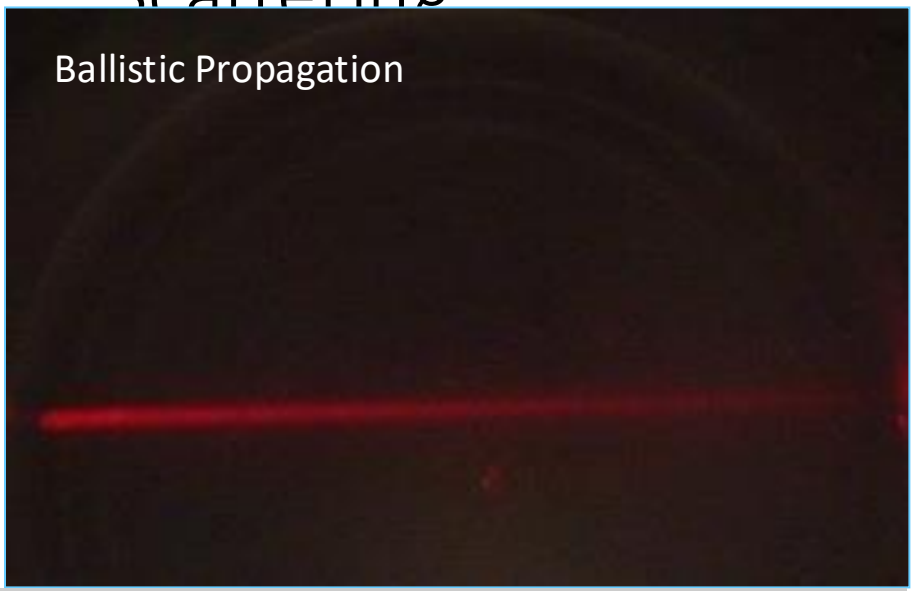
Scattering Experiment II



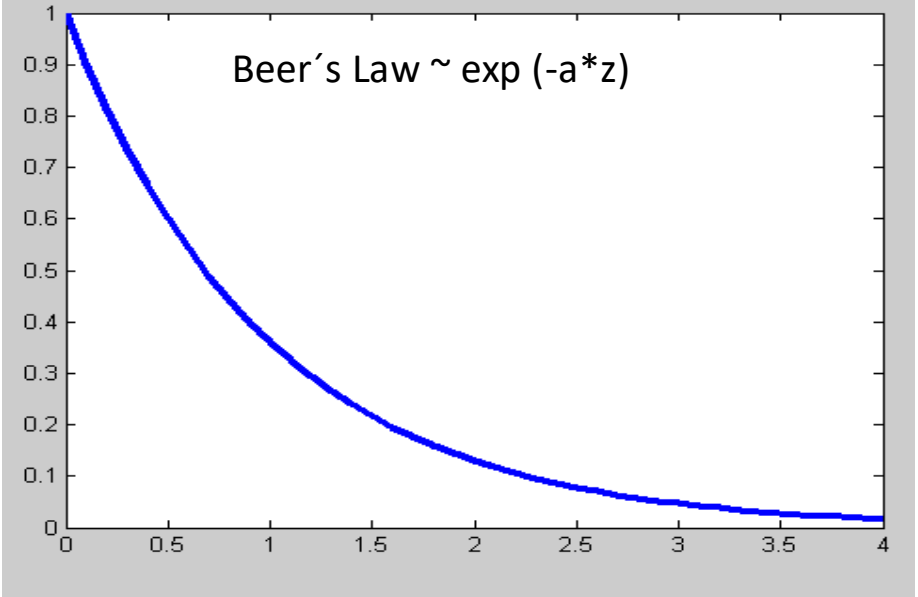
Volume Scattering

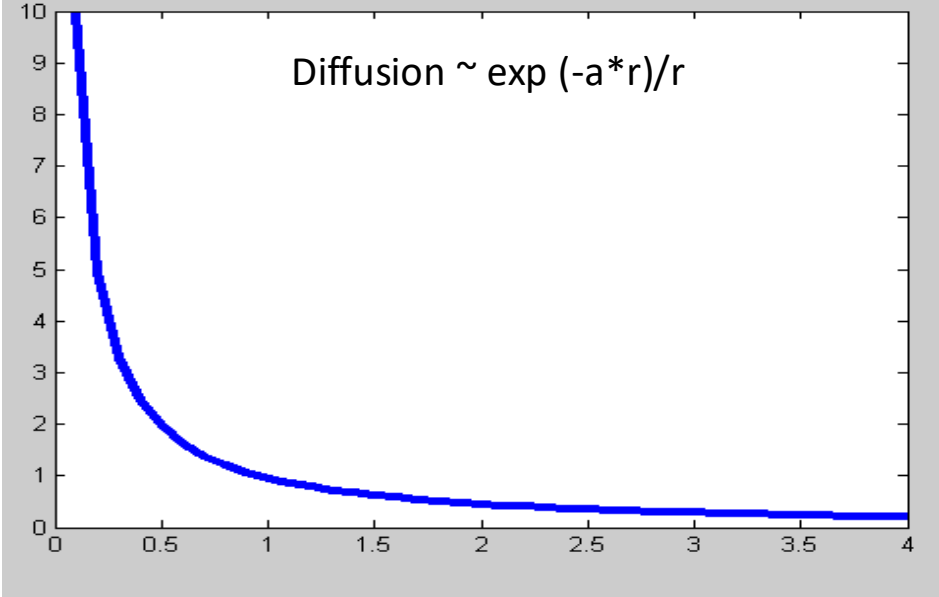


Scattering


Ballistic Propagation


Highly Scattering

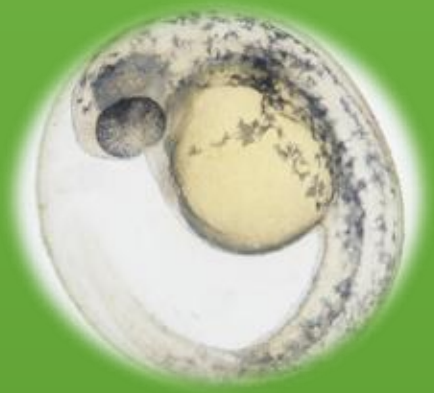

Beer´s Law ~ exp (-a*z)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.5
1
1.5
2
2.5
3
3.5
4


Diffusion ~ exp (-a*r)/r
10
9
8
7
6
5
4
3
2
1
0
0
0.5
1
1.5
2
2.5
3
3.5
4

Scattering



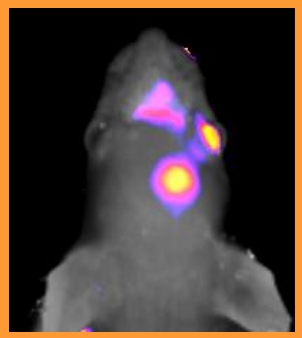
Transparent Samples
(Beer's Law)



C. elegans, P. hawaiiensis, zebrafish (early), ...

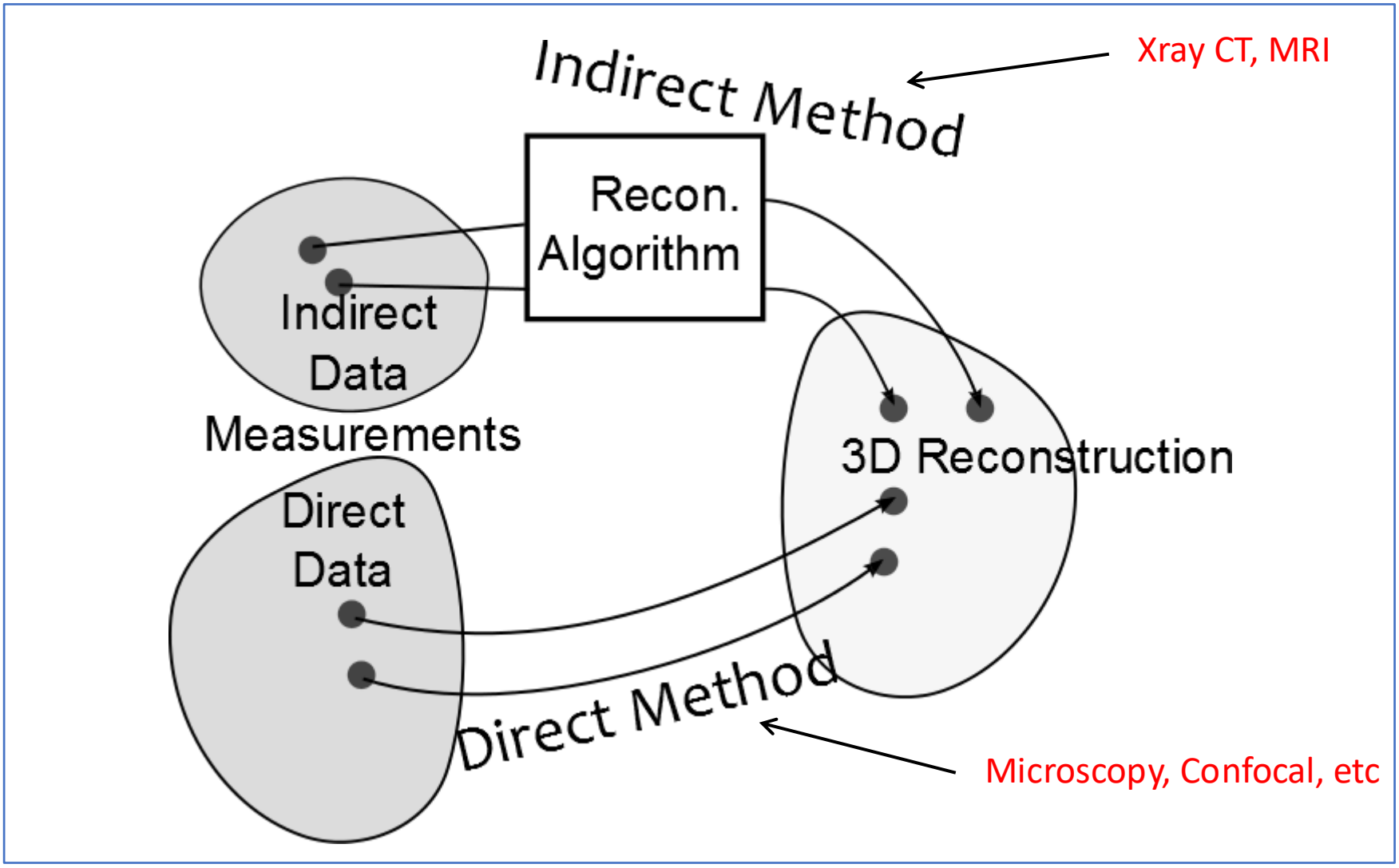


Highly Scattering Samples
(Diffusion Approximation, Fick's law)

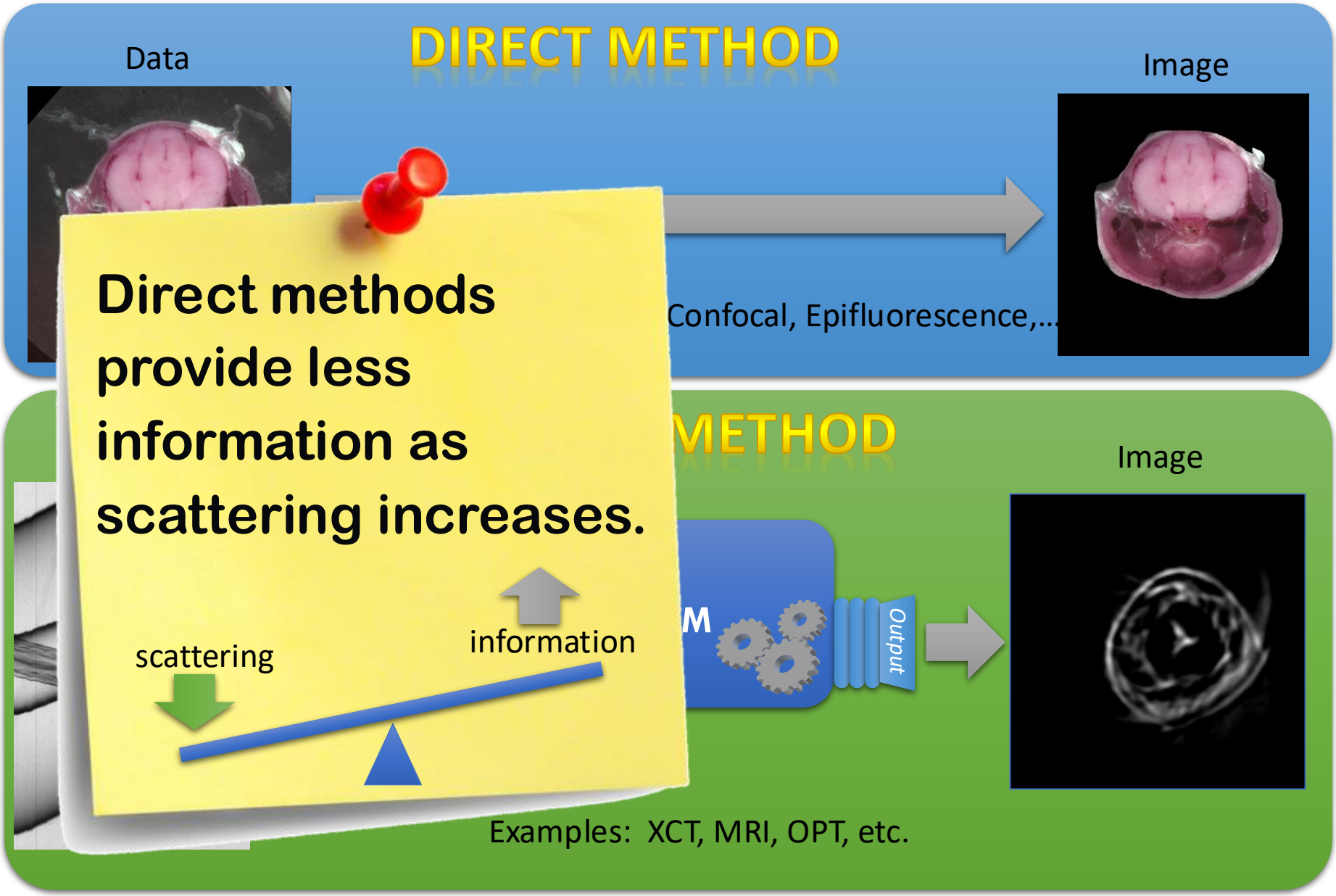


Most tissues >1mm, mice, humans

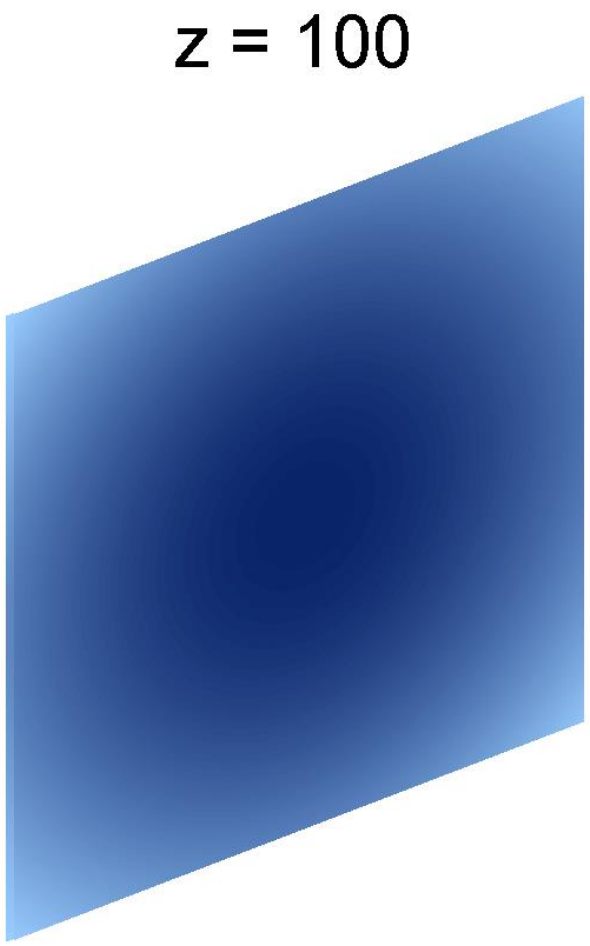
What are indirect methods?

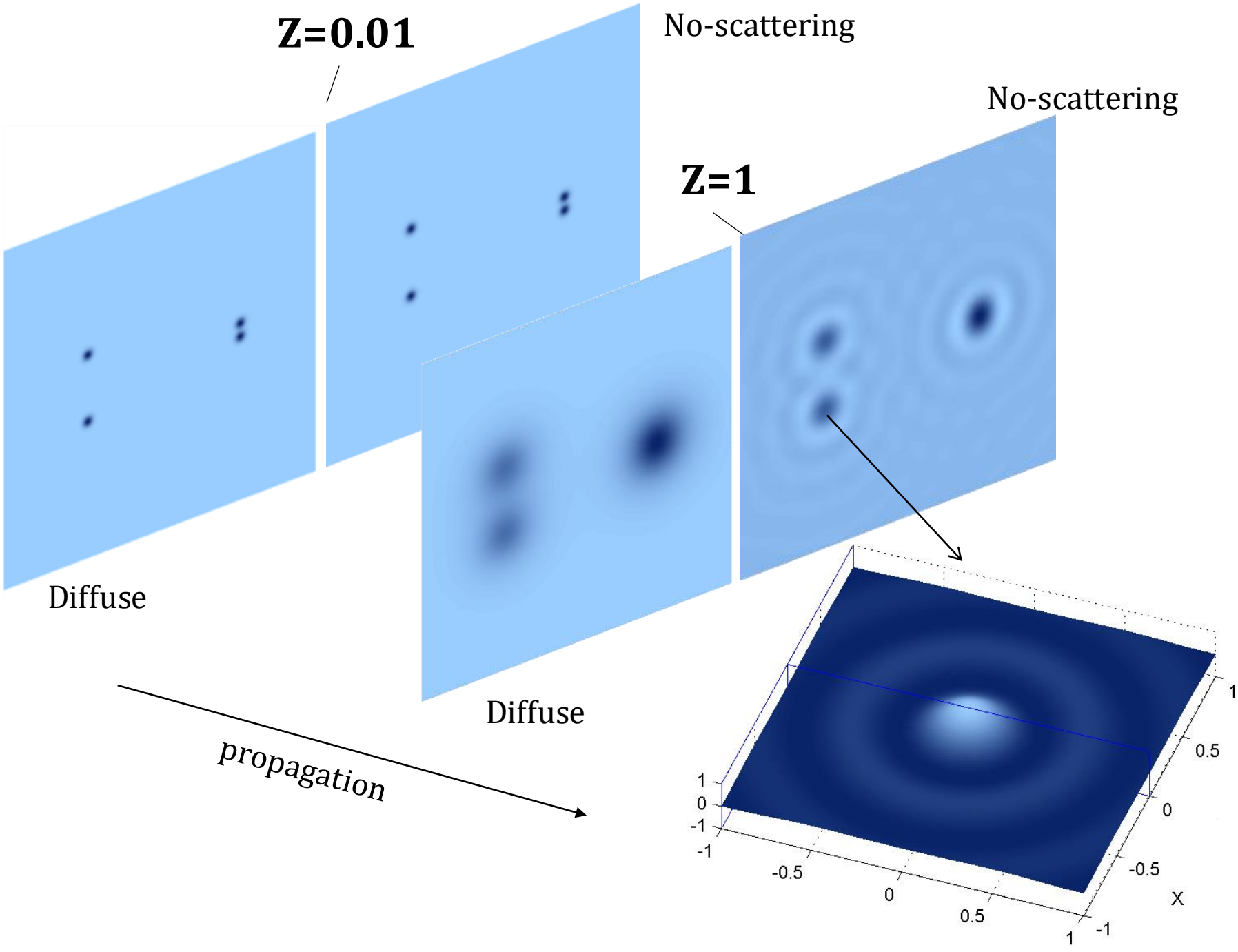


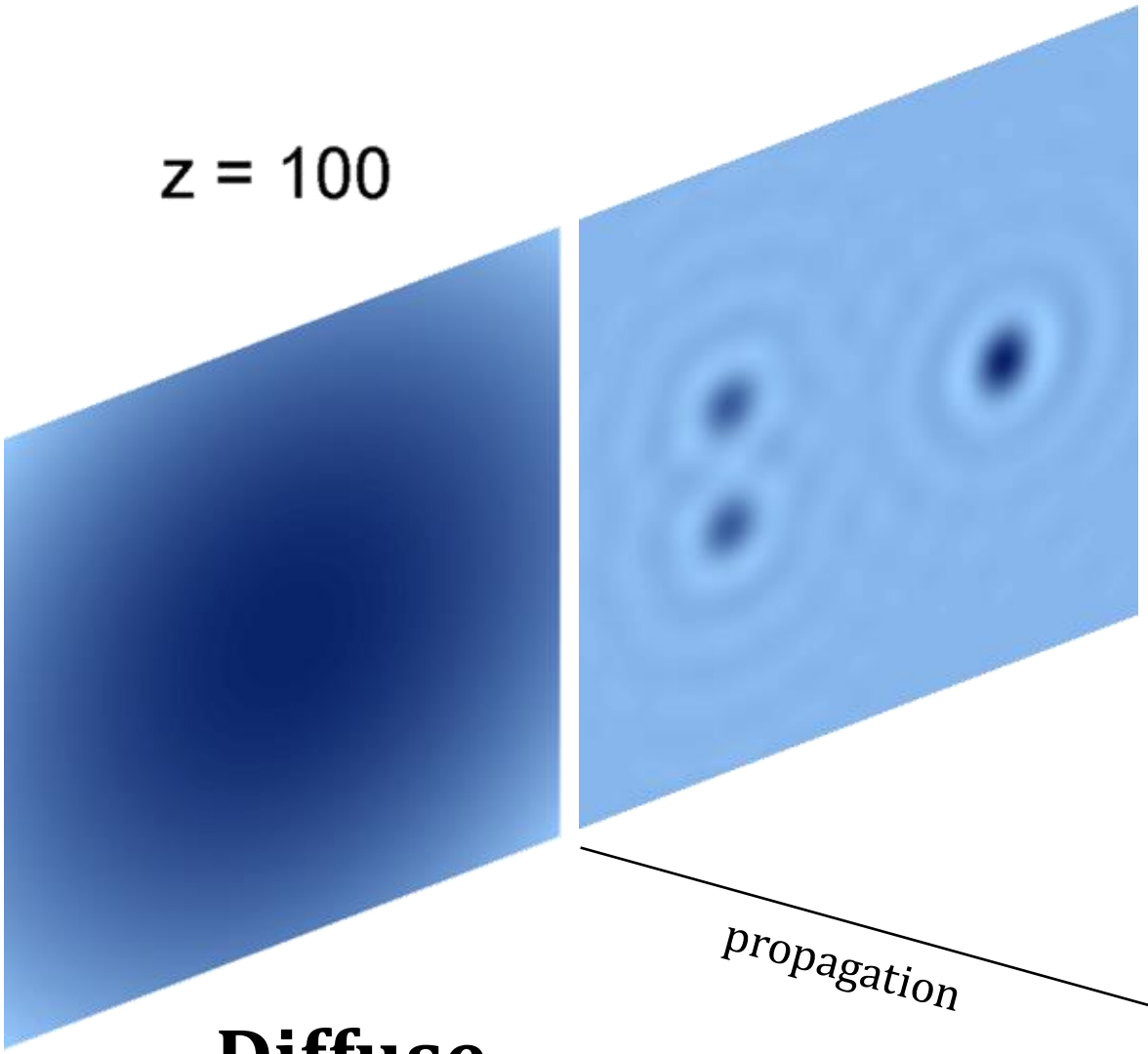
Direct vs Indirect Methods



In the case of diffuse light…




Z=0.01
No-scattering
No-scattering
Z=1
Diffuse
Diffuse
propagation
1
0
-1
-1
-0.5
0
0.5
1
1
0.5
0
-0.5
-1
X



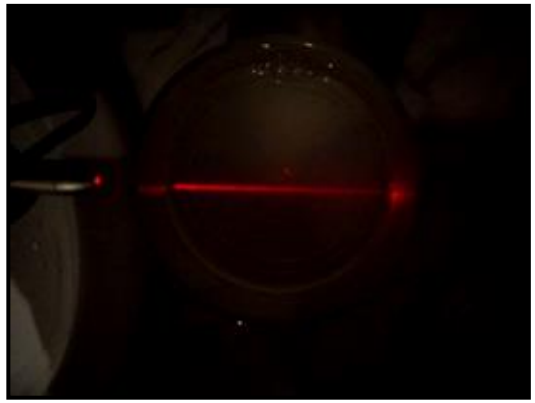
No-scattering

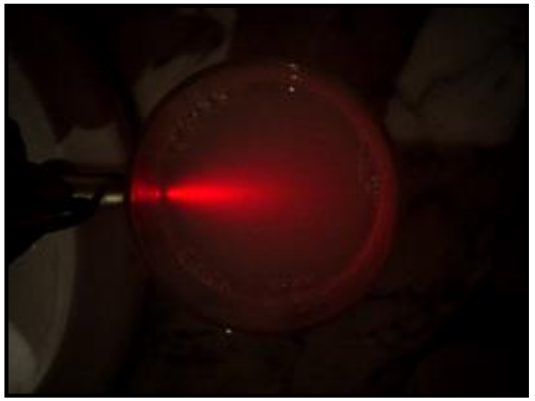
Diffuse

Low scattering

Optical Tomography in Microscopy

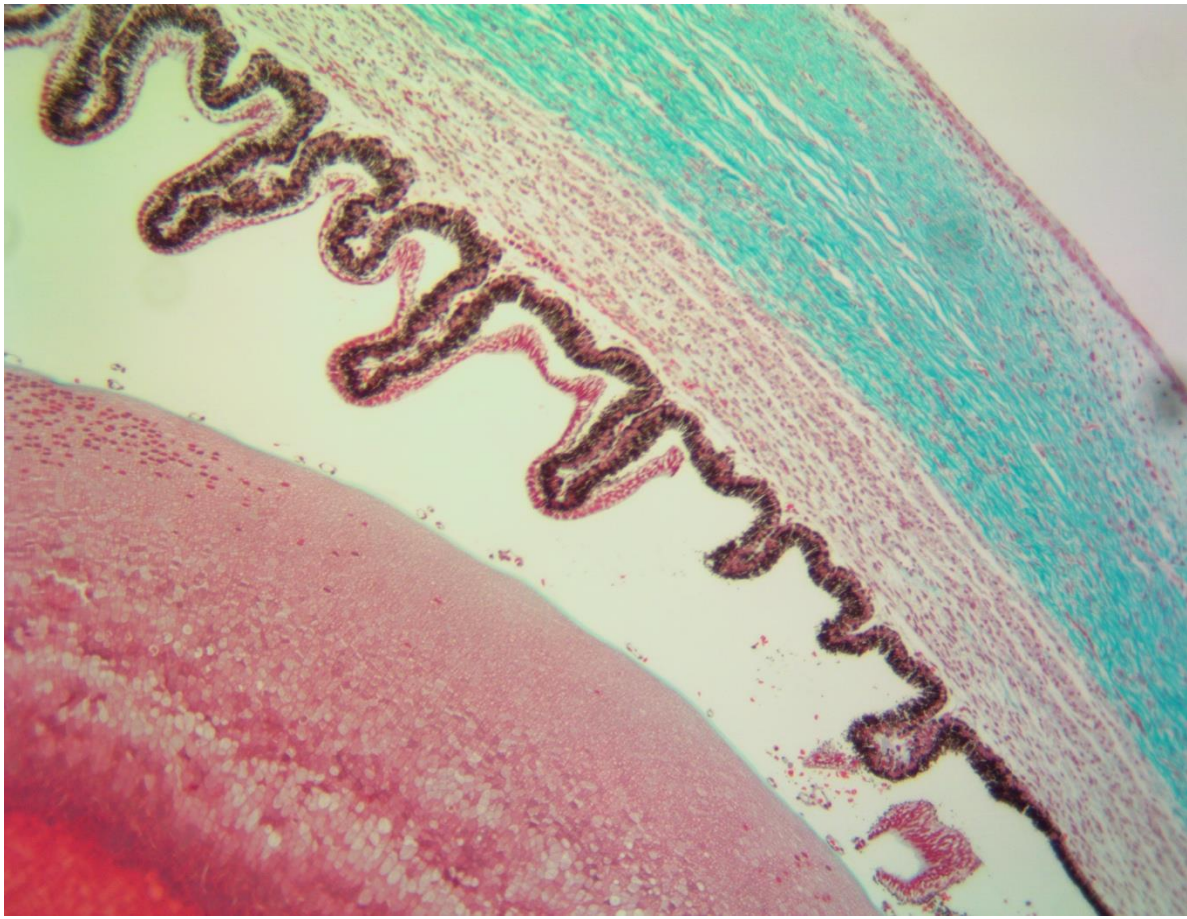
What we saw in the previous lectures





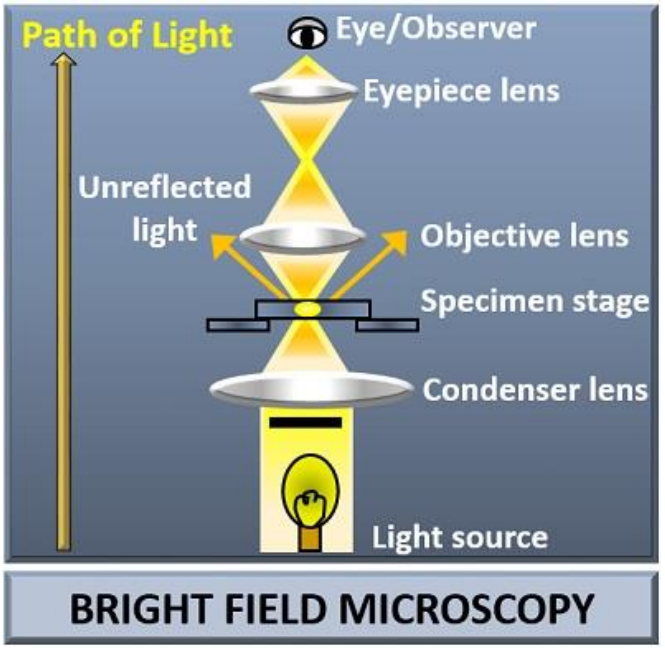


Widefield Microscopy



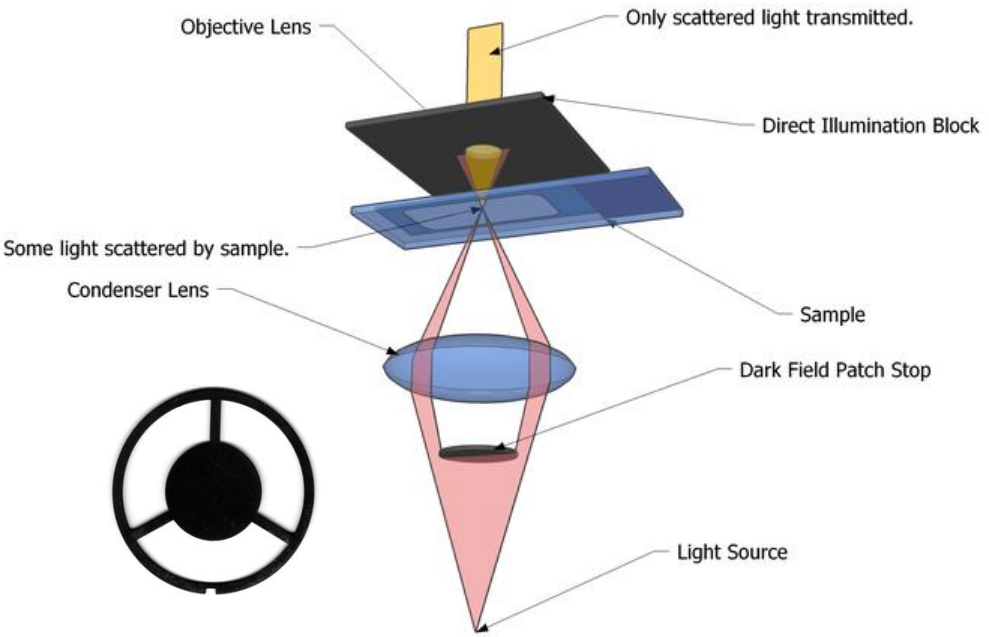
Bright Field Microscope



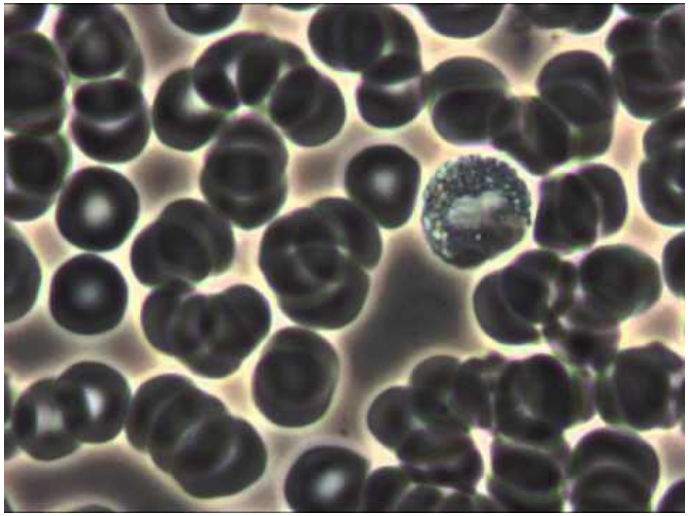


Dark Field Microscope





Phase Contrast Microscope



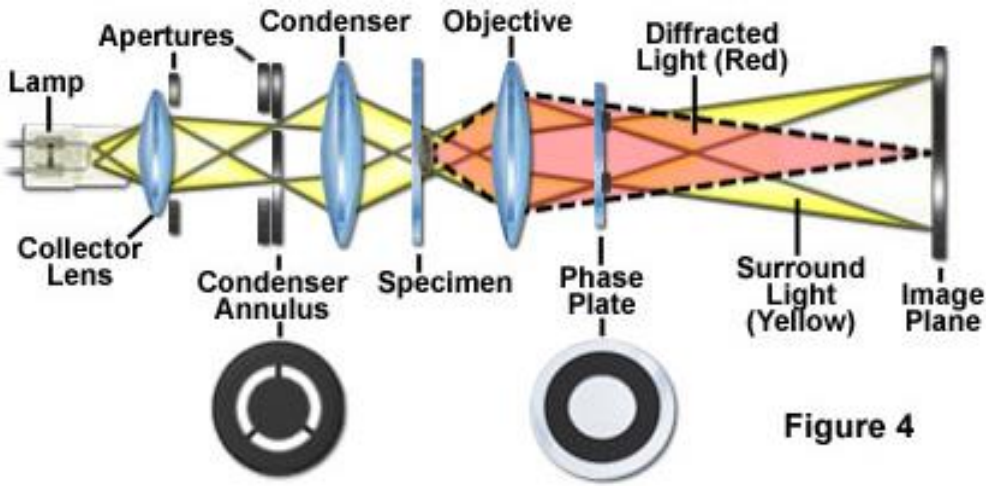
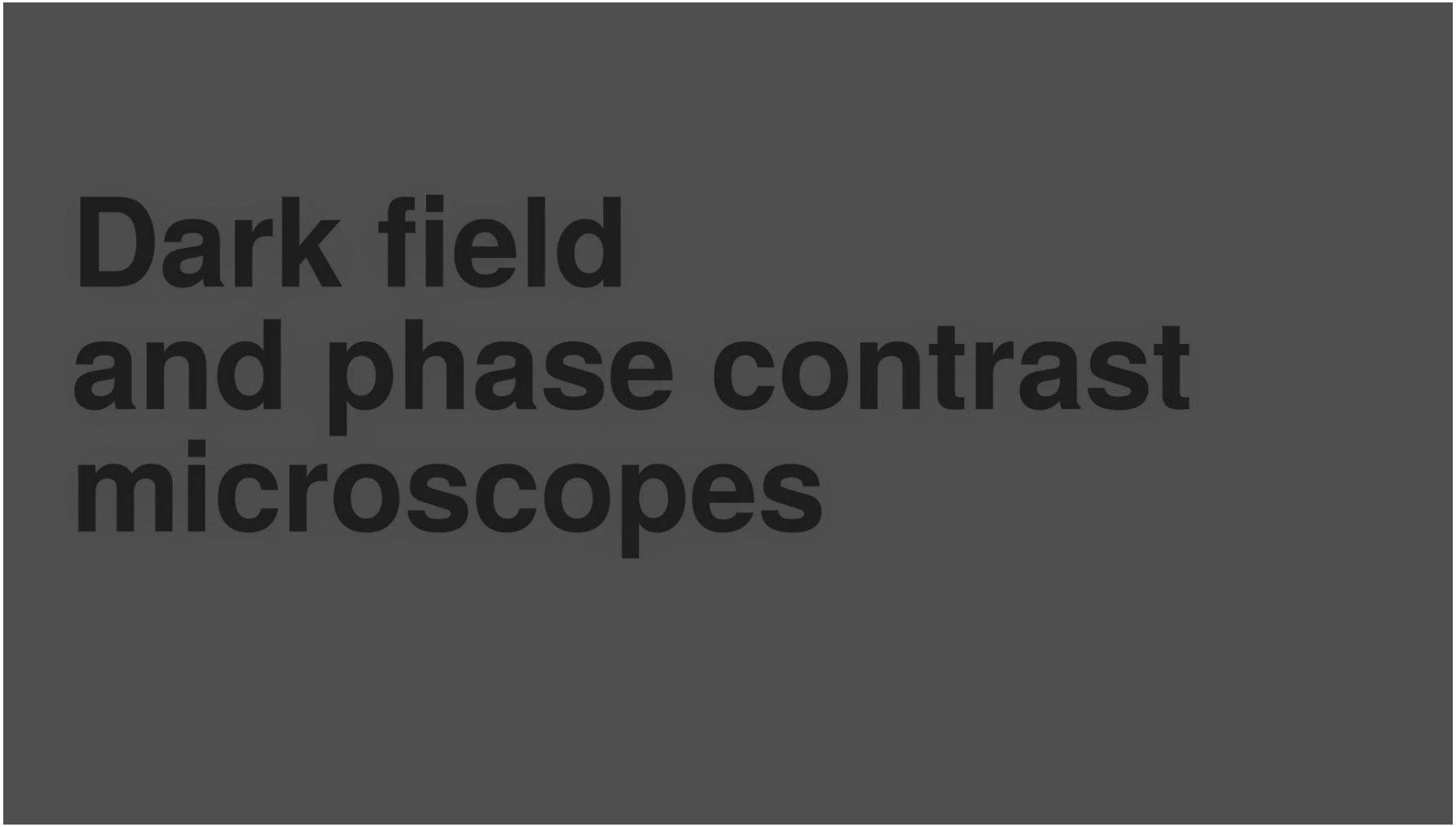
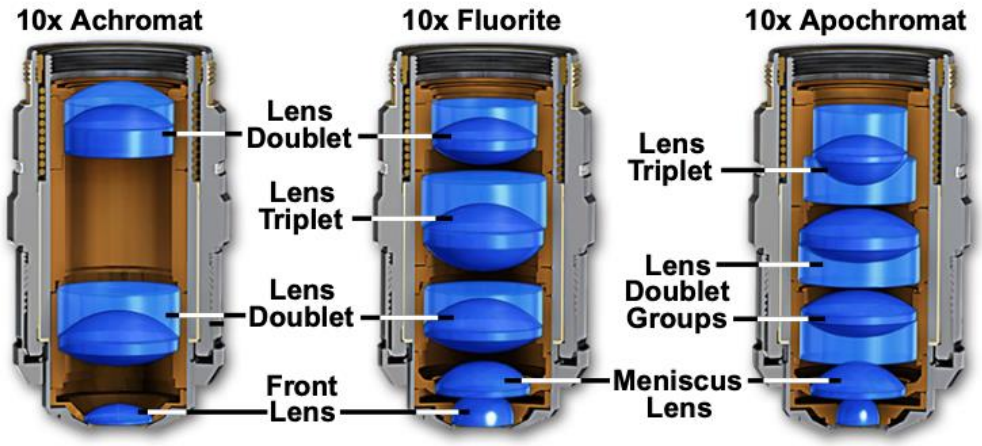


Figure 4

Dark Field and Phase Contrast Microscopes

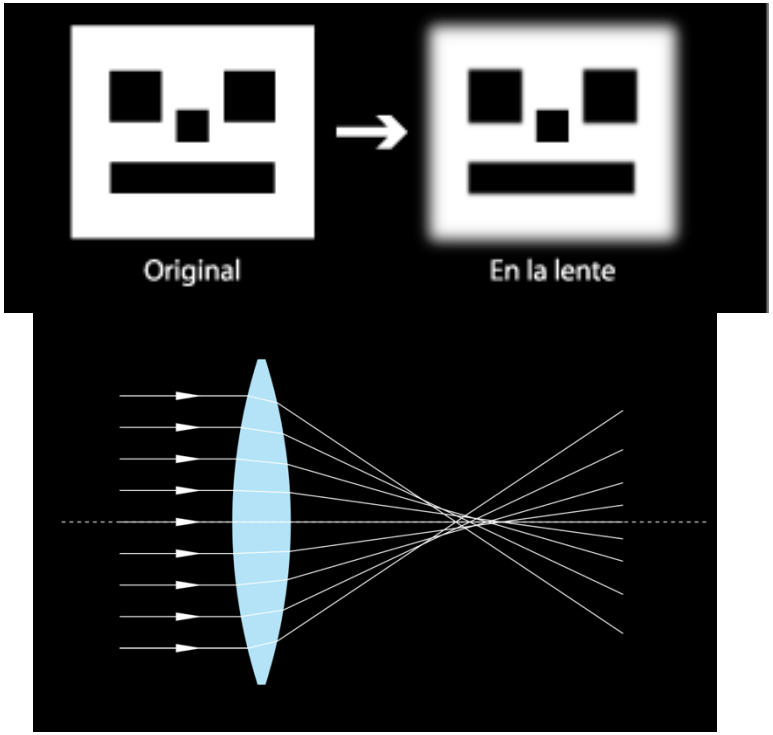


Objectives

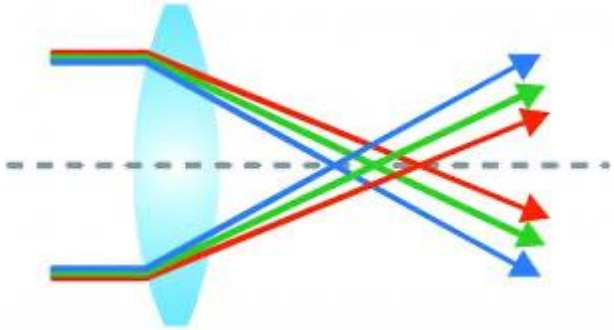


Objective Type	Spherical Aberration	Chromatic Aberration	Field Curvature
Achromat	1 Color	2 Colors	No
Plan Achromat	1 Color	2 Colors	Yes
Fluorite	2-3 Colors	2-3 Colors	No
Plan Fluorite	3-4 Colors	2-4 Colors	Yes
Plan Apochromat	3-4 Colors	4-5 Colors	Yes

Aberrations

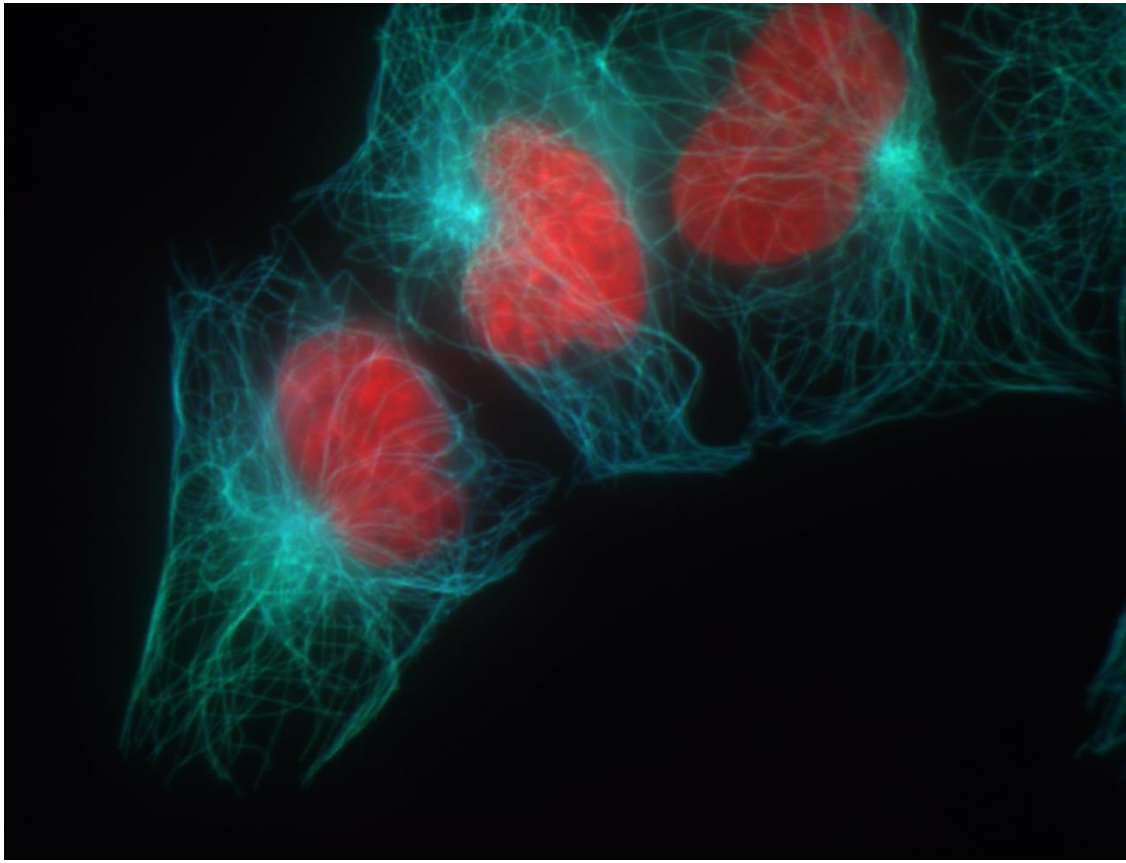






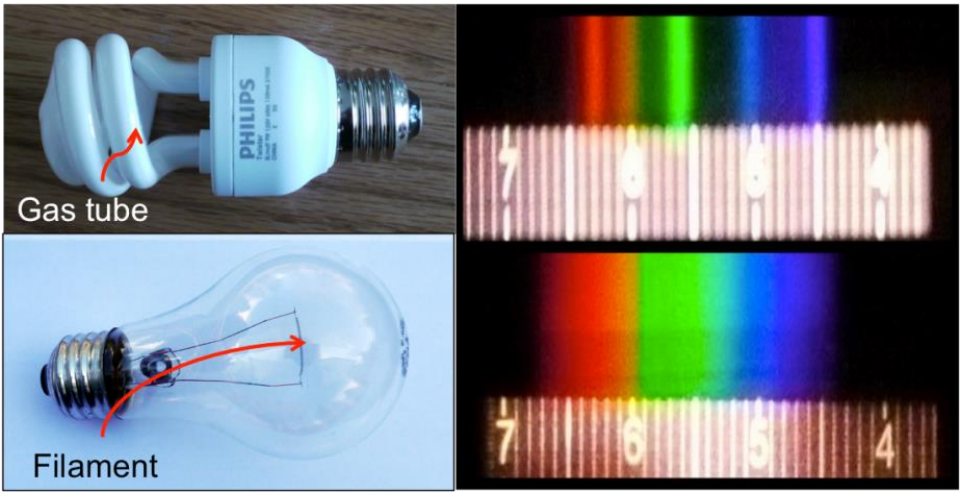
uc3m | Universidad Carlos III de Madrid

Fluorescence Microscopy



Fluorescence Microscopy

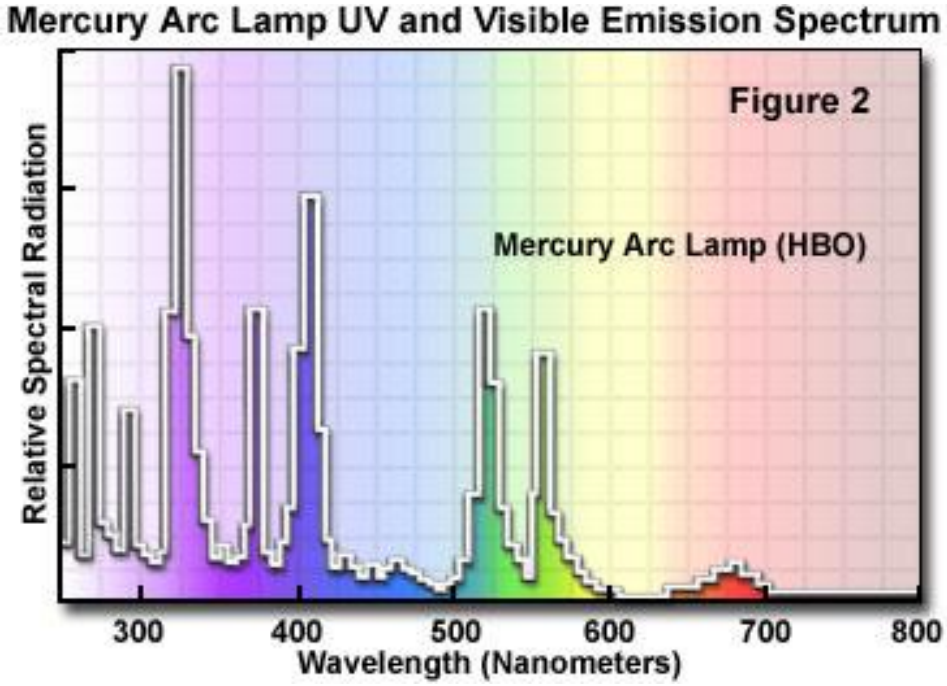
- The light source is a mercury arc lamp.
- The lamp emits over a wide range of wavelengths.





Fluorescence Microscopy

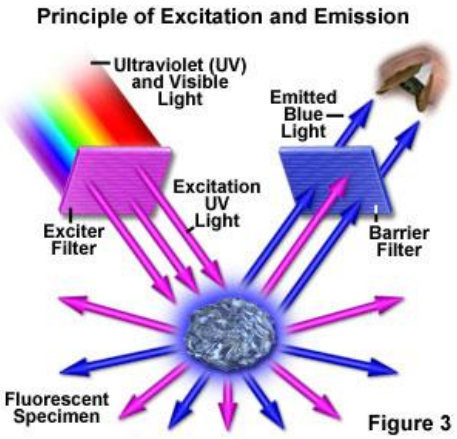
- The light source is a mercury arc lamp.
- This lamp emits over a wide range of wavelengths.

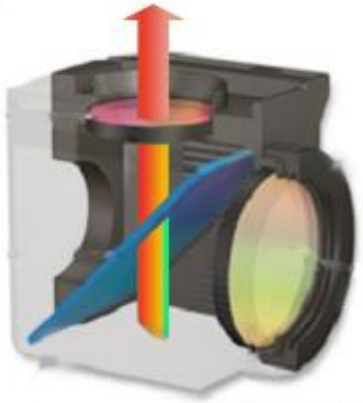


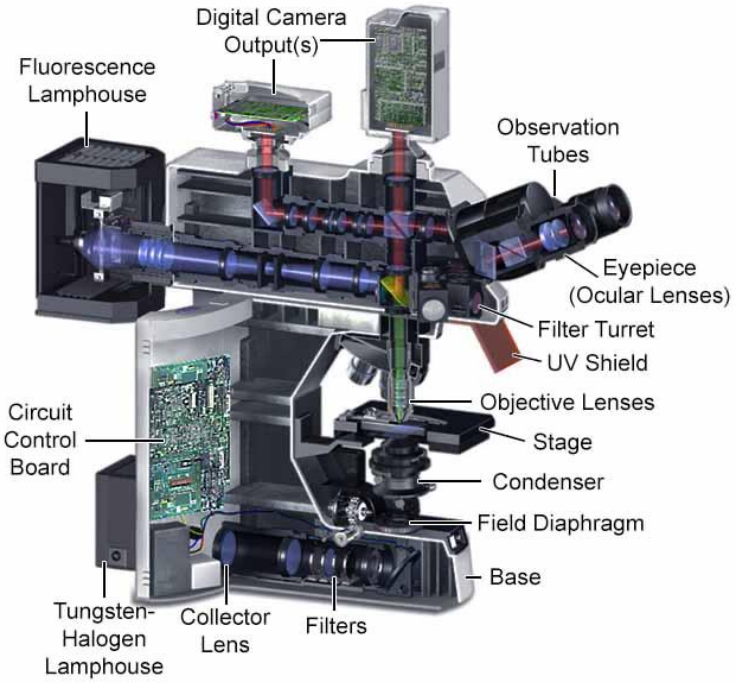


Fluorescence Microscopy

- Using filters, we can select specific wavelength ranges for excitation and emission.
- We need:
  - Excitation filters to illuminate the sample at a specific wavelength.
  - The dichroic mirror directs the excitation light and separates fluorescence emission from reflected excitation light.
  - Emission filters select a specific wavelength range.



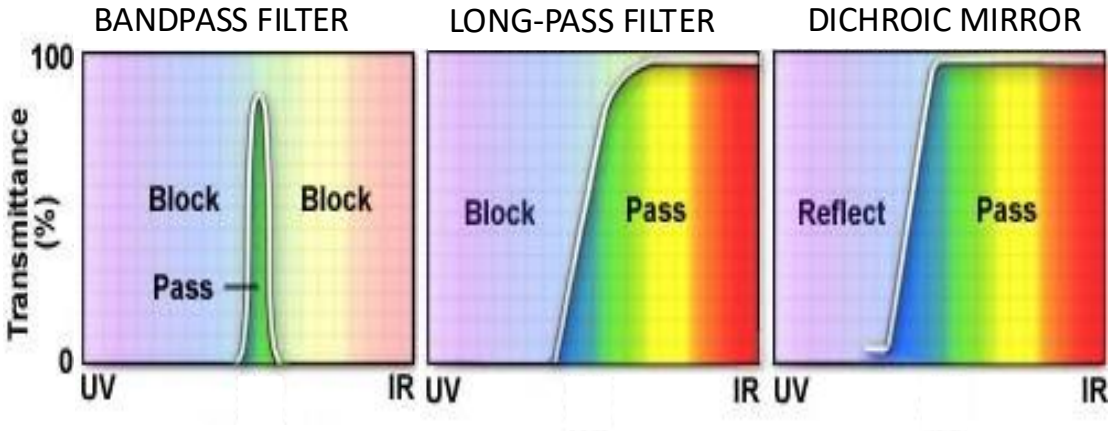




Fluorescence Microscopy

The excitation light intensity is generally several orders of magnitude greater than the fluorescence emission. We need to isolate the signal efficiently.





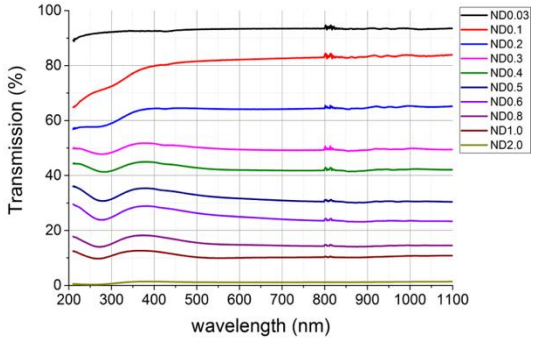
Fluorescence Microscopy

Neutral-density filters

They uniformly attenuate light intensity across the visible wavelength range (also used in other imaging techniques such as photography).



https://www.olympus-lifescience.com/zh/microscope-resource/primer/java/filters/absorption/



Fluorescence Microscopy

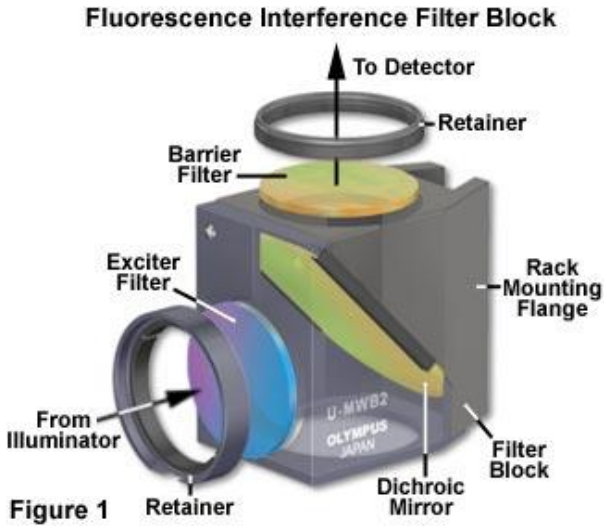
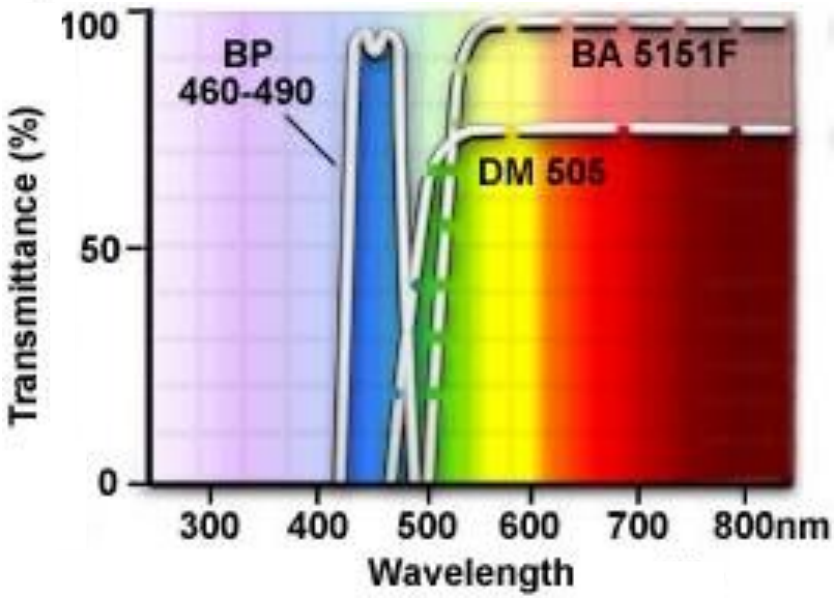
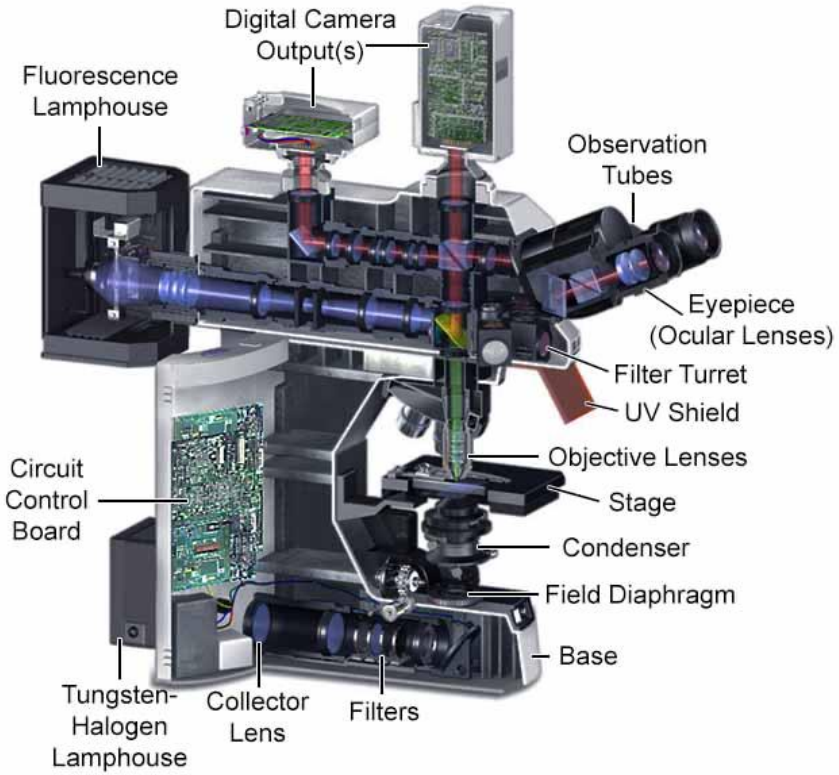
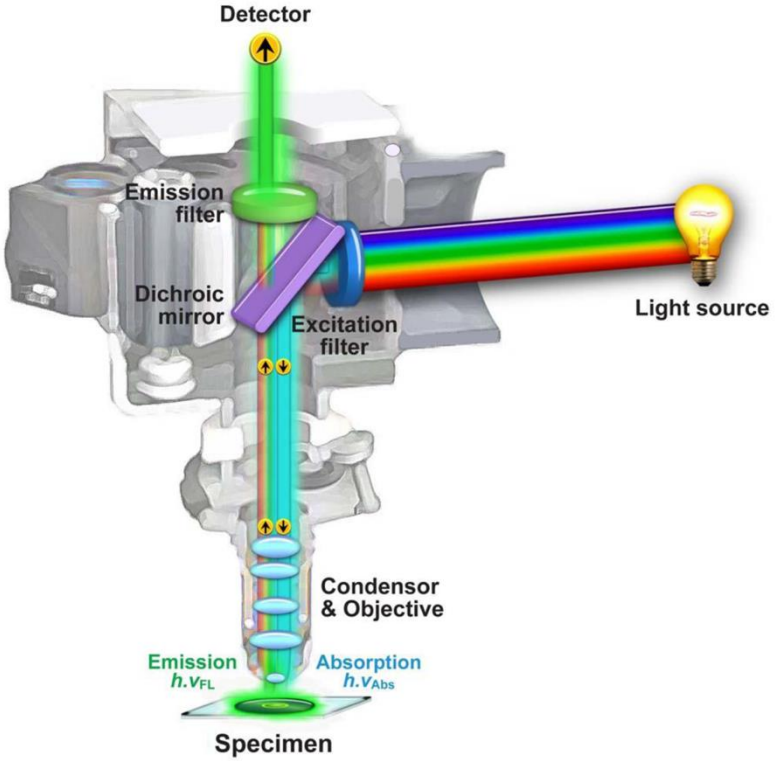


Figure 1

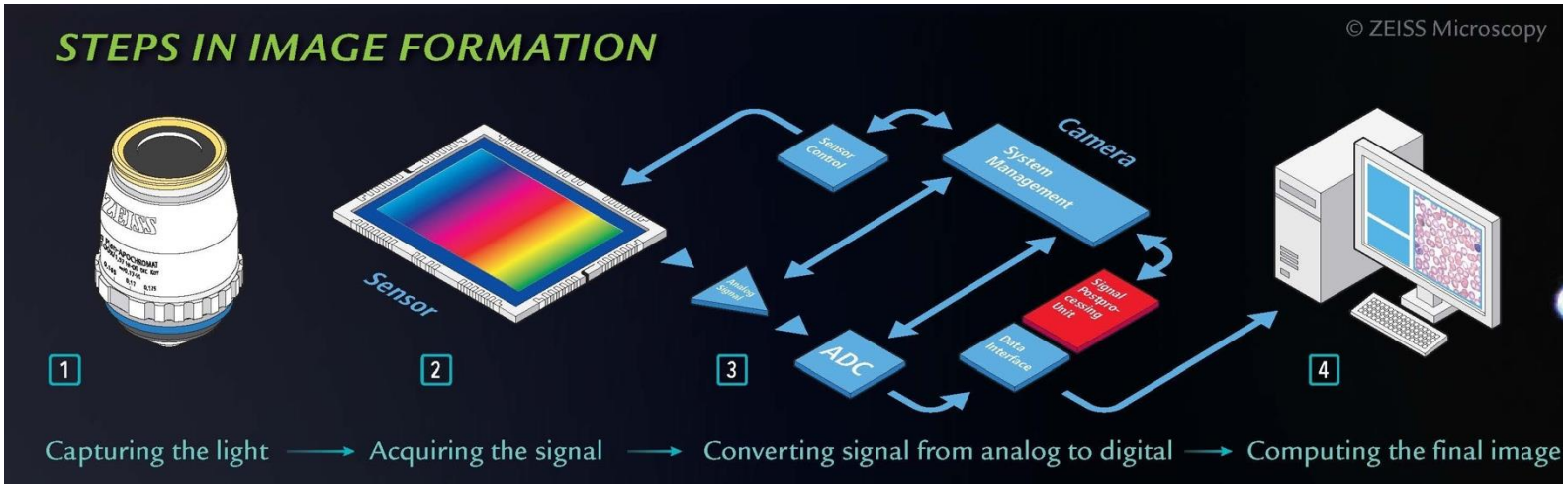


Fluorescence Microscopy

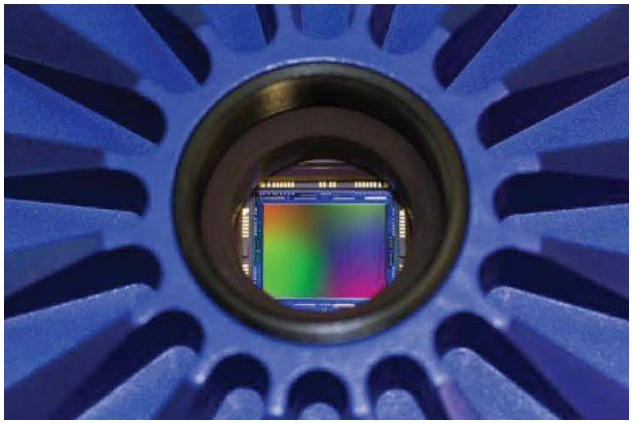


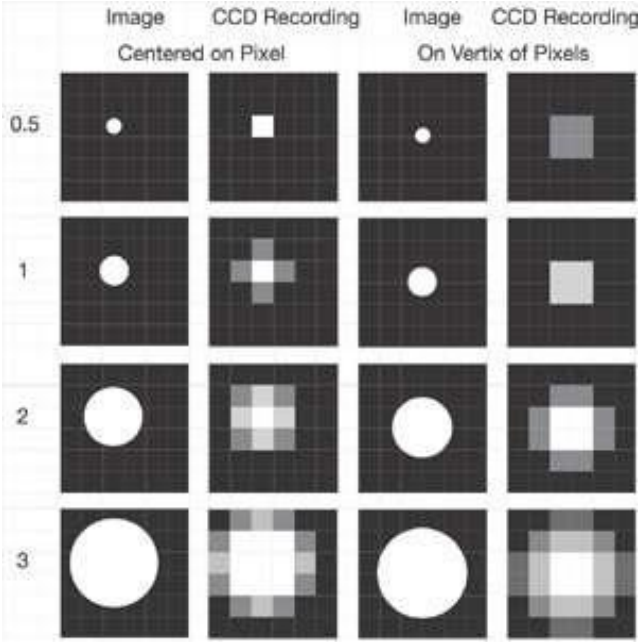


Fluorescence Microscopy

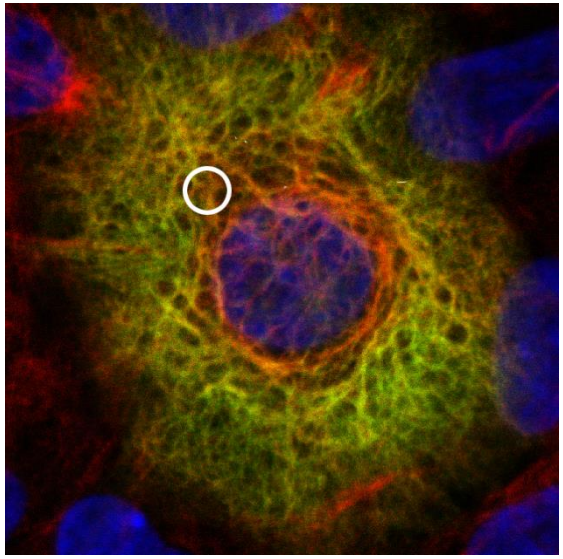


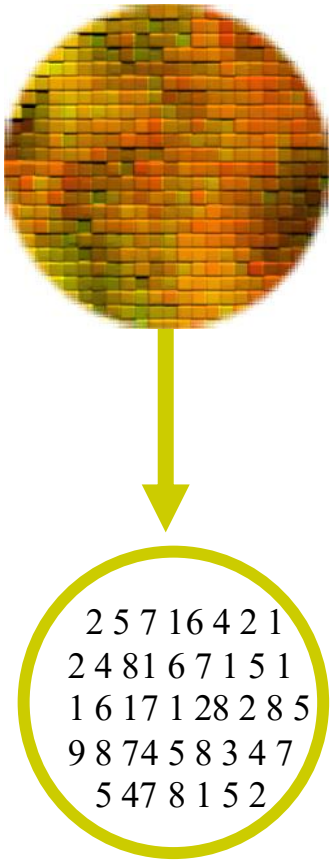
Fluorescence Microscopy





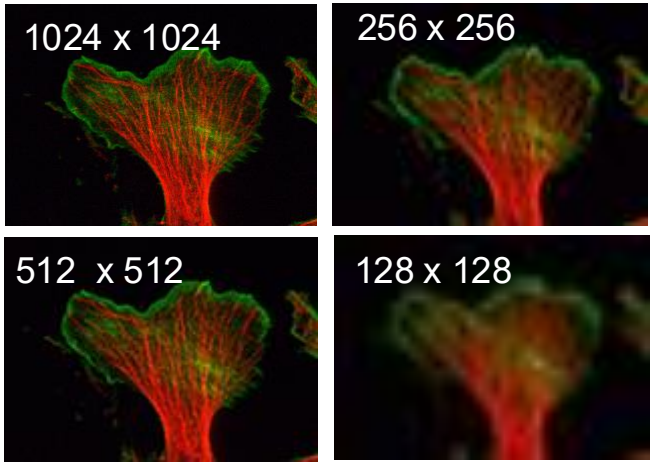
Fluorescence Microscopy

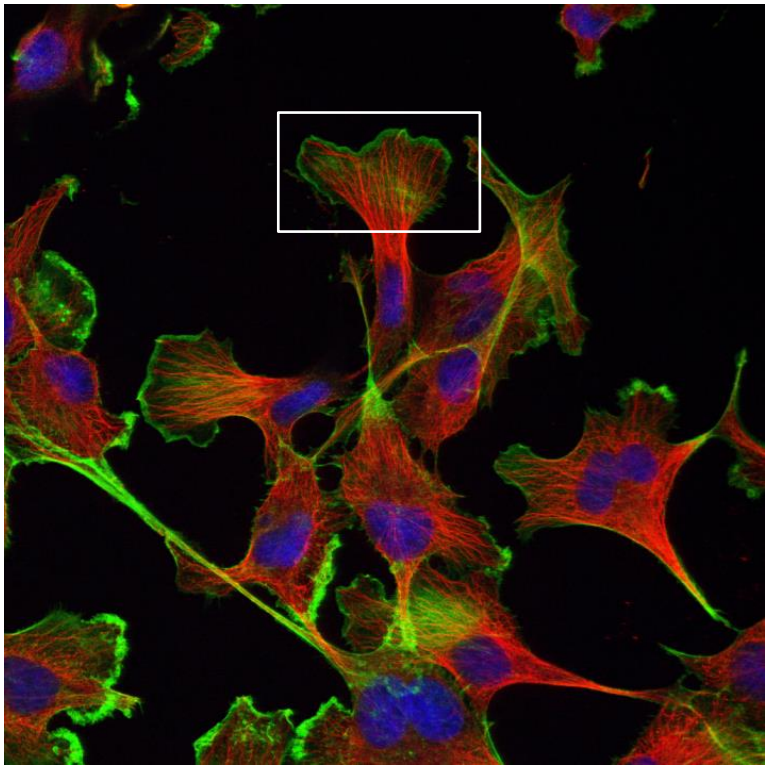




Fluorescence emission is encoded using values from 0 to 255 (in an 8-bit image), 0 to 4095 (in a 12-bit image), or 0 to 65535 (16 bits), representing different signal intensity levels.

Fluorescence Microscopy

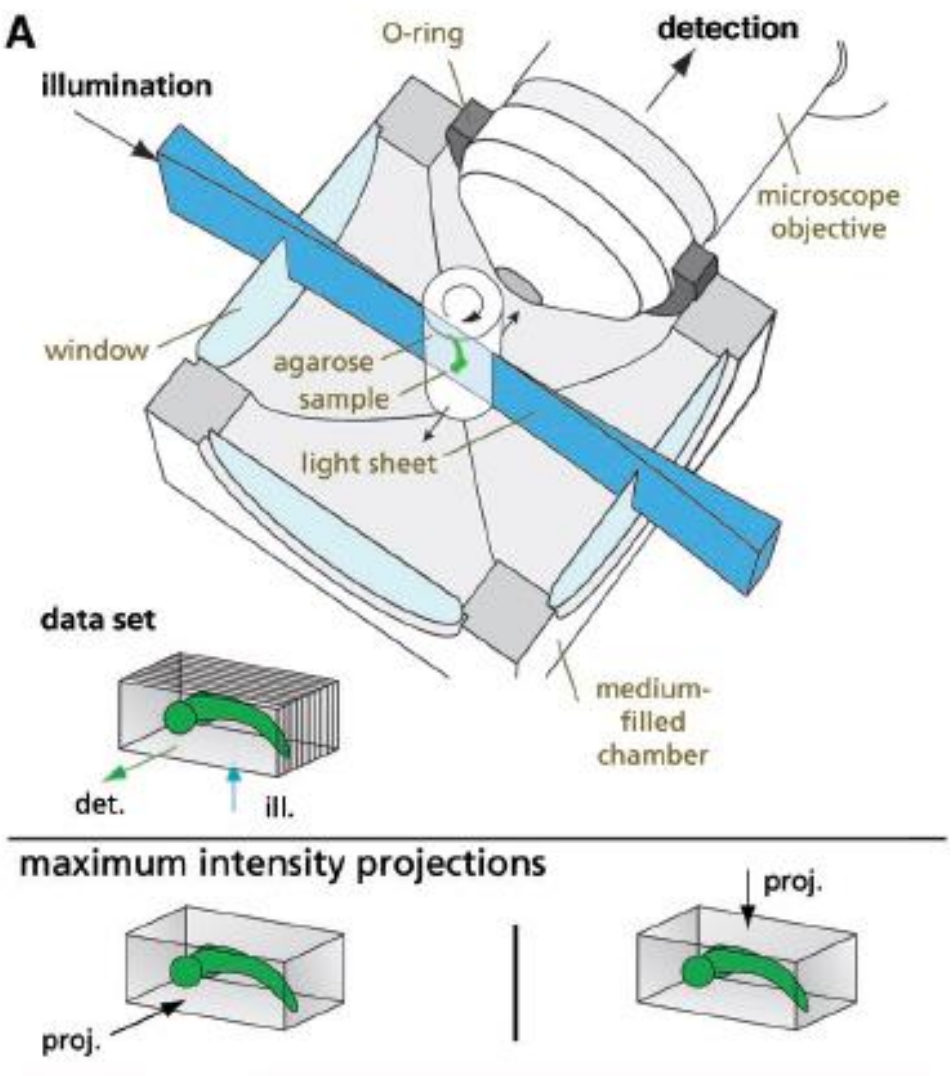


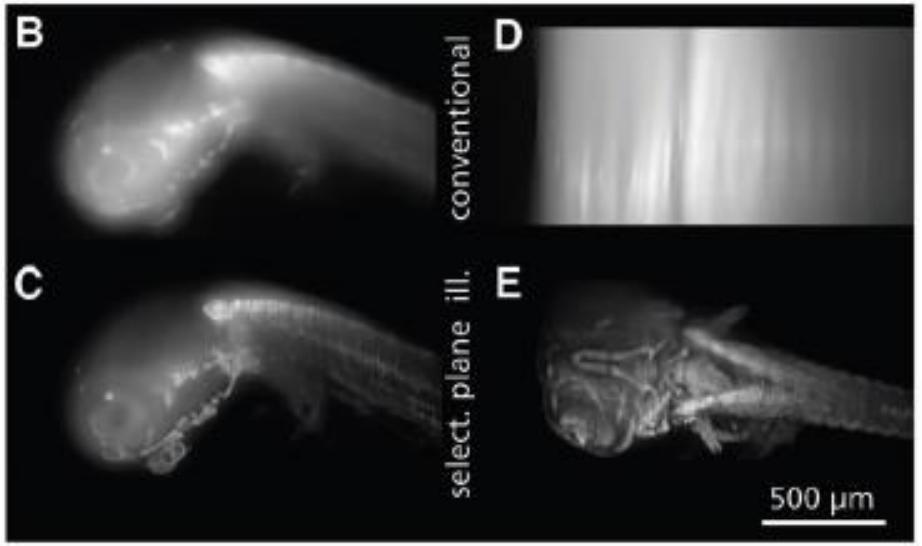


Light Sheet Microscopy

SPIM

Light Sheet Techniques





Optical Sectioning Deep Inside Live Embryos by Selective Plane Illumination Microscopy

Jan Huisken,* Jim Swoger, Filippo Del Bene, Joachim Wittbrodt, Ernst H. K. Stelzer*

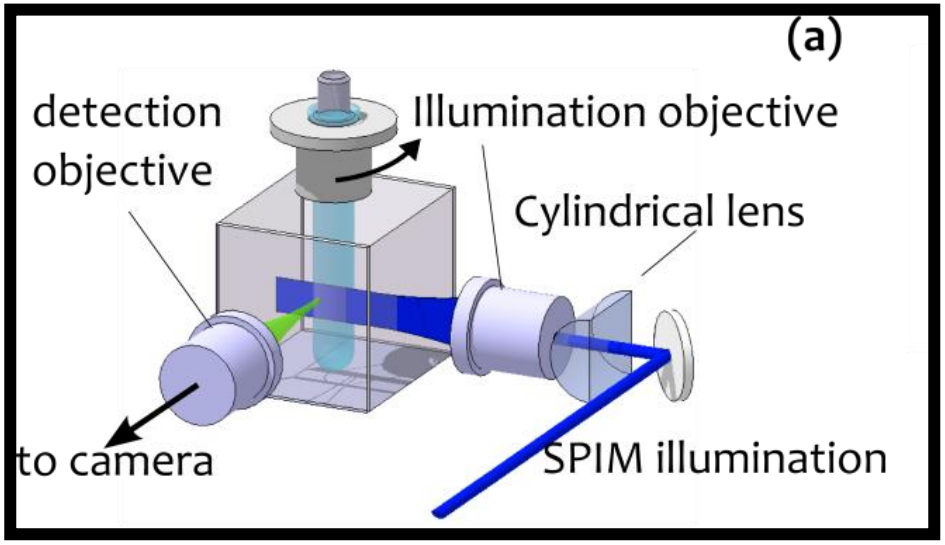
SCIENCE VOL 305 13 AUGUST 2004

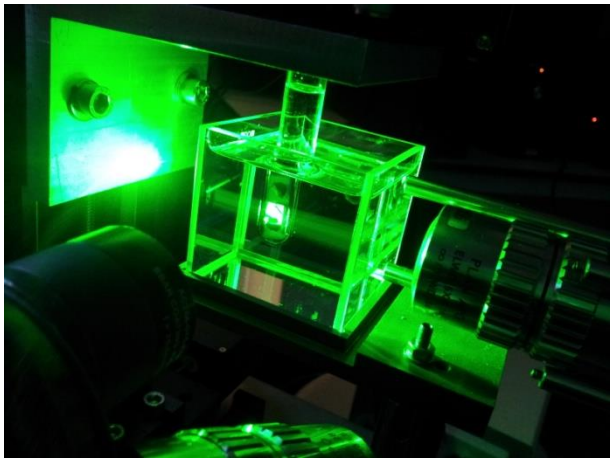
SPIM

Science 2004

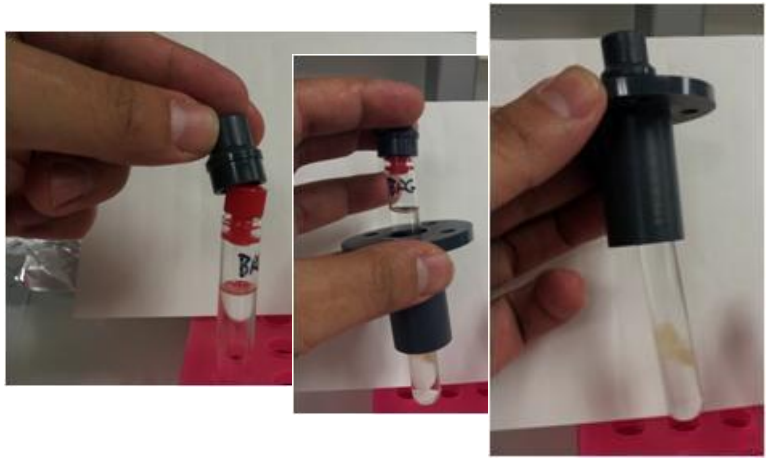
Light Sheet Microscopy

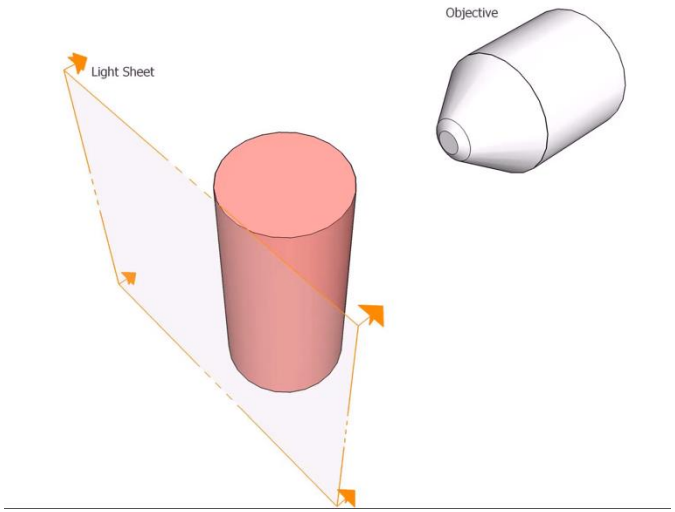
1) We generate a Sheet of Light using a cylidrical lens





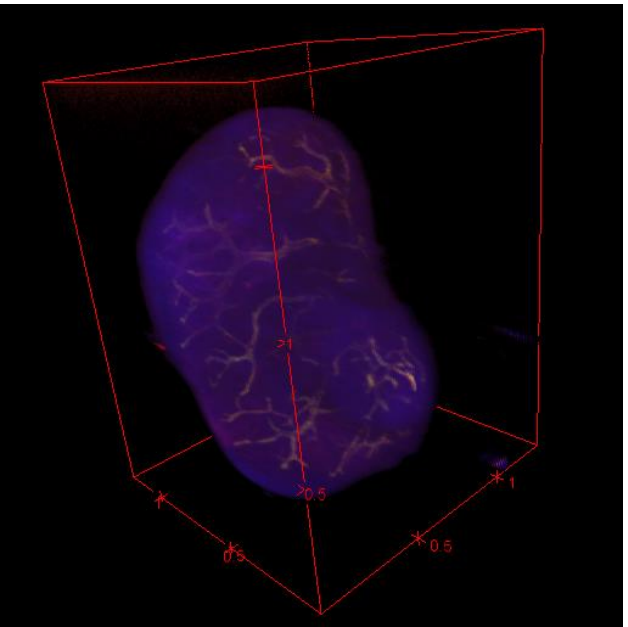
2) We scan by moving the object across the sheet

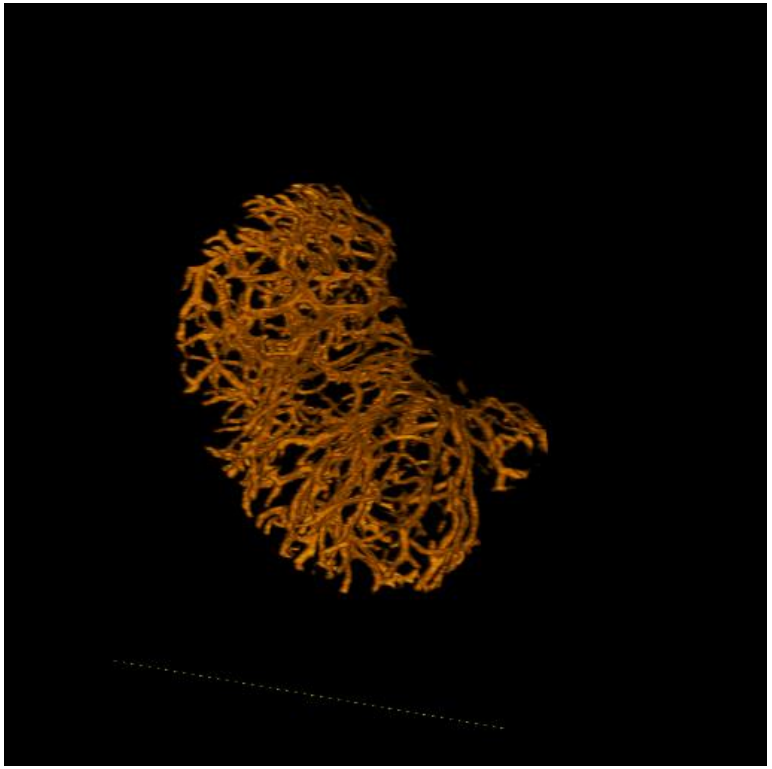




SPIM

- SPIM of Lymph Node (courtesy of Ola Ozga & Jens Stein)

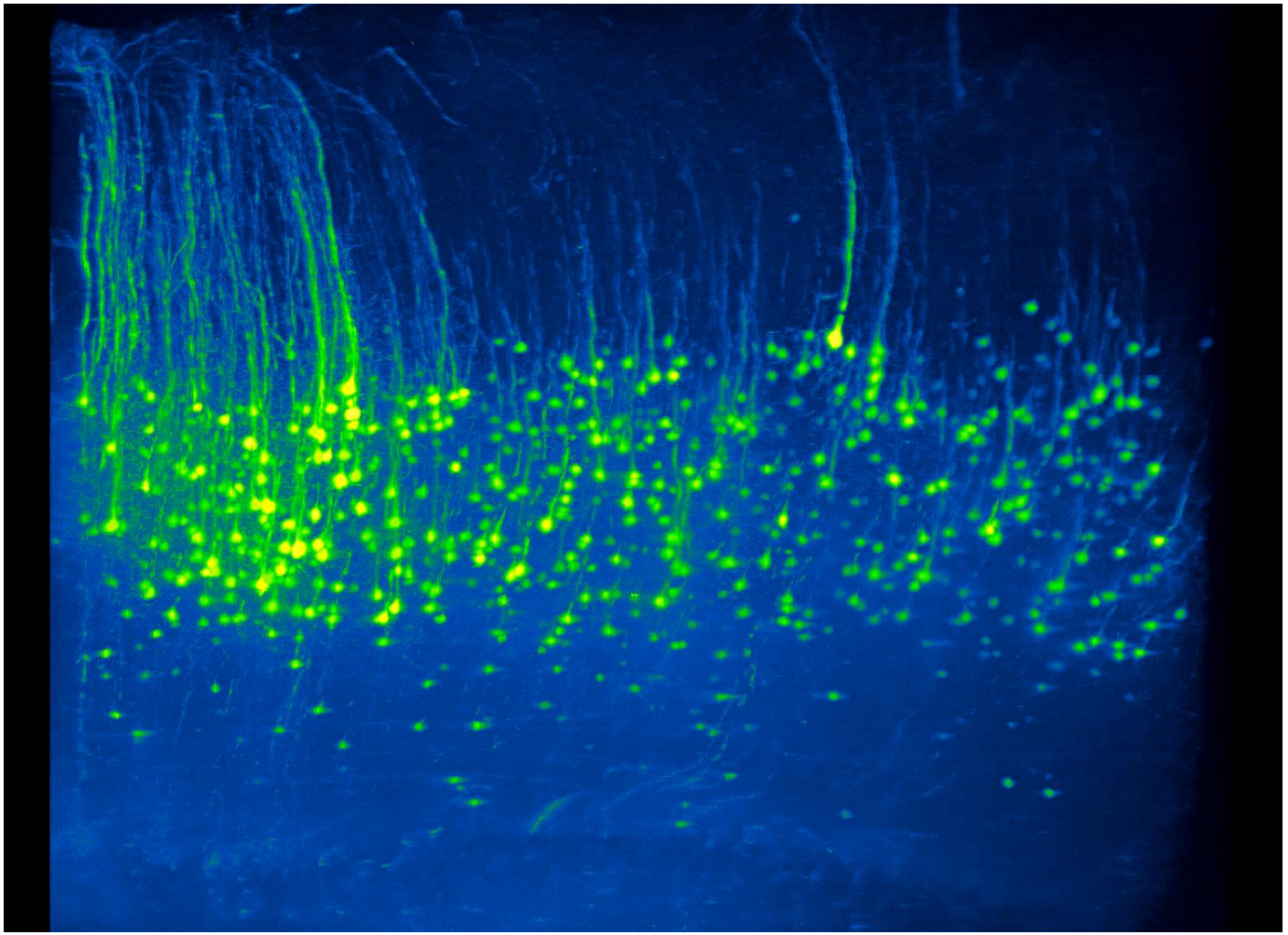




Blue = Cell type 1
Green = Cell Type 2
Orange= HEV*
Red = Cell Type 3

**High Endothelial Venules*

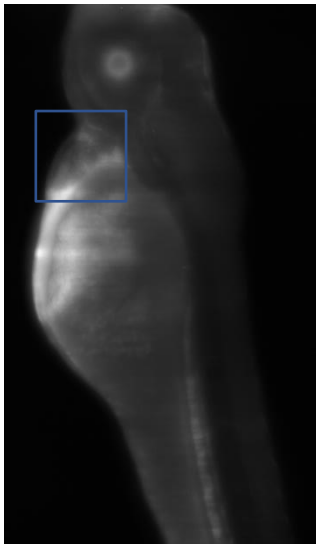
Examples: Brain with Scale A2



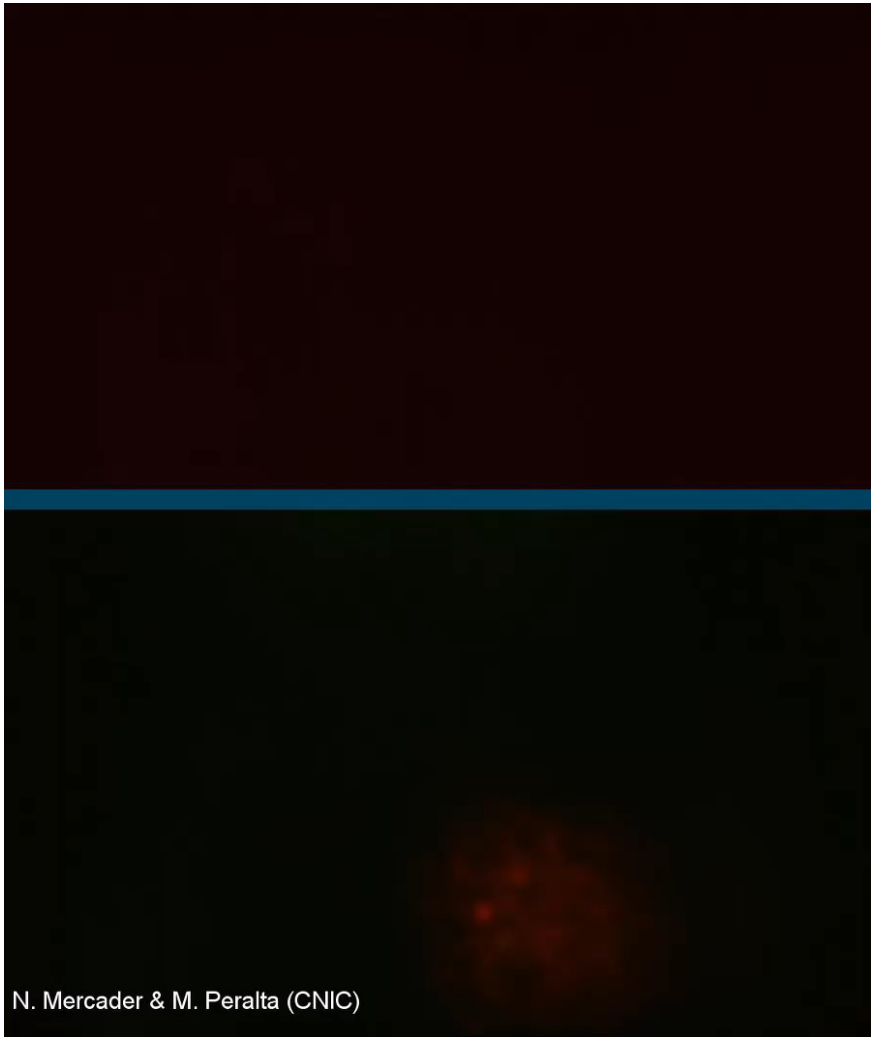
GFP-expressing Neurons cleared with Scale A2. F. Debarbieoux and C. Ricard (Univ-AMU)

Other Advances

- Multicolor In-vivo Imaging



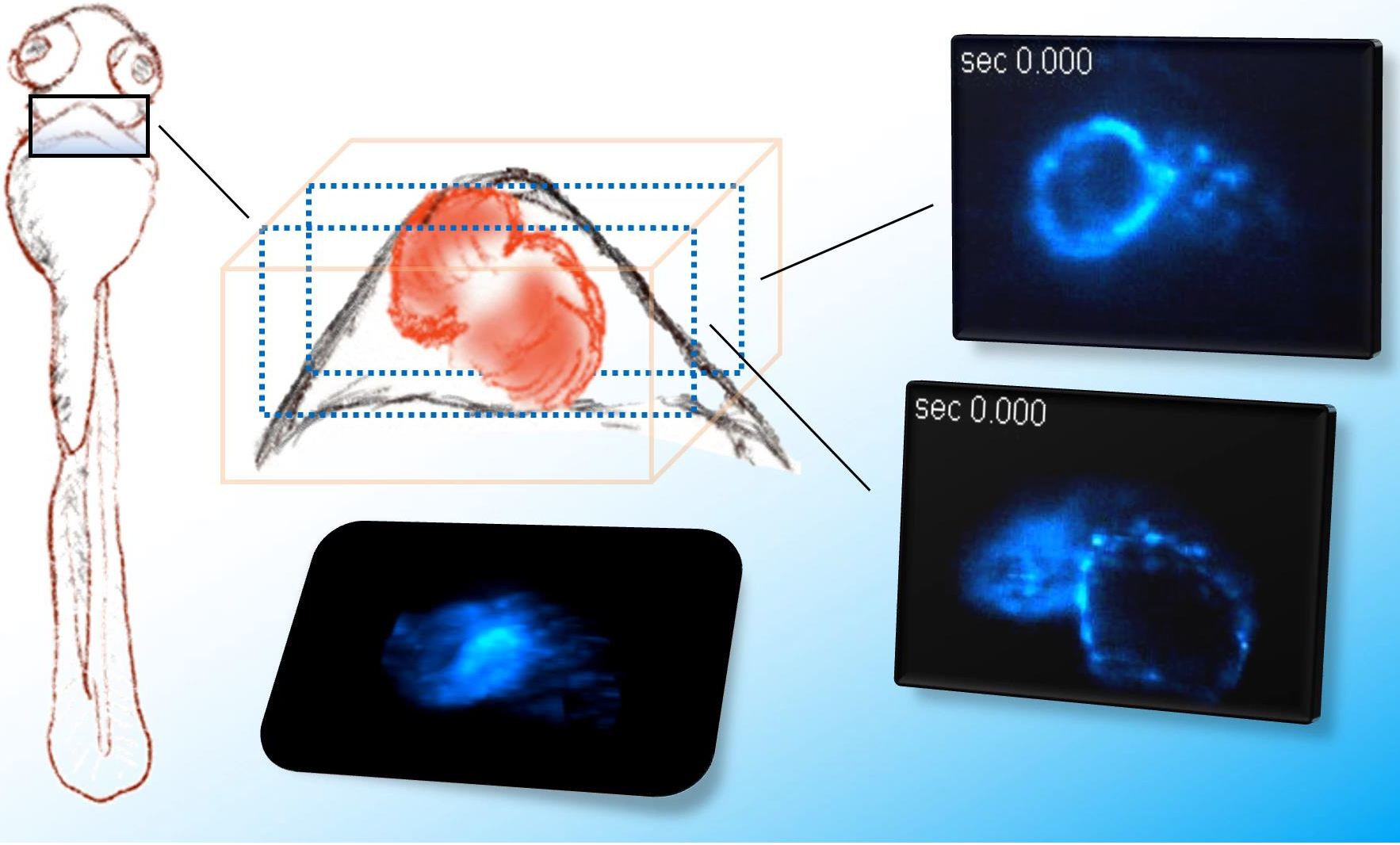
20x Objective

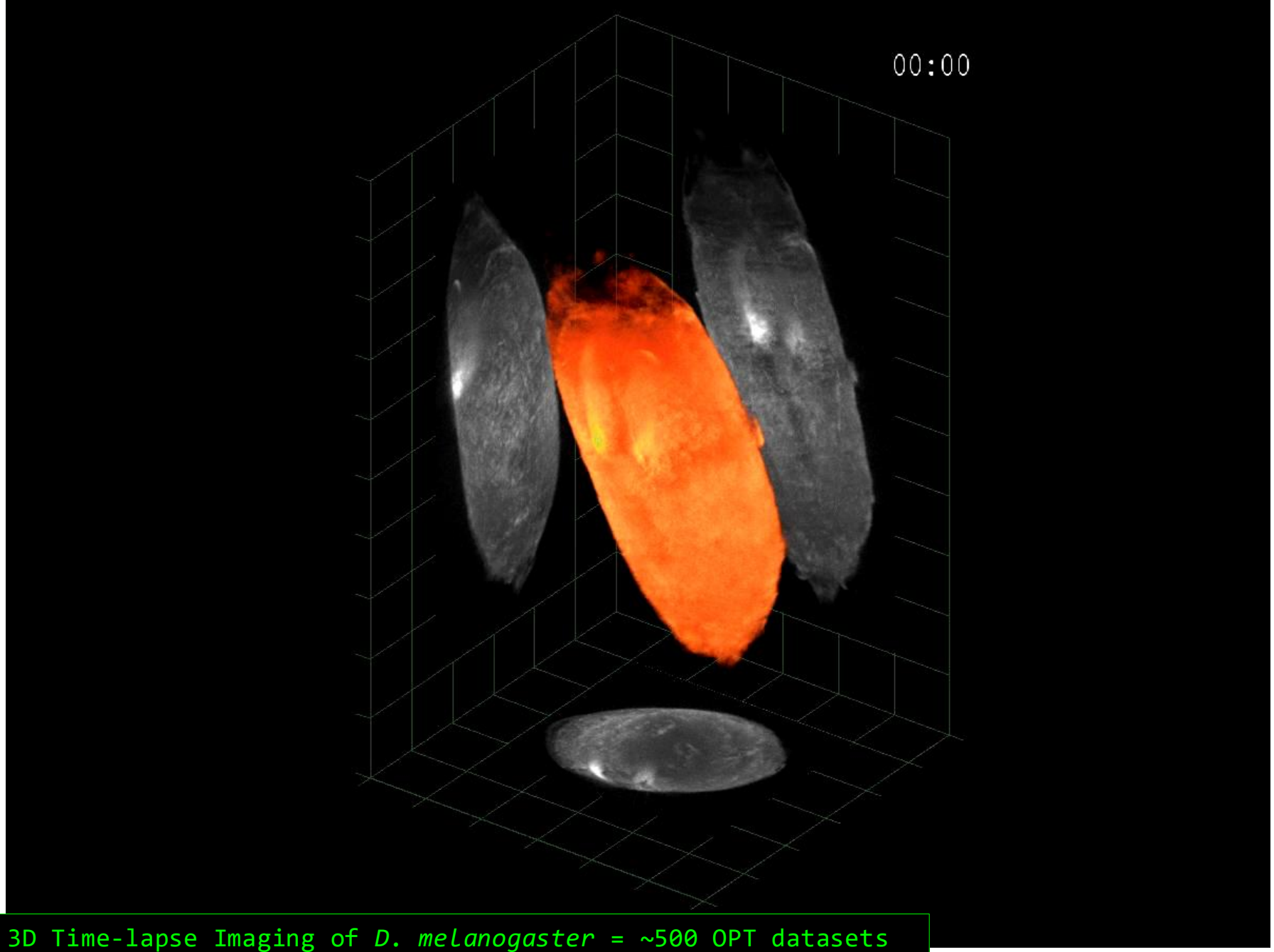


Different Z-positions of the sample

Nadia Mercader, Marina Peralta, CNIC

Dynamic Focusing SPIM

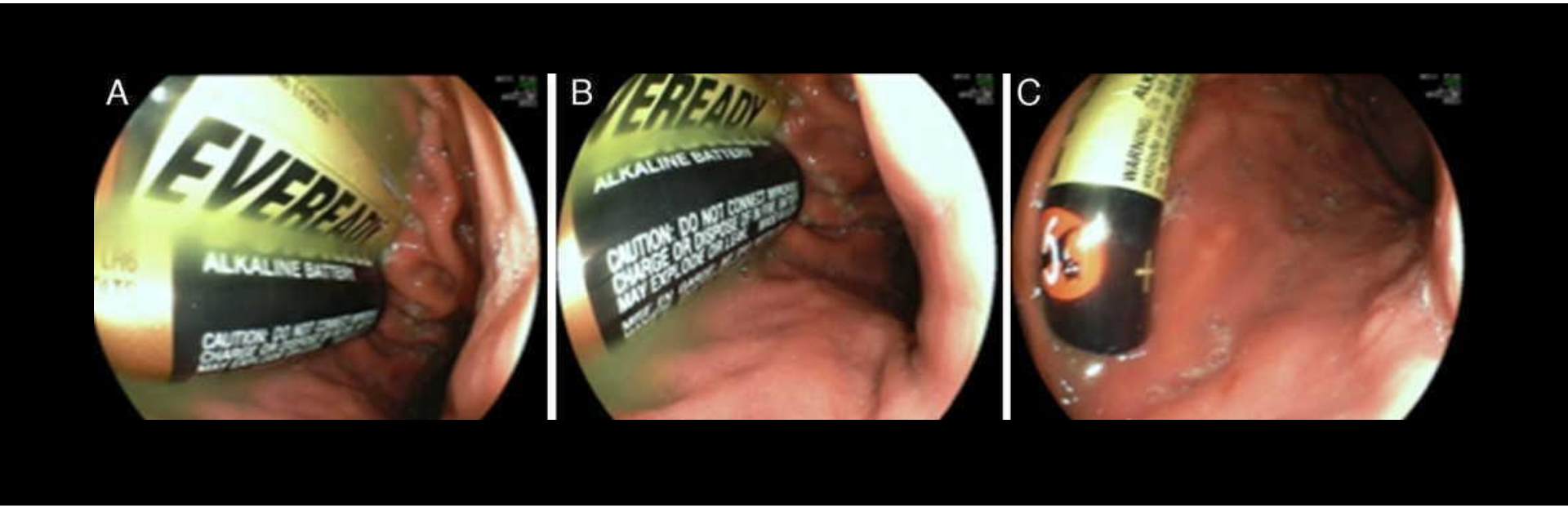




3D Time-lapse Imaging of *D. melanogaster* = ~500 OPT datasets
CUDA Implementation: Dong Di and Shouping Zhu.
Specimen: S. Oehler and B. Savakis (Fleming)

Endoscopy

- Low scattering (surface image)



Endoscopy



a

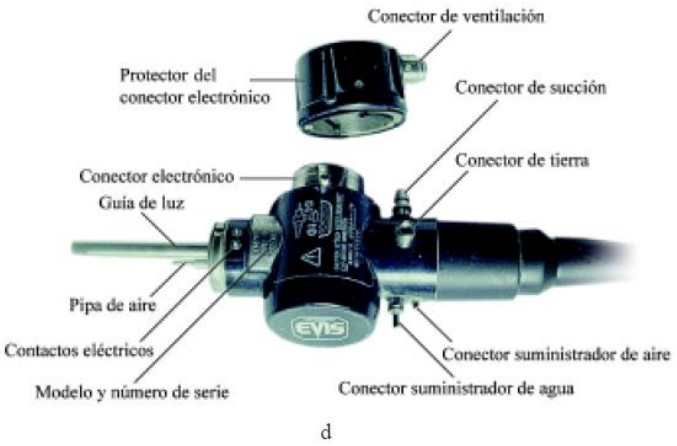


c



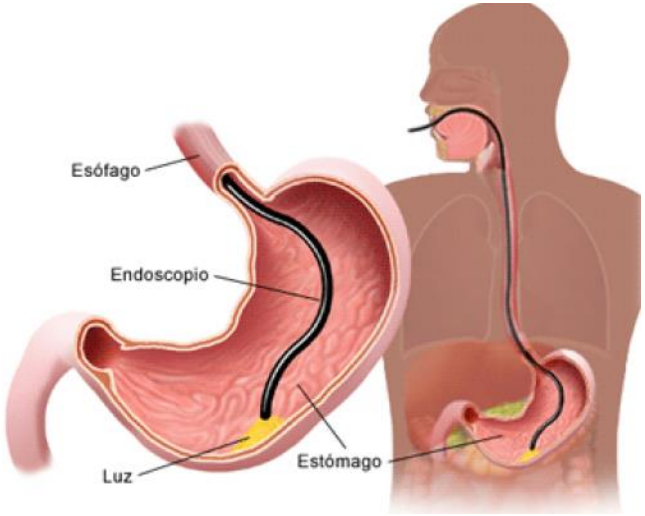


b



d

Endoscopy







Endoscopy

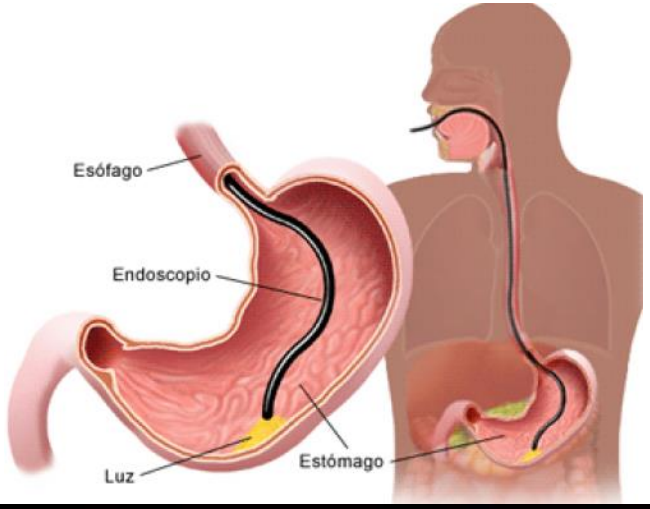








Image-guided surgery

- Surface imaging in tissue



Image-guided surgery

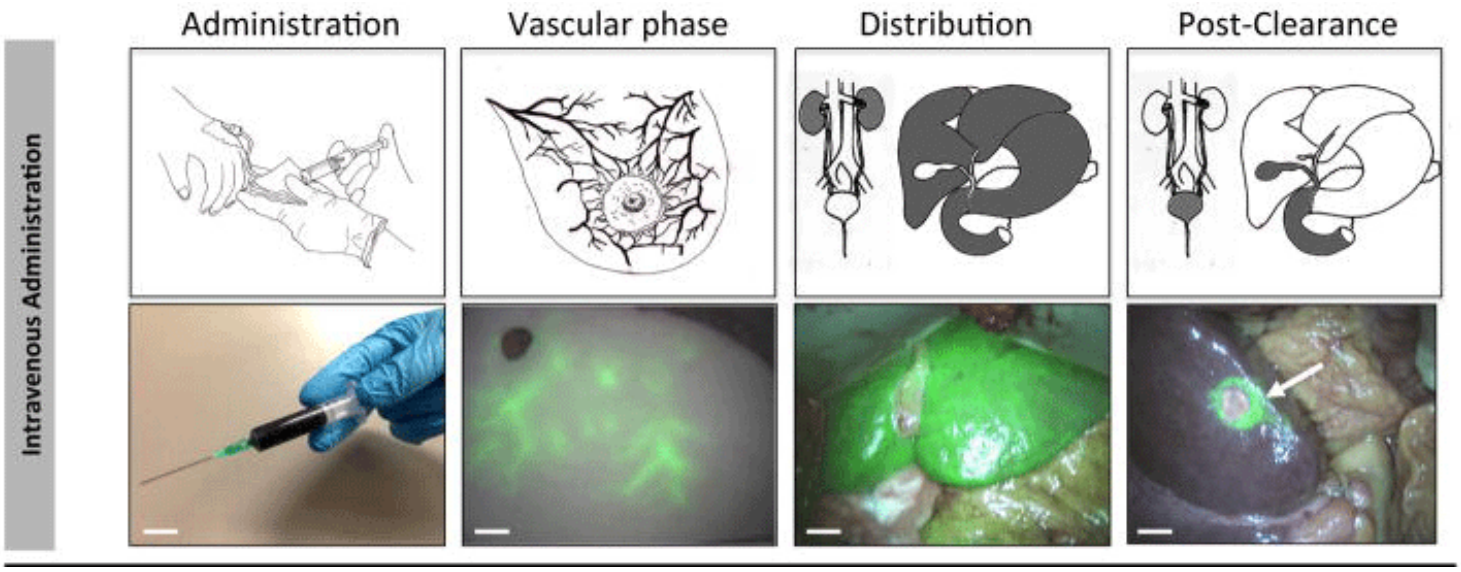


Image-guided surgery

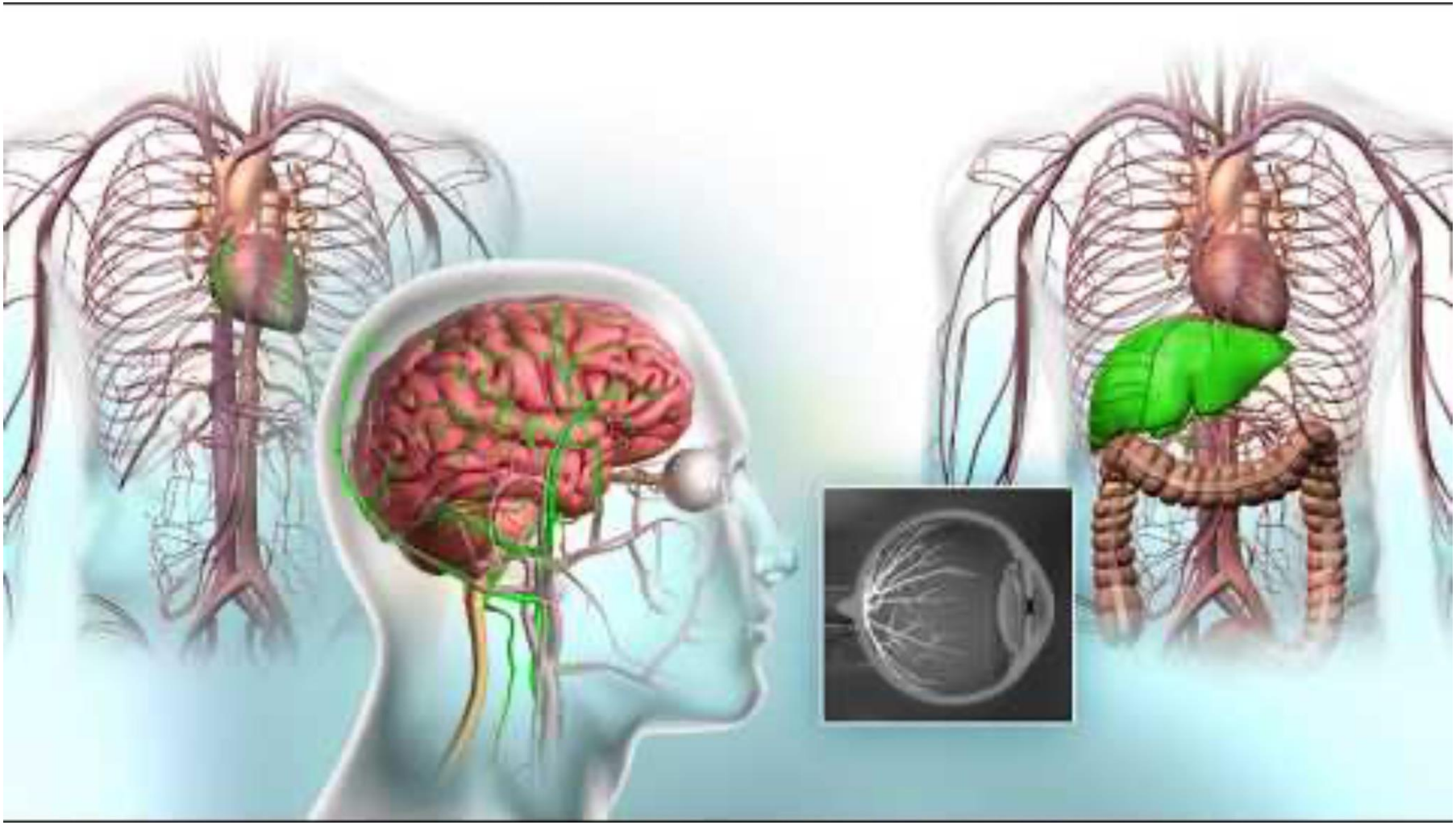
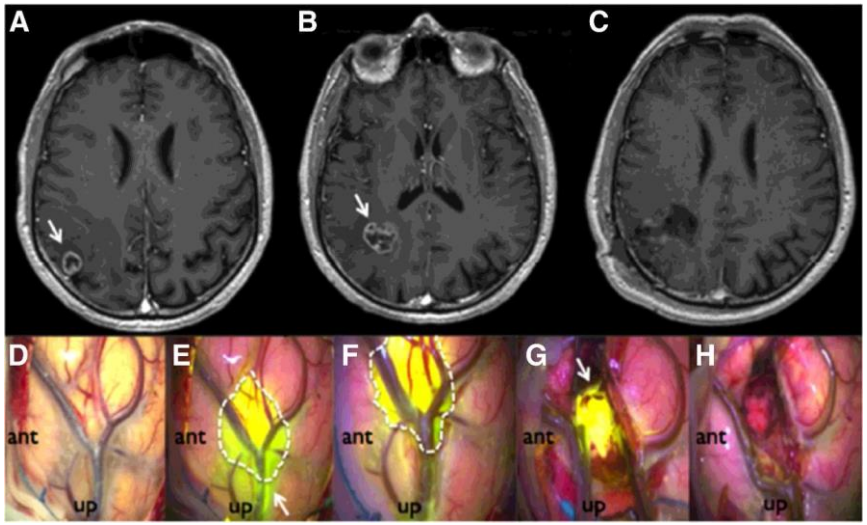
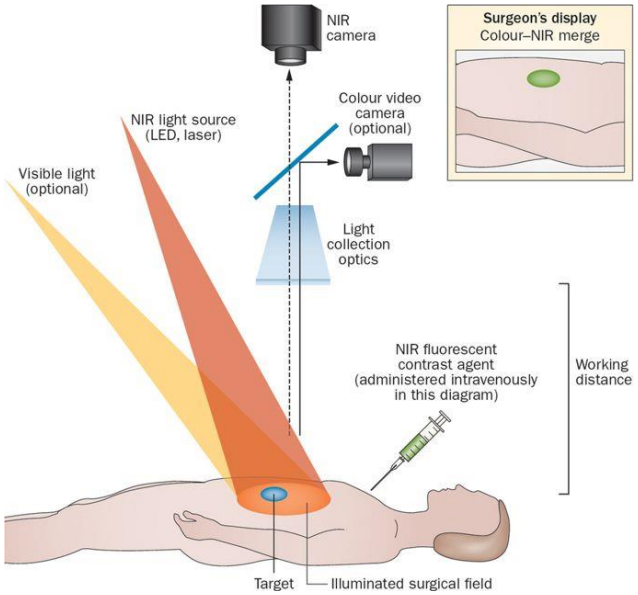
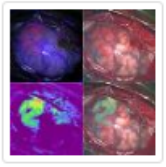


Image-guided Sugery





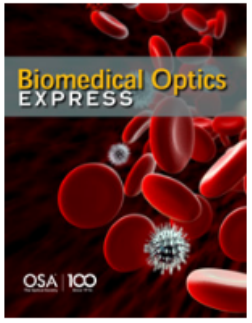
https://www.osapublishing.org/boe/abstract.cfm?uri=boe-6-10-3765



Review of fluorescence guided surgery visualization and overlay techniques

Jonathan T. Elliott, Alisha V. Dsouza, Scott C. Davis, Jonathan D. Olson, Keith D. Paulsen, David W. Roberts, and Brian W. Pogue

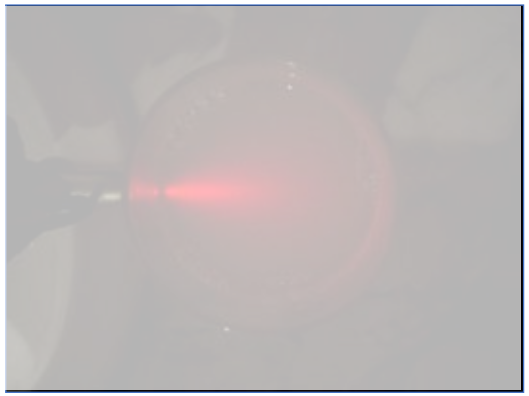
Author Information ▾ Find other works by these authors ▾

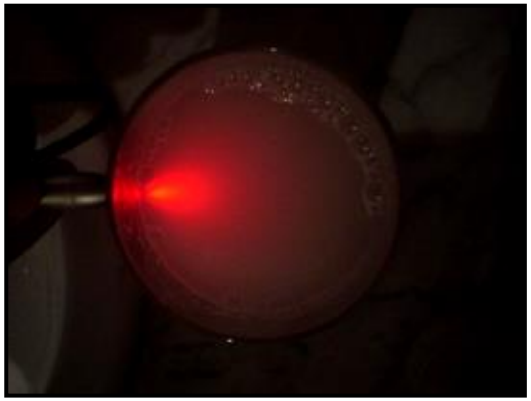


HIGH scattering

Optical Tomography in Preclinical

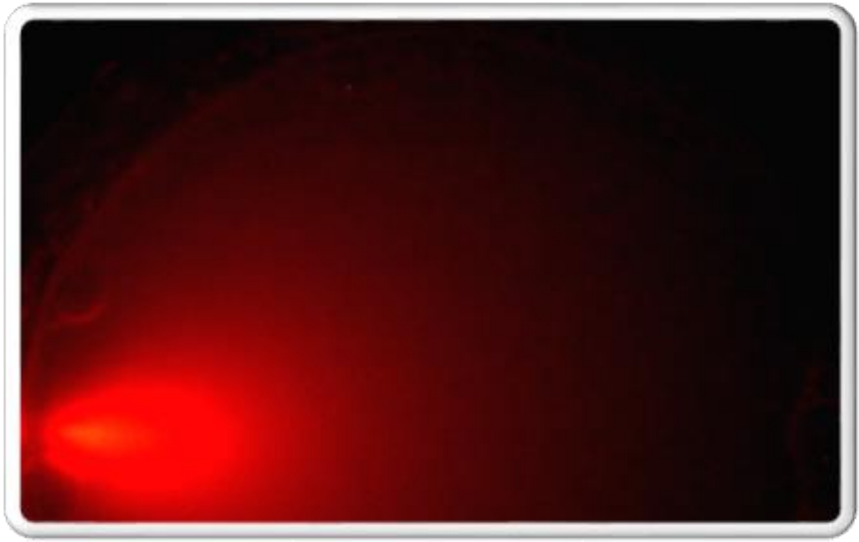






Preclinical Imaging
FLUORESCENCE MOLECULAR TOMOGRAPHY (FMT)

(Highly Scattering Tissues)



Fluorescence Molecular Tomography (FMT)

- Developed by *V. Ntziachristos* in the context of Molecular Imaging as FMT in 2002.
- Models light propagation through the diffusion equation

NEW TECHNOLOGY

Fluorescence molecular tomography resolves protease activity *in vivo*

VASILIS NTZIACHRISTOS, CHING-HSUAN TUNG, CHRISTOPH BREMER & RALPH WEISSLEDER
Center for Molecular Imaging Research, Massachusetts General Hospital and Harvard Medical School, Charlestown, Massachusetts, USA
Correspondence should be addressed to V.N.; email: vasilis@helix.mgh.harvard.edu

Published online 24 June 2002, doi: 10.1038/nm729

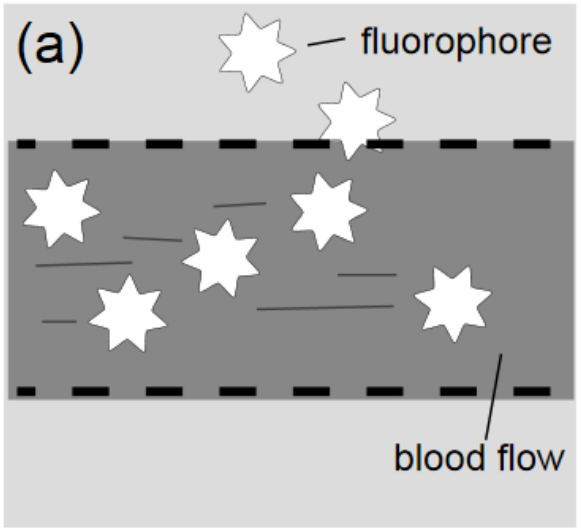
24 years ago, almost to the day!

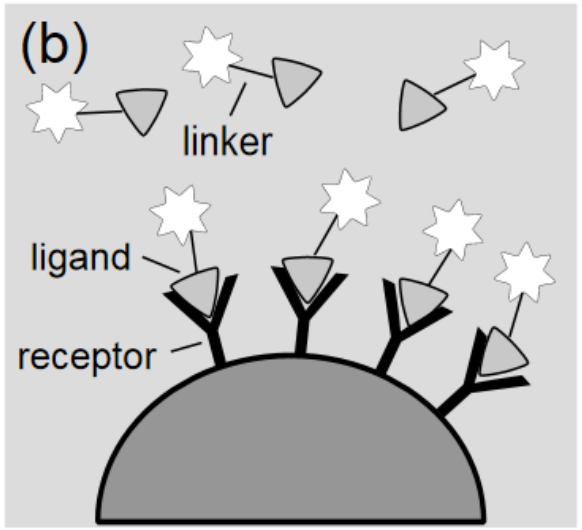
Probes

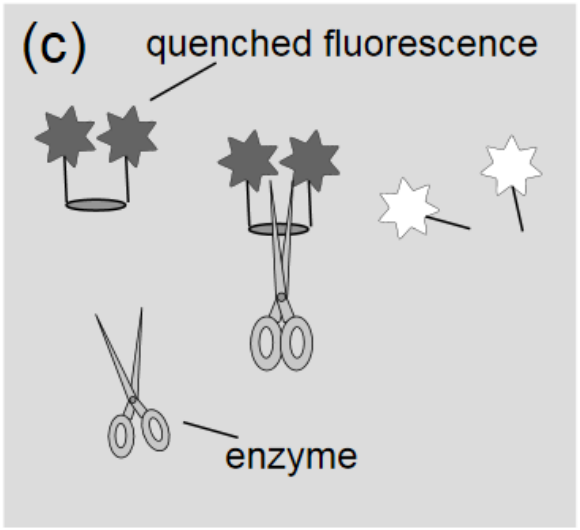
Non specific probes

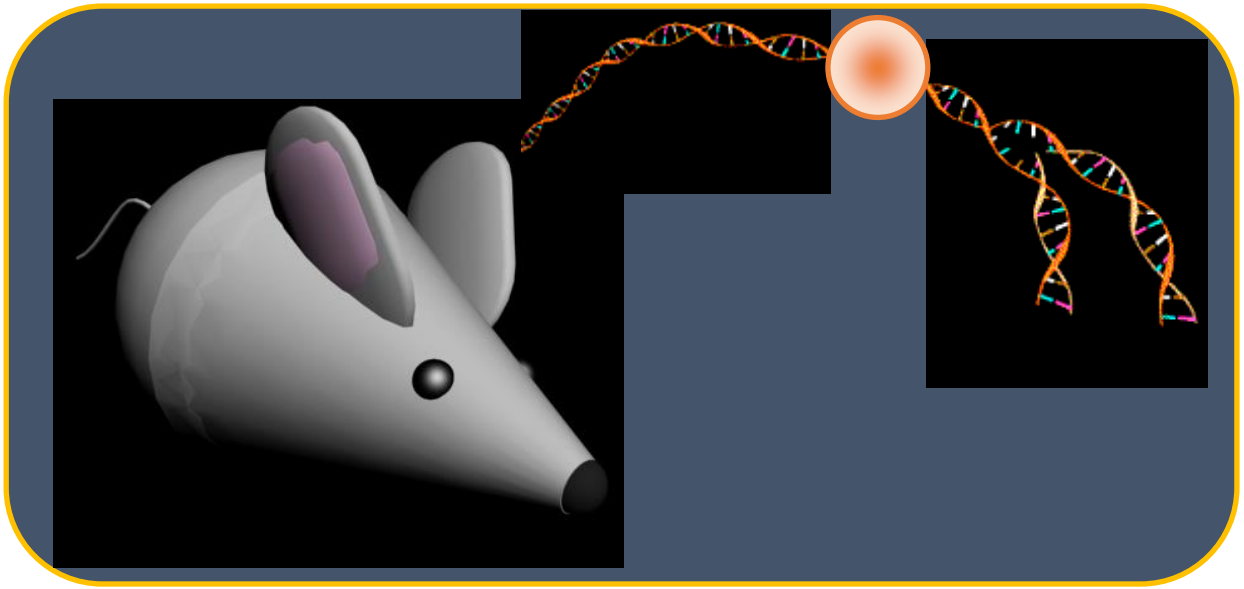
Targeted probes

Activatable probes









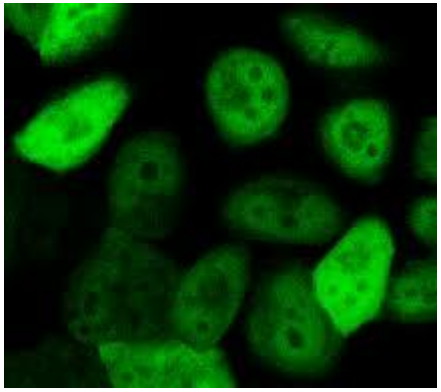
Transgenically Express a Fluo. protein

Image Contrast

- Image Contrast will depend on:
  - Tissue Absorption
  - Autofluorescence
  - Probe Specificity
  - Signal to Noise

Fluorescence

Fluorescence, is capable of explaining*:*



Green Fluorescent Protein (GFP) -expressing cells



GFP-expressing mice

And here is the culprit:

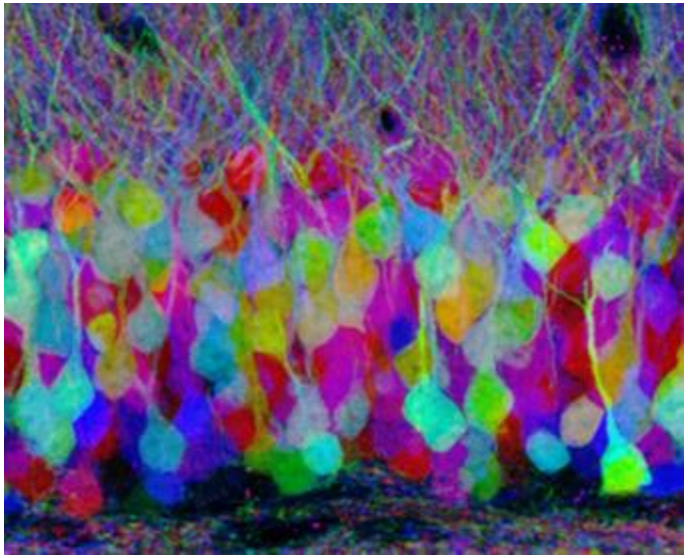




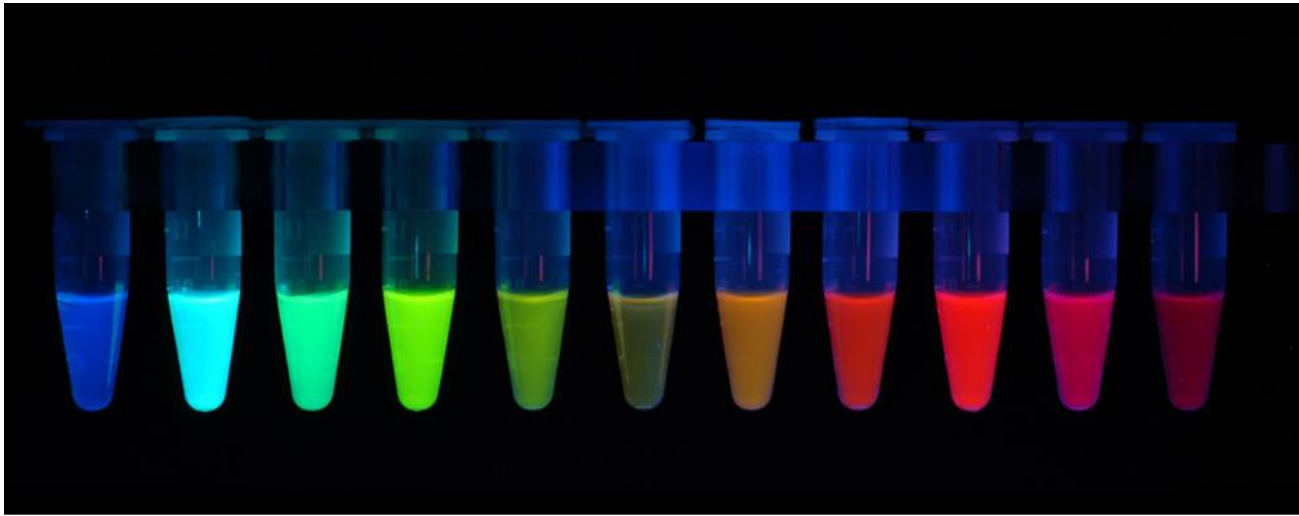
The Nobel Prize in Chemistry 2008
Osamu Shimomura, Martin Chalfie, Roger Y. Tsien

Fluorescence

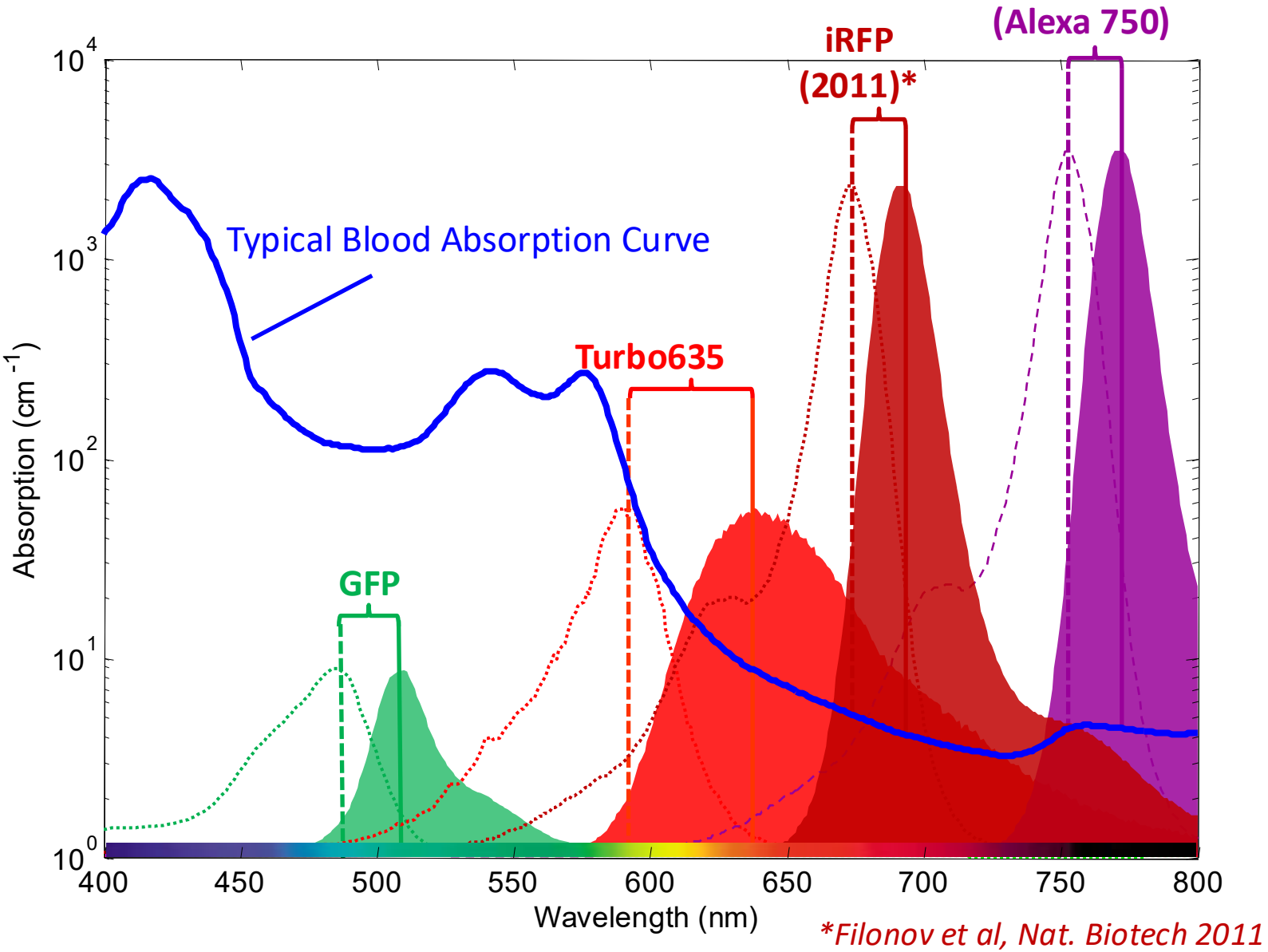
Thanks to the cloning of the GFP we have now a great number of fluorescent proteins to choose from:



Development of nerve cells. Each nerve cell expresses a combination of fluorescent proteins. Each one, needs to be visualized with its own filter.

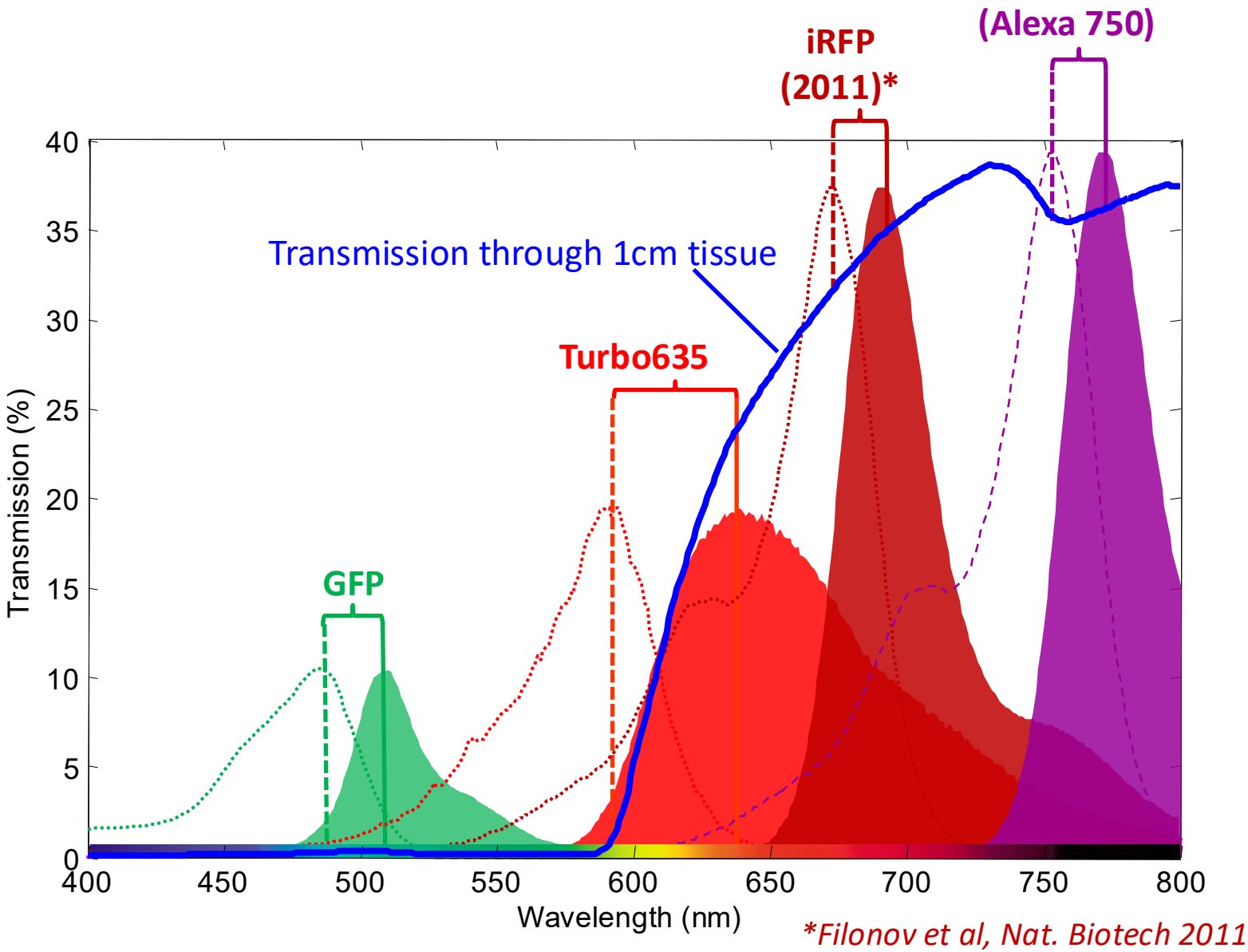


Absorption



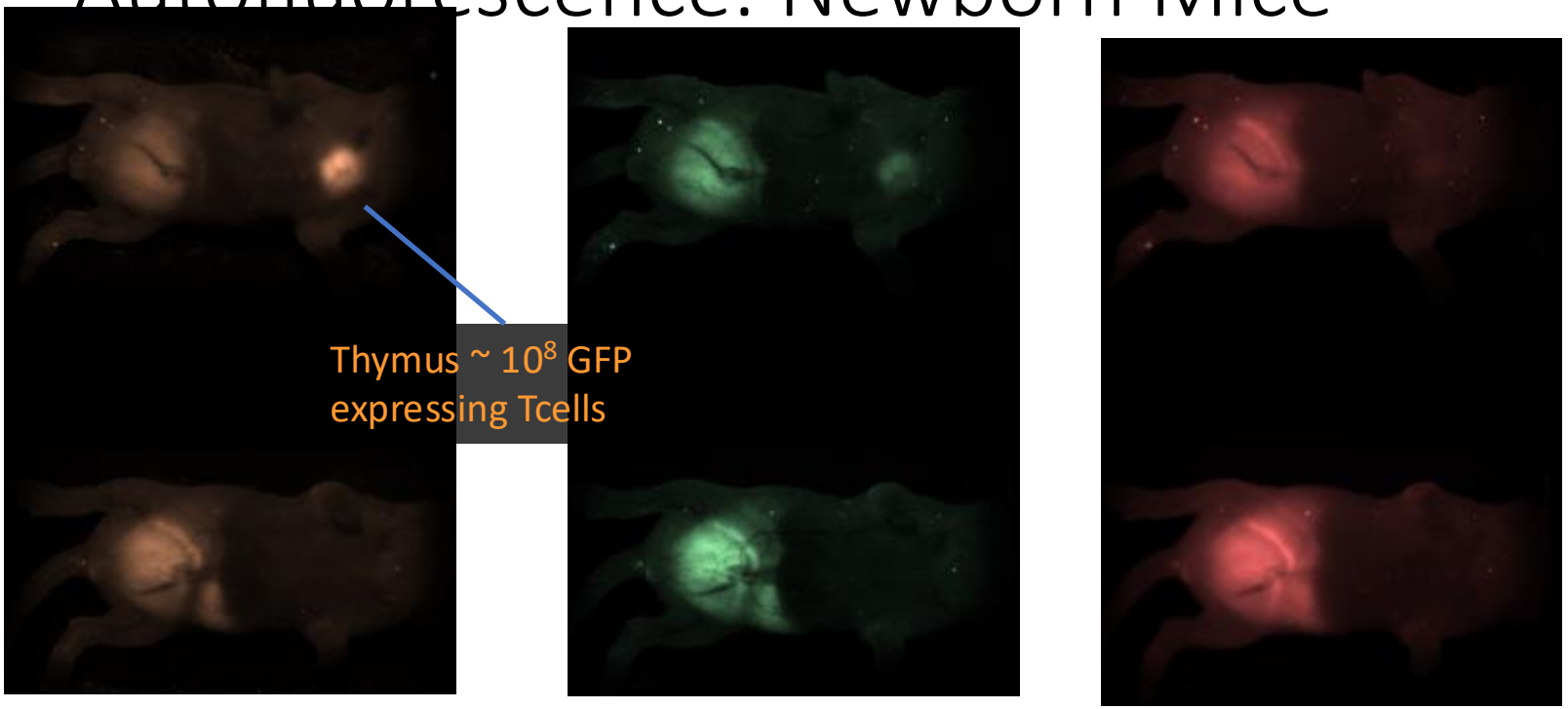
*Filonov et al, Nat. Biotech 2011

Absorption



*Filonov et al, Nat. Biotech 2011

Autofluorescence: Newborn Mice



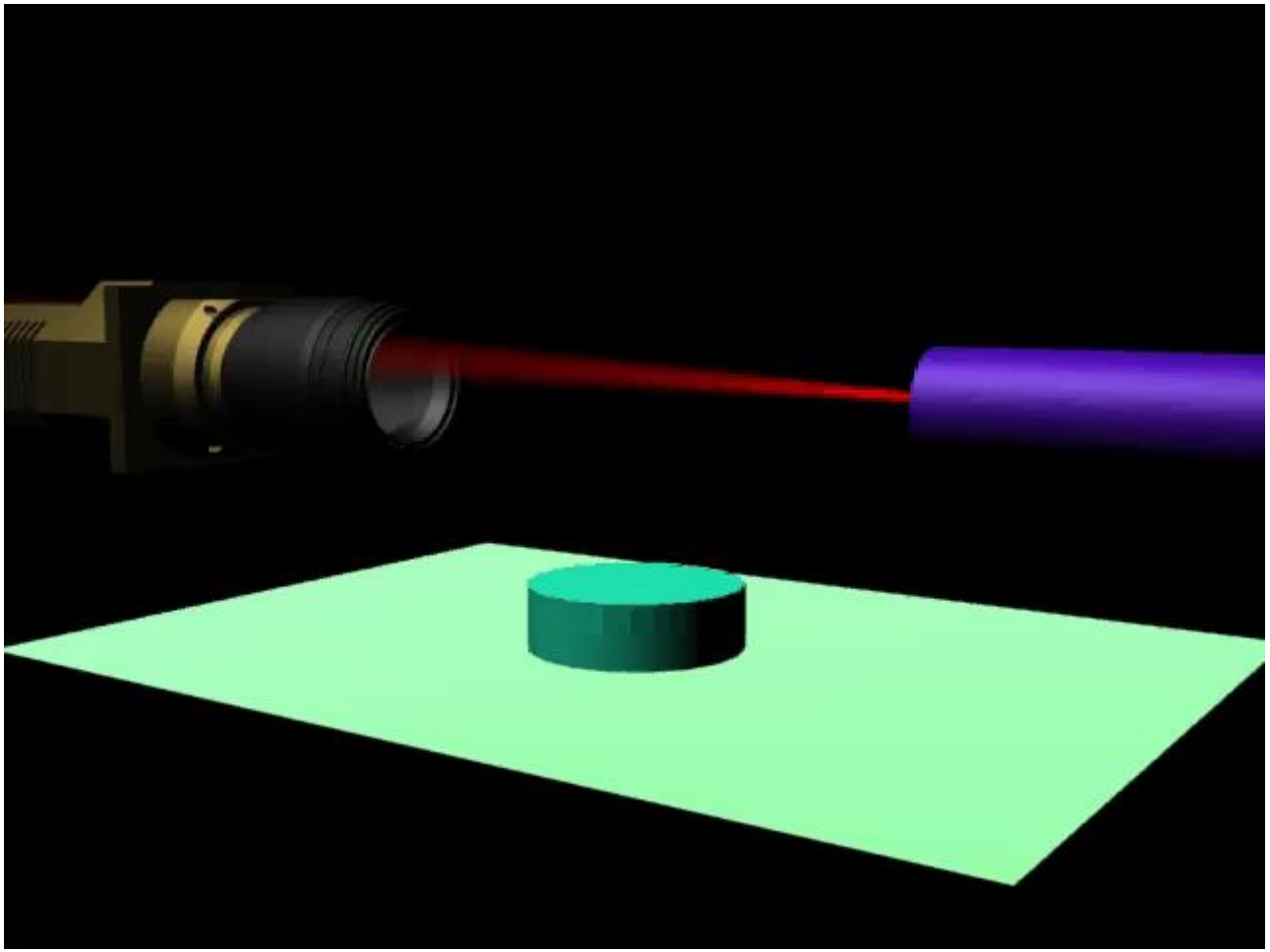
Exc: 488nm
Emm: 510nm

Exc: 488nm
Emm: 540nm

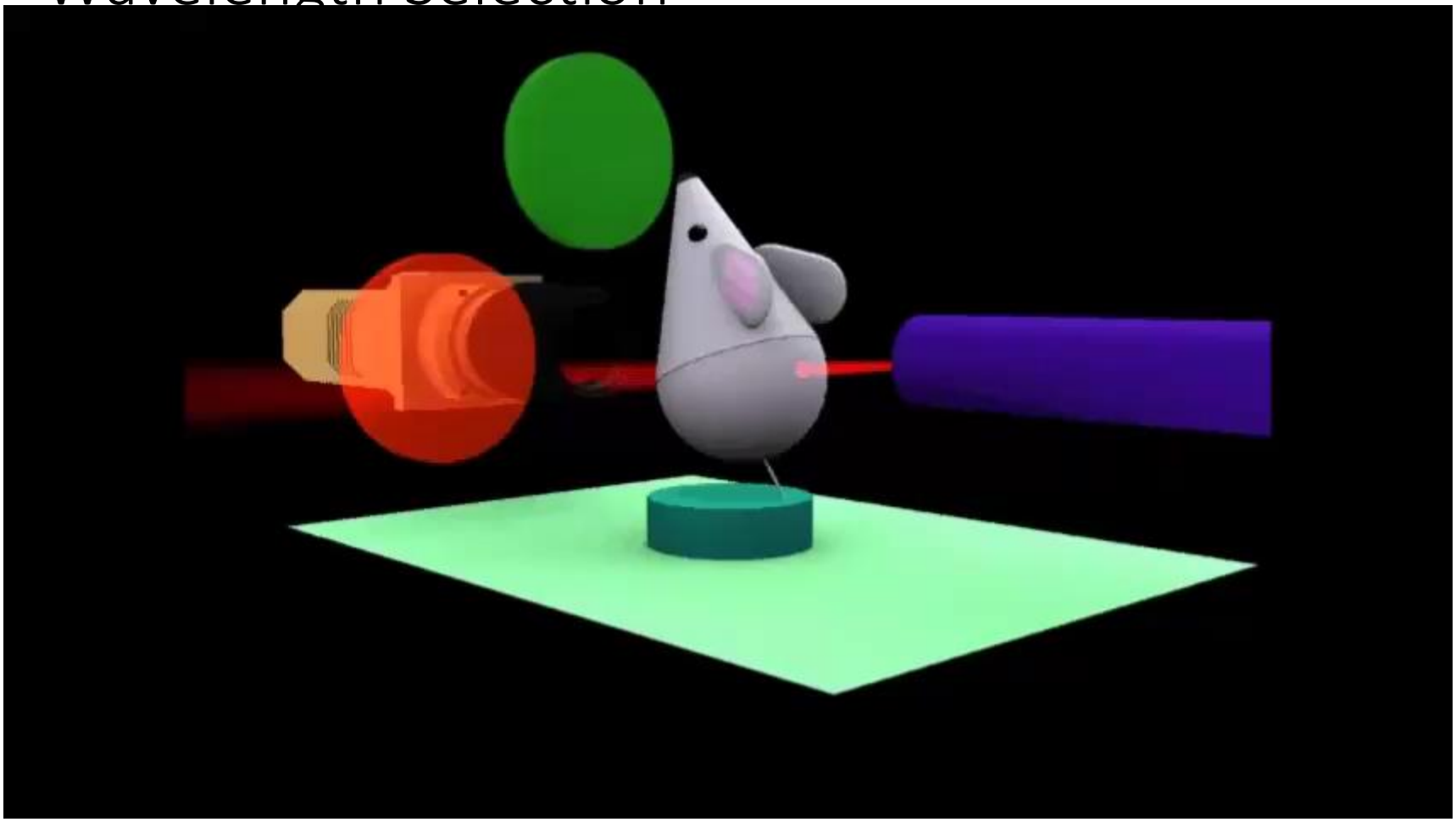
Exc: 488nm
Emm: 615nm

Emission Wavelength

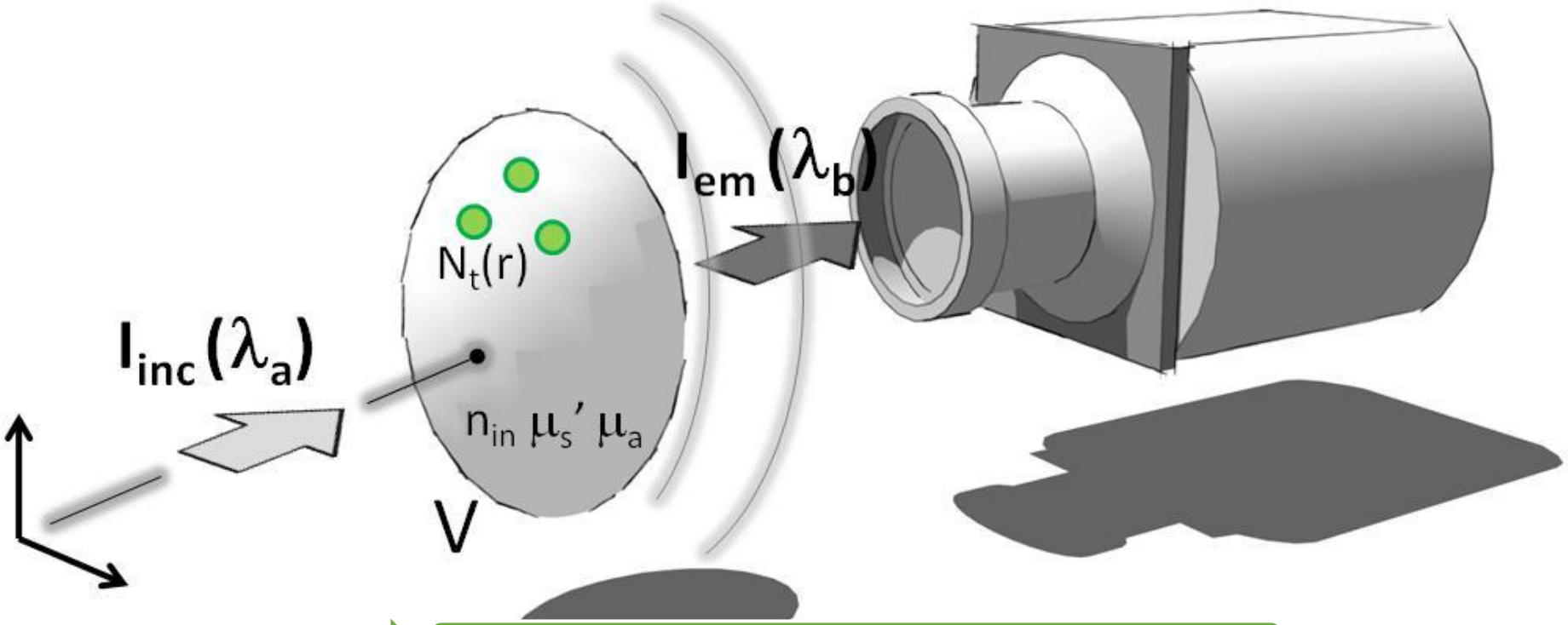
Non-Contact FMT



Wavelength Selection



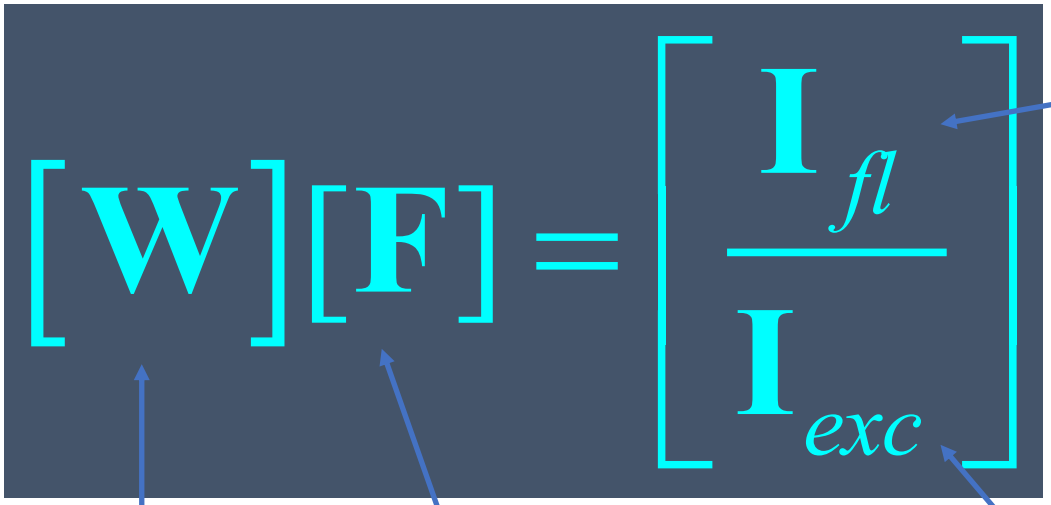
FMT Principles



FMT recovers Probe Concentration

Inverse Problem: Using the measurements as a starting point, what distribution of fluorophores or fluorescent proteins caused them?

Inverse Problem



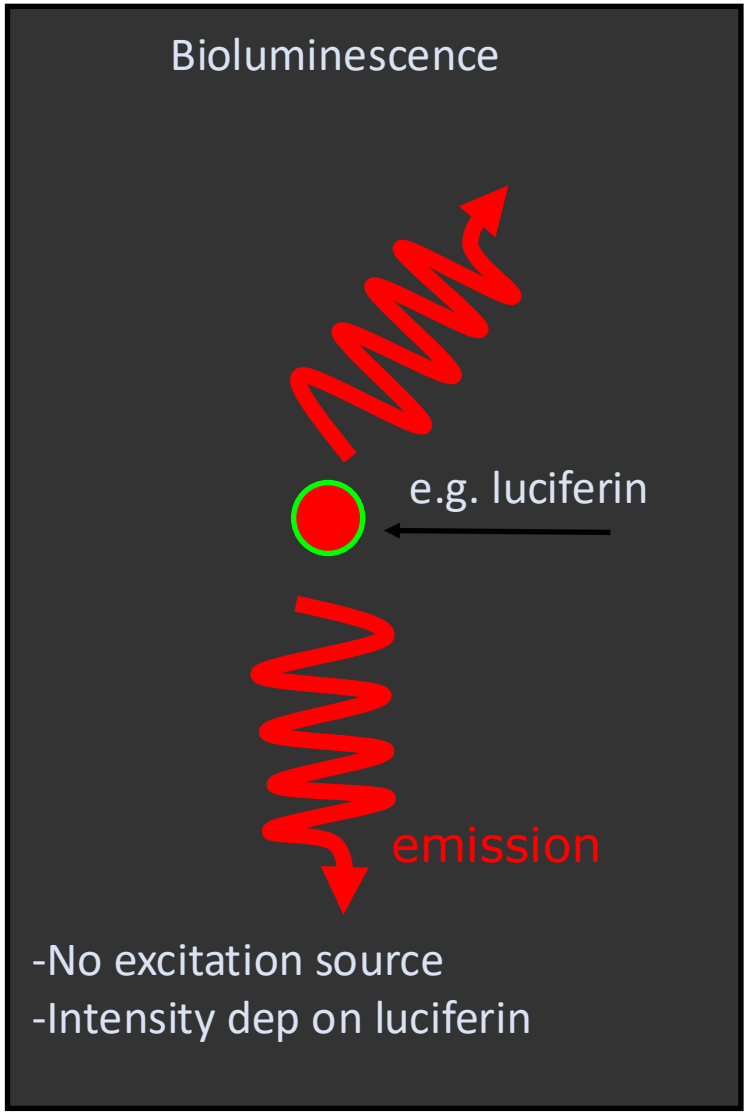
Inverse Problem

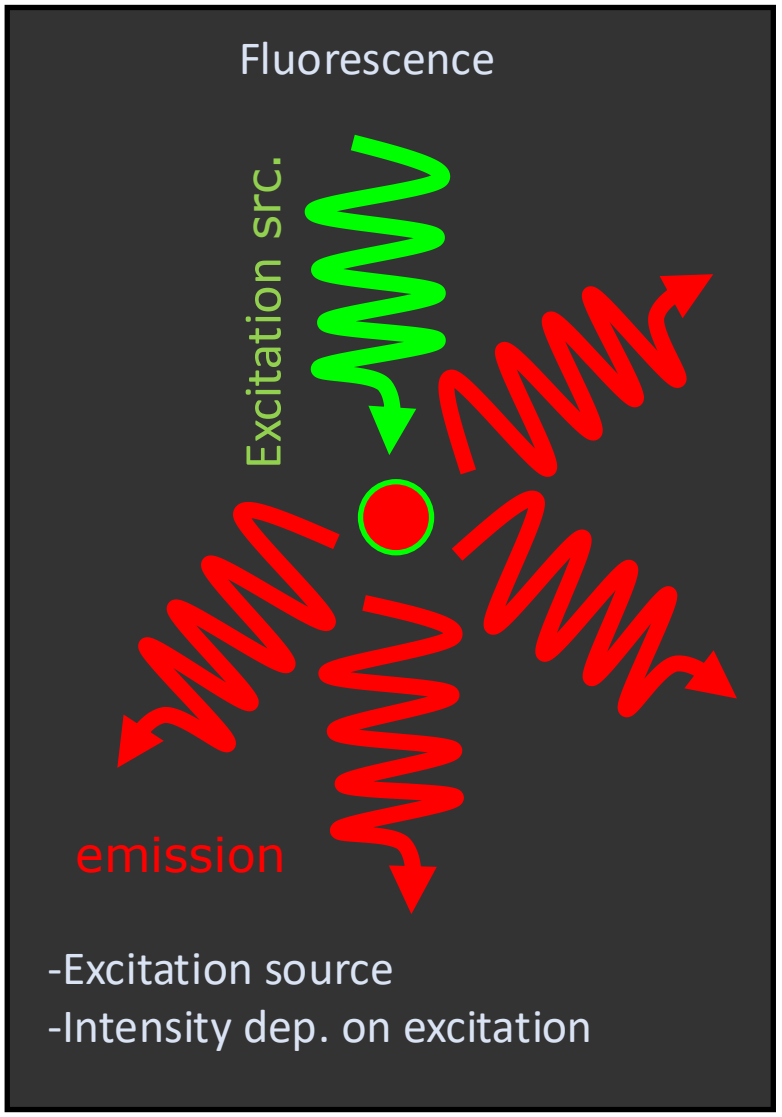
$$[\mathbf{F}] = [\mathbf{W}]^{-1} \left[\frac{\mathbf{I}_{fl}}{\mathbf{I}_{exc}} \right]$$

3D Fluorescence Map

Ill-Posed Problem → Iterative Methods, SVD

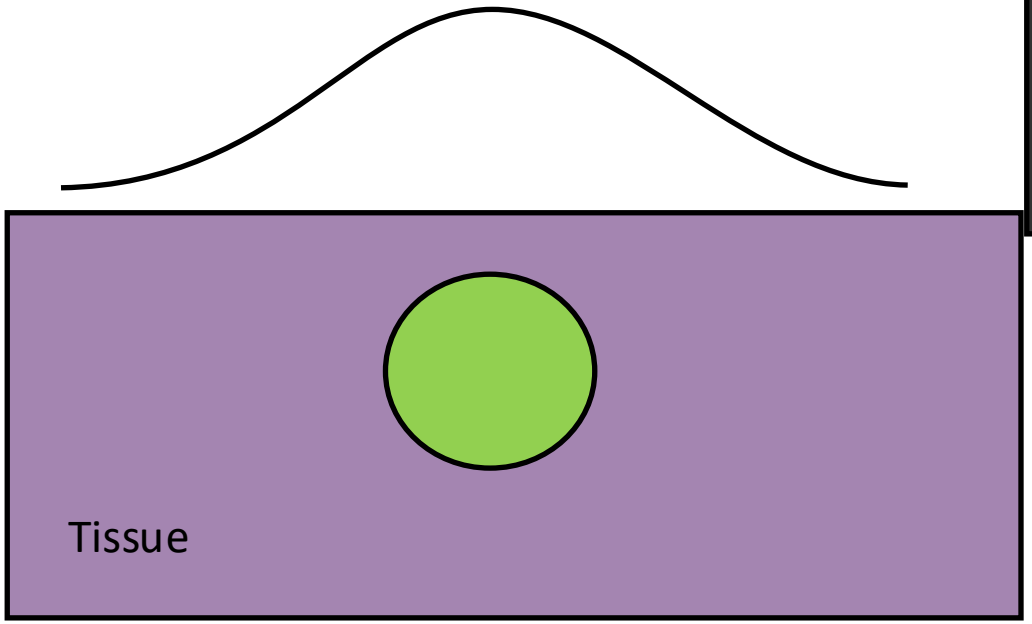
Bioluminescence vs Fluo.

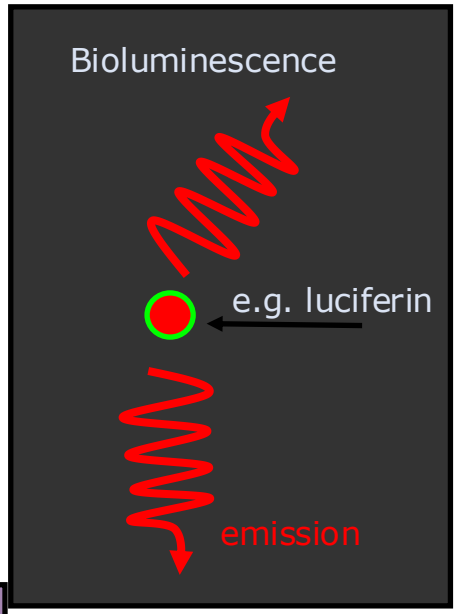




Bioluminescence vs Fluo.

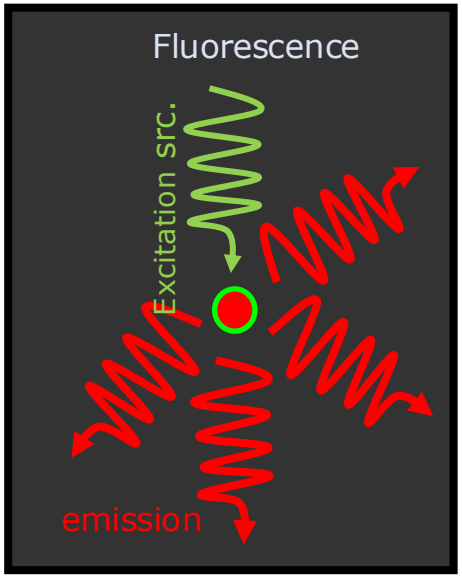
BIOLUMINESCENCE

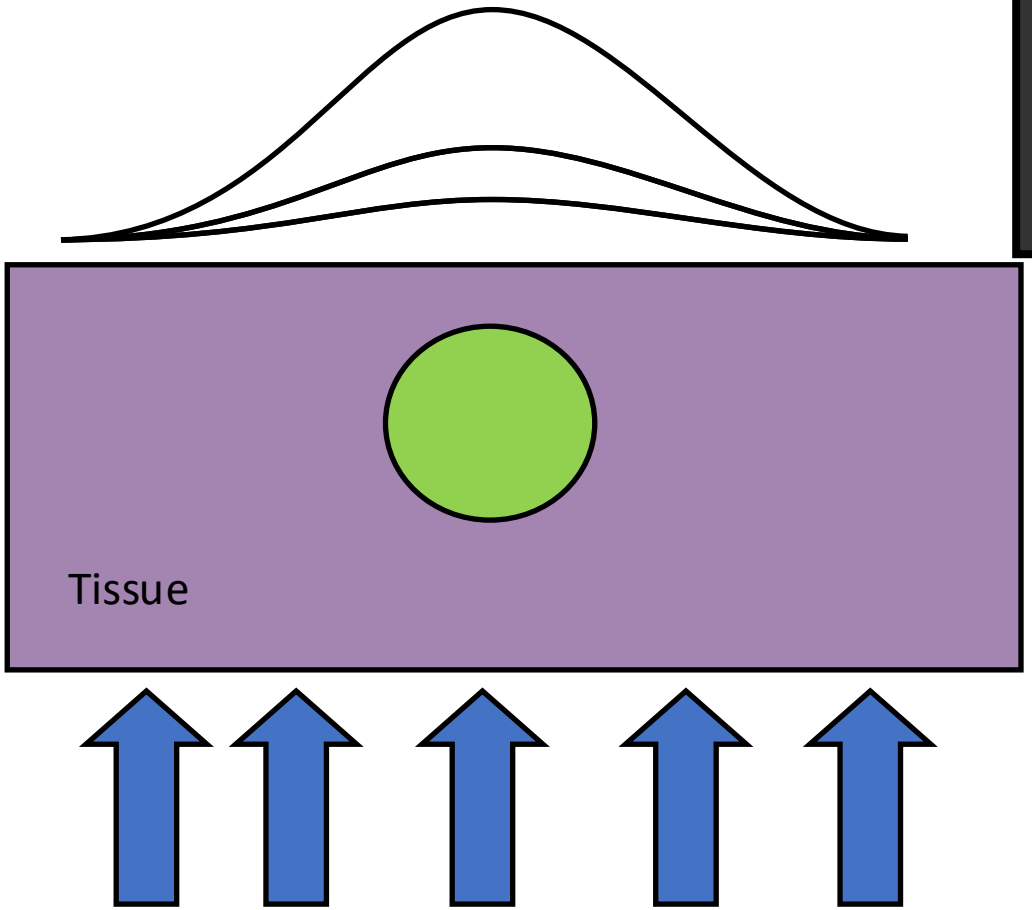




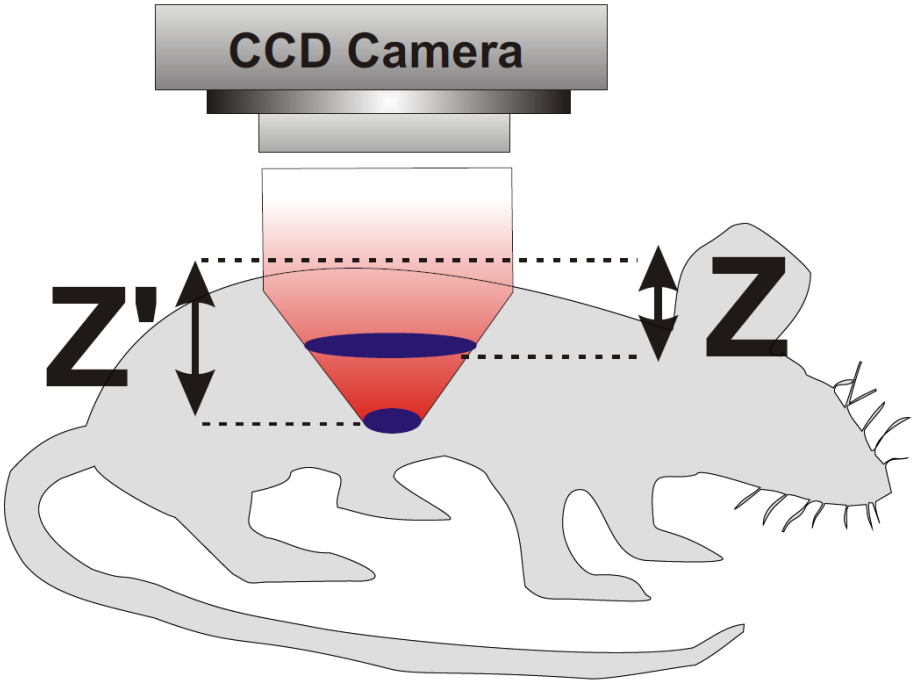
Principles of Tomography

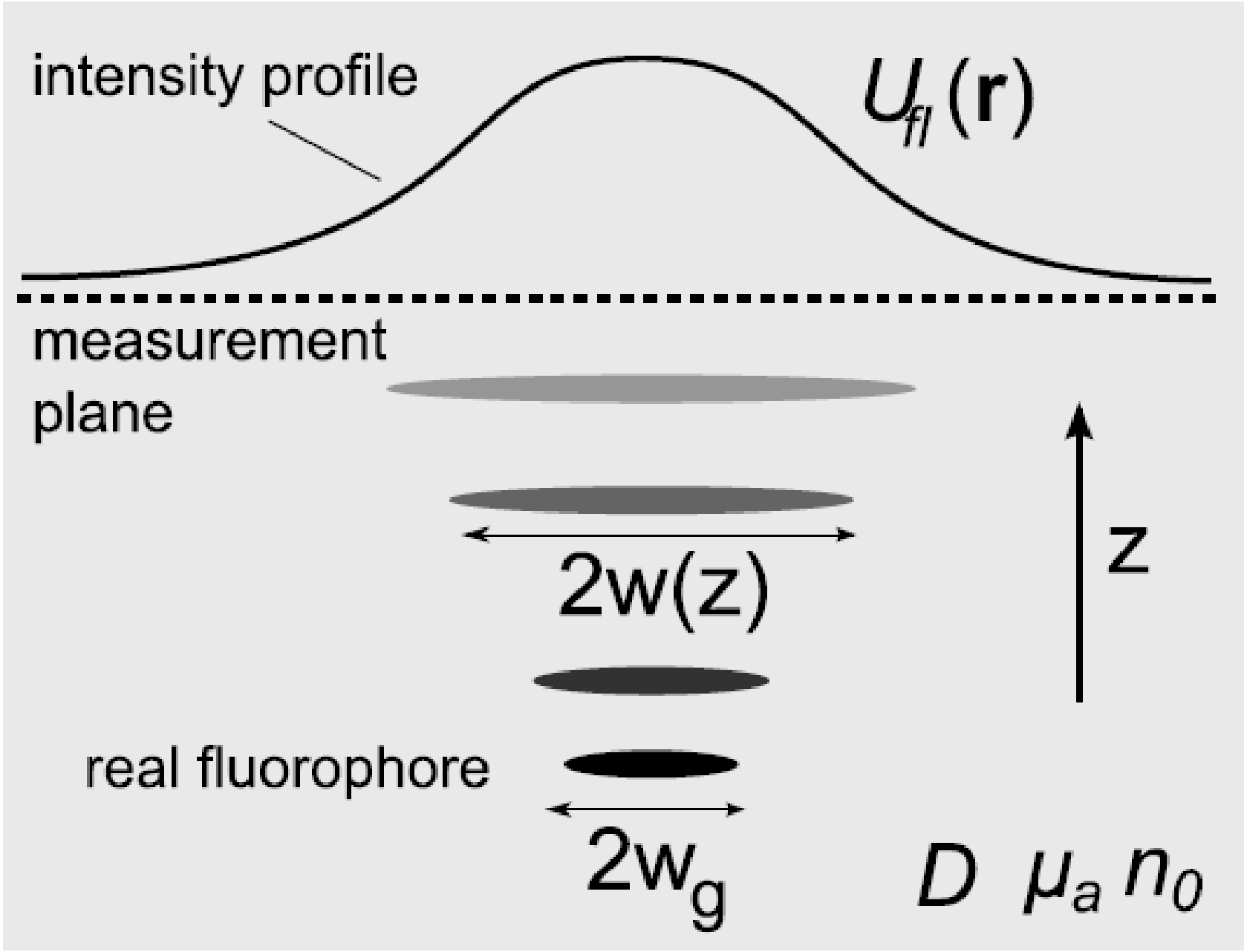
FLUORESCENCE





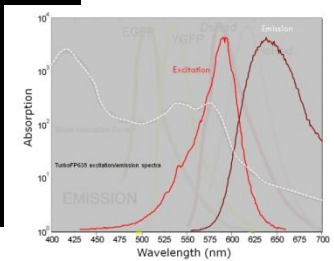
Relationship with Measurements



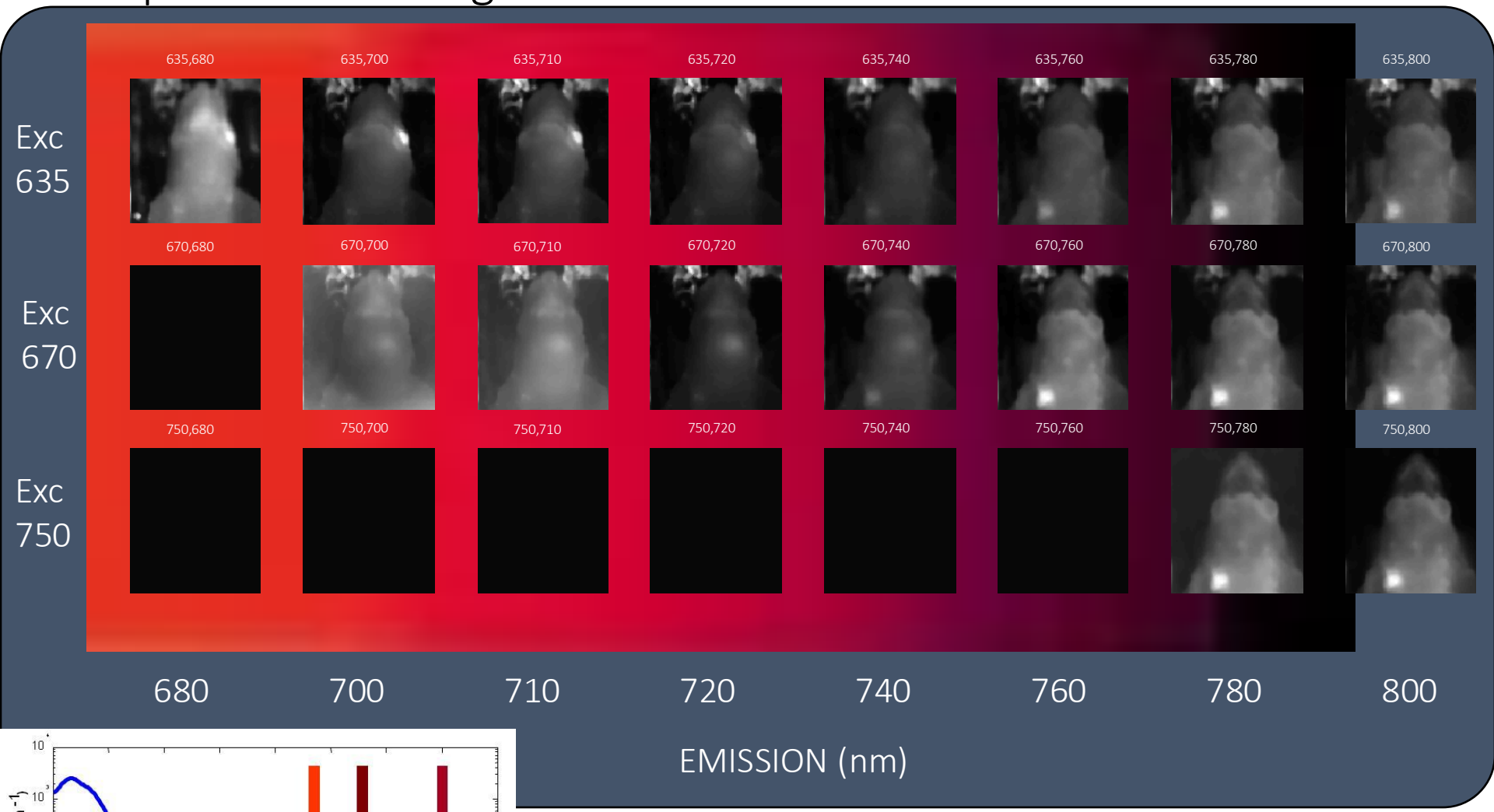

intensity profile
$U_{fl}(\mathbf{r})$
measurement plane
2w(z)
z
real fluorophore
$2w_g$
D μ_a n_0

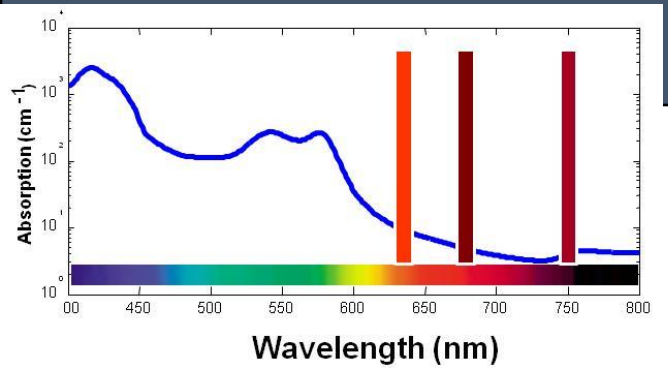
Spatial Information



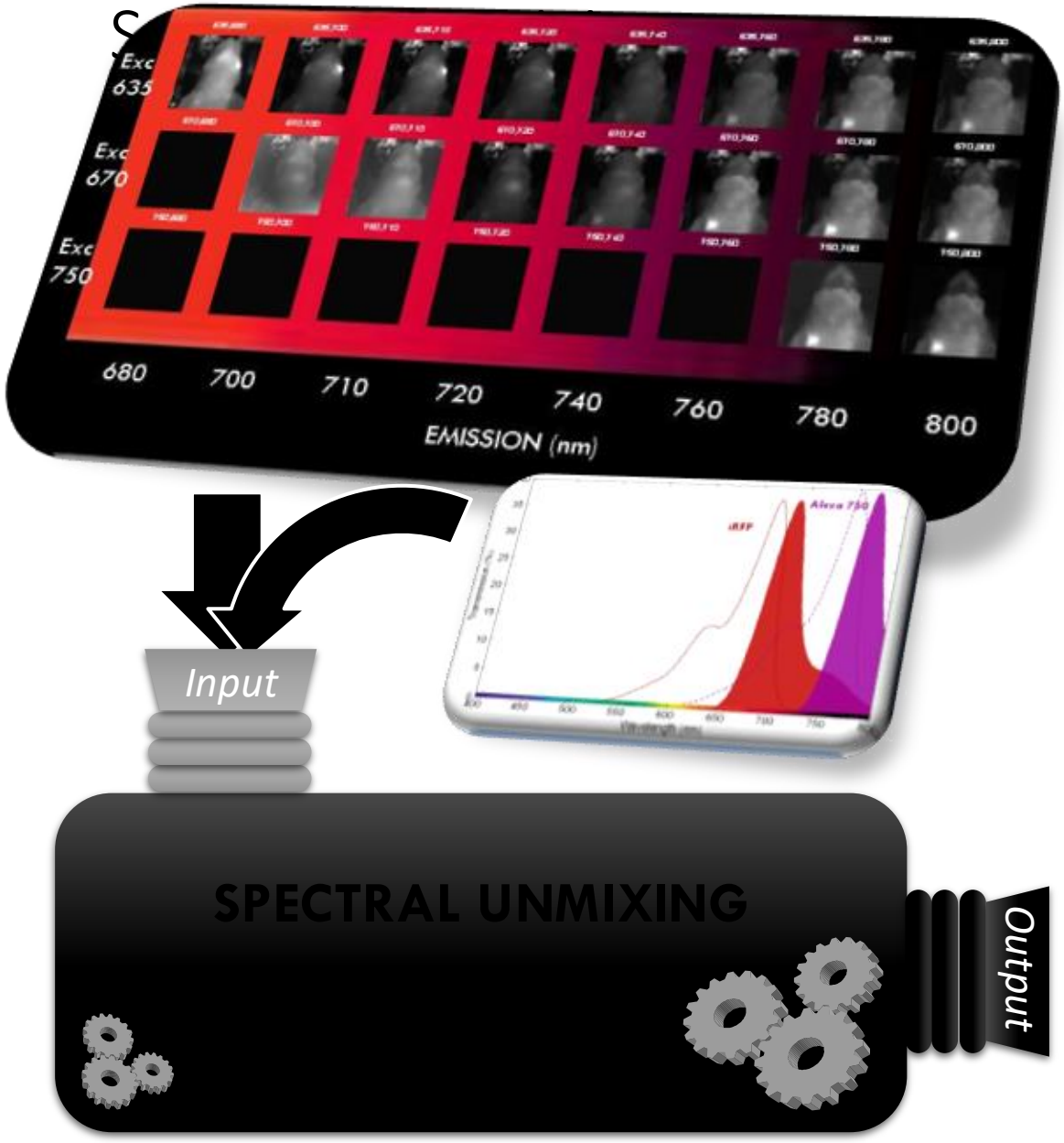


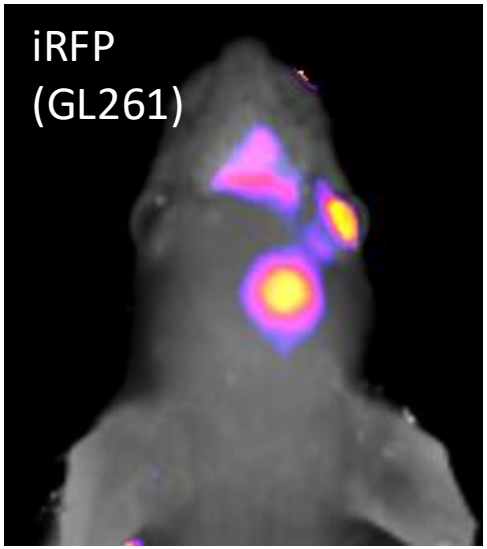
Spectral Unmixing

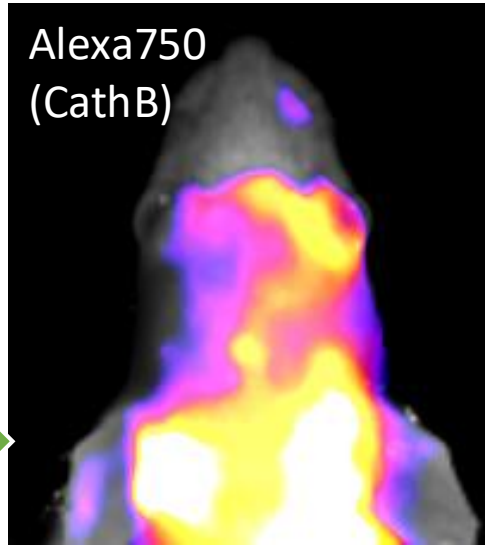




A. Arranz


Exc
635
Exc
670
Exc
750
680
700
710
720
740
760
780
800
EMISSION (nm)
iRFP
Alexa 750
Input
SPECTRAL UNMIXING
Output

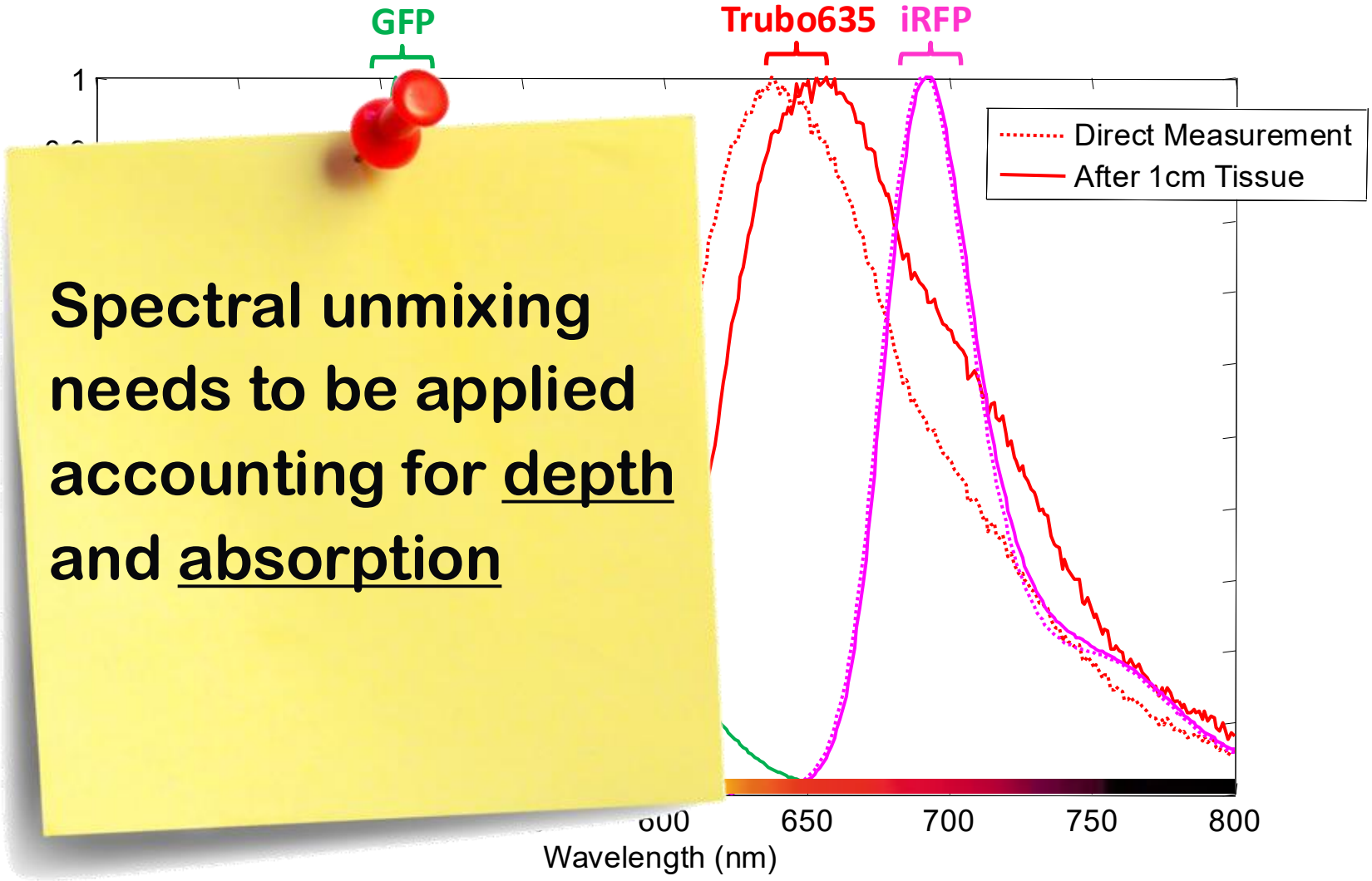

iRFP
(GL261)


Alexa750
(CathB)

A. Arranz

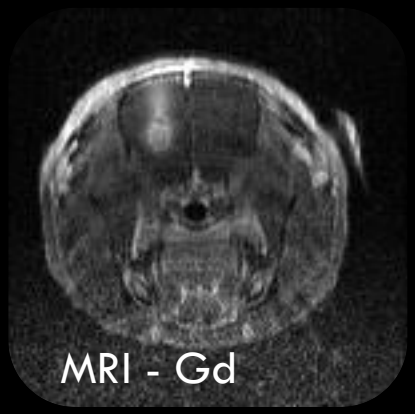
Spectral Unmixing

Spectra change after propagation through tissue due to wavelength-dep. absorption

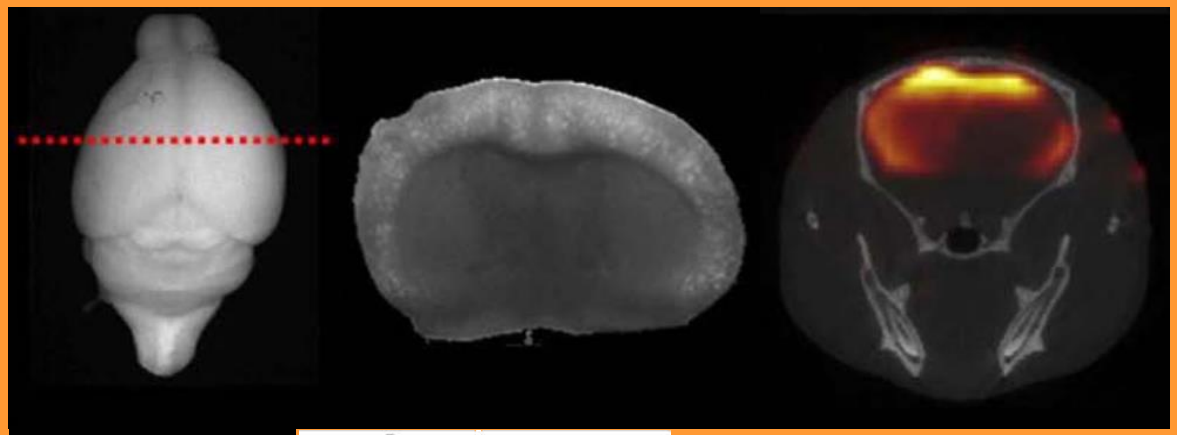


Spectral unmixing needs to be applied accounting for depth and absorption

Accounting for anatomy…



FMT Surfac





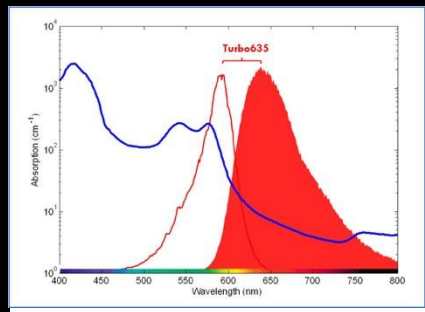


FMT + CT

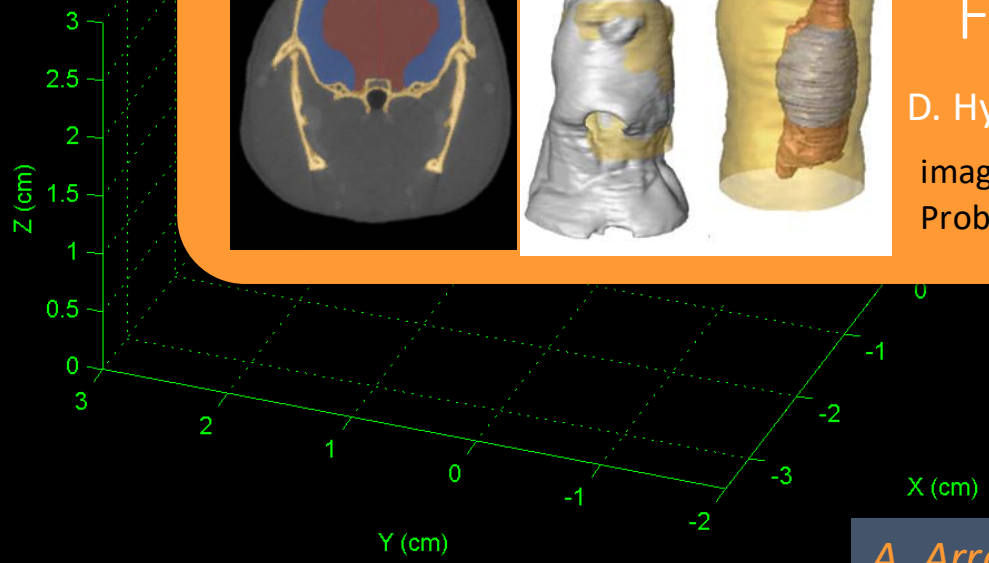
D. Hyde, NeuroImage 2009

imaging of amyloid-β plaques
Probe: AO1987 (Novartis)



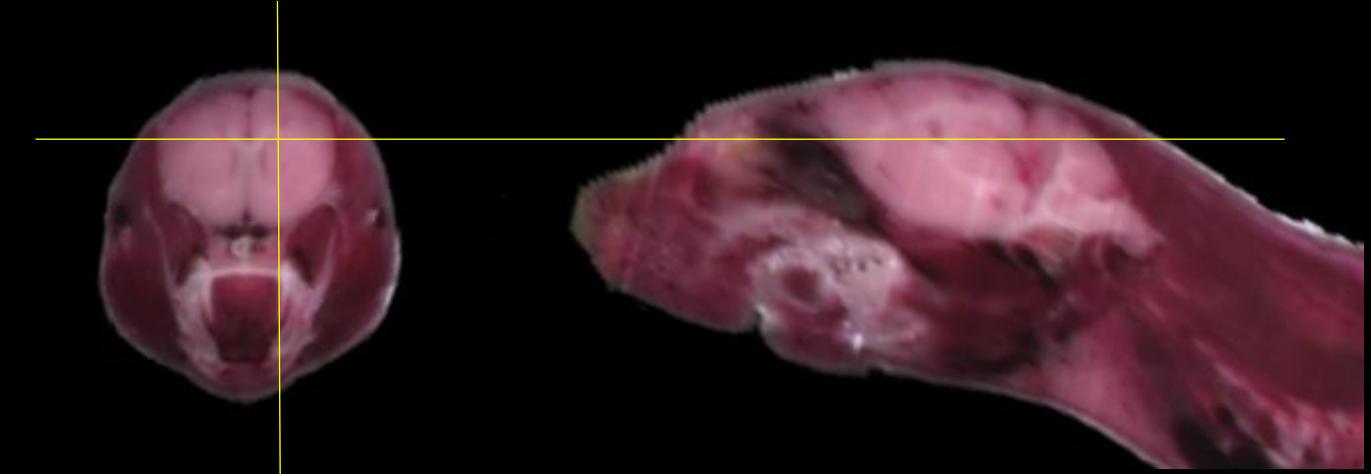


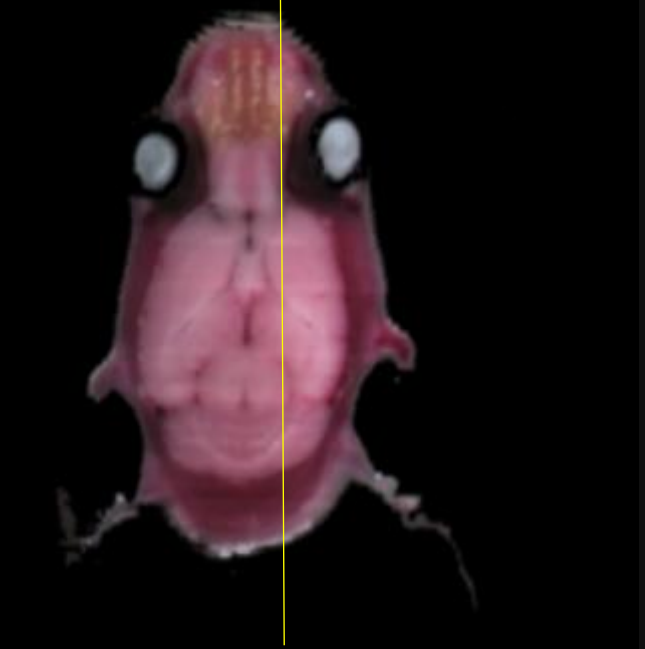
GL261-glioma cell line
Katushka expressing cells

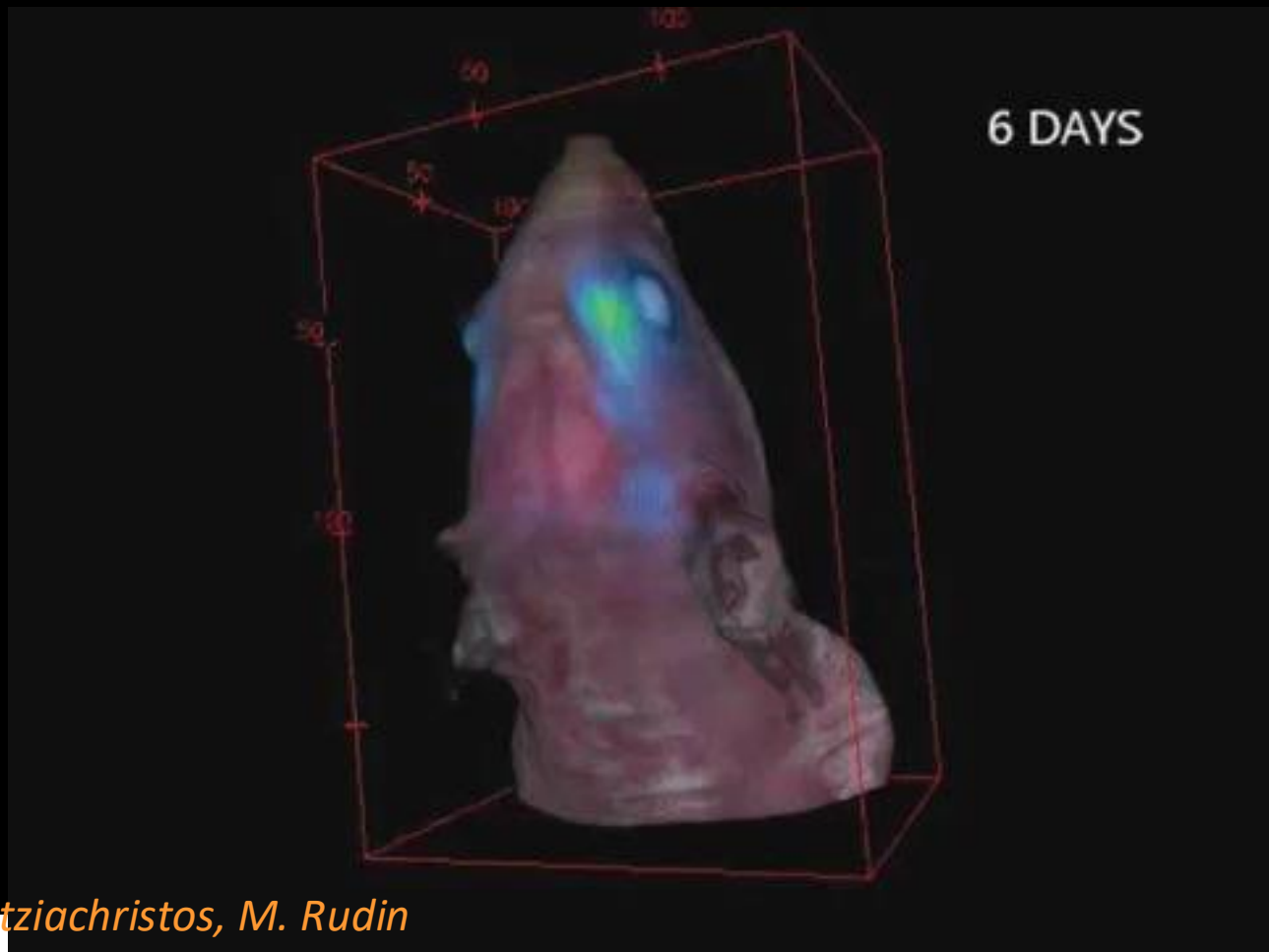


A. Arranz, J. Grandjeaean

ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich





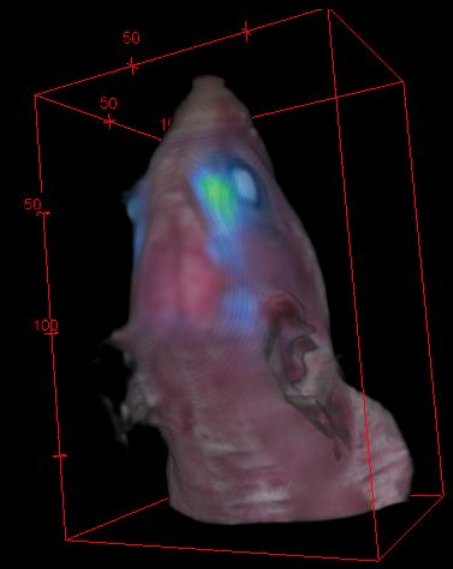


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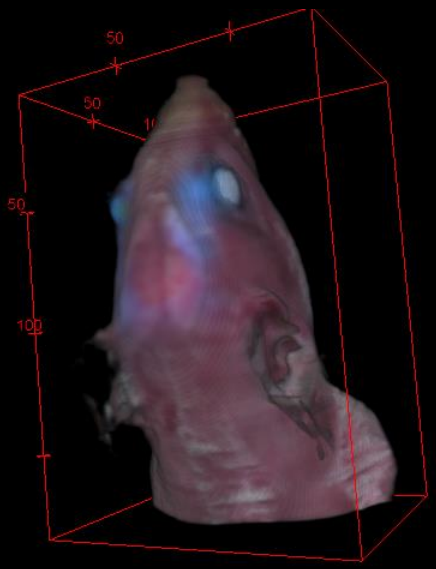
TECHNISCHE UNIVERSITÄT MÜNCHEN

Cryoslicing done at TUM
V. Ntziachristos

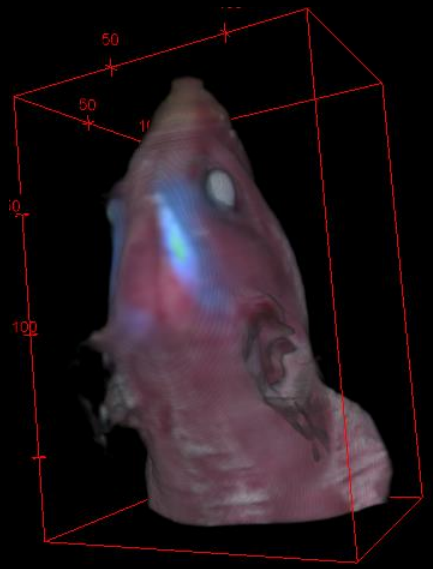
Alicia Arranz, J. Grandjean, V. Ntziachristos, M. Rudin



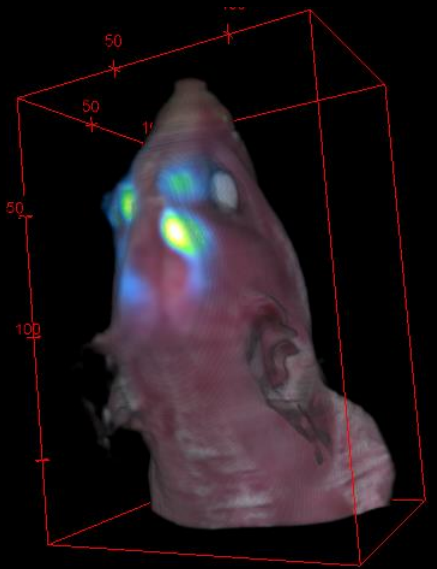
Day 6



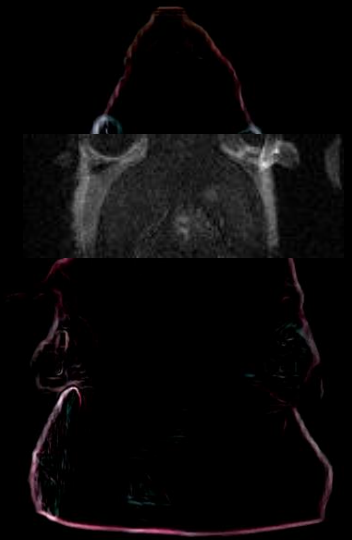
Day 13

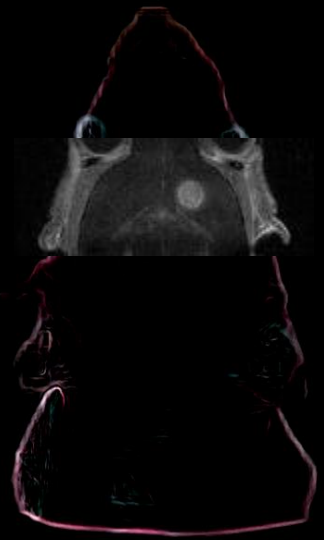


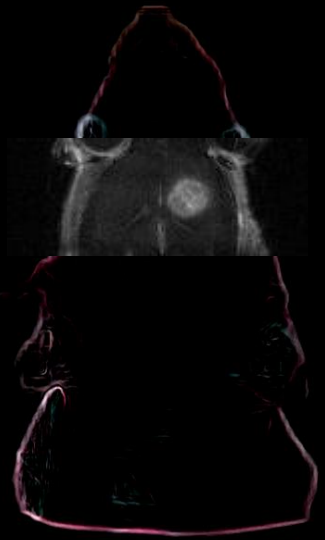
Day 20

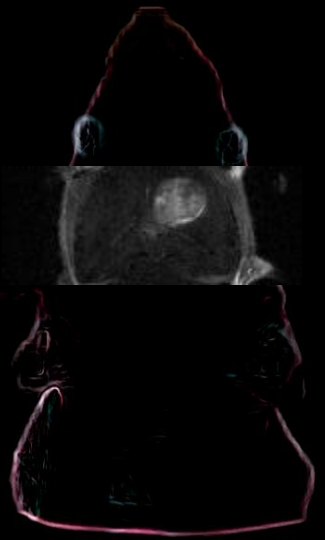


Day 27









Alicia Arranz, J. Grandjean, V. Ntziachristos, M. Rudin

GL261-glioma cell line
iRFP expressing cells

ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich